33 Finite composite insertions beyond the scalar action sector

The Yang–Mills target includes a renormalized family of gauge-invariant local curvature polynomials and covariant derivatives, with its short-distance structure [3, section 4]. A scalar action-density reconstruction does not establish that family. For example, flat torus configurations with different cycle holonomies have the same elementary plaquettes but different winding loops. In SU(3) there is also a local symmetry obstruction: the scalar action is charge-conjugation even, while an appropriately real multiple of $\mathrm{Tr}(F_{12}^3)$ is odd and nonzero. The closure of an even observable algebra cannot automatically supply that odd operator.

At fixed lattice the bounded invariant algebra can nevertheless be specified constructively. Polynomials in link matrix entries and their conjugates form a unital self-adjoint algebra separating points of

the compact link space, hence are dense by Stone–Weierstrass. Haar averaging over site gauges is a norm-one projection onto the invariant continuous functions. The averaged polynomials are therefore a dense gauge-invariant algebra. This finite statement supplies no shrinking-support normalization or continuum limit.

33.1 Explicit local representatives and their real action bound

Set

$$C_N(V) = \frac{V - V^{-1}}{2} - \frac{\operatorname{Tr}(V - V^{-1})}{2N} I, \qquad (\delta_\mu^U X)(x) = \operatorname{Ad}(U_\mu(x)) X(x + a\hat{\mu}) - X(x).$$

The first is holomorphic on $\mathrm{SL}(N, \mathbb{C})$ and conjugation covariant. Repeated raw covariant differences of $C_N(U_p)$, transported to a common base point and contracted with invariant color tensors, give explicit gauge-invariant holomorphic lattice functions. Assign weight $r + 2$ to a curvature with r differences. A polynomial of total dimension d has a representative

$$O_{\alpha,a}(x) = a^{-d} P_\alpha(C_N(U_p), \delta^U C_N(U_p), \ldots). \tag{129}$$

Its fixed determining carrier has a number of lattice cells depending on the chosen composite. Symmetrizing paths and stencils assigns its tensor and reflection transformation law. On a smooth classical connection its leading expansion is the intended curvature-and-derivative polynomial.

For real U, write $a_p = 1 - \Re \operatorname{Tr} U_p / N$. Eigenvalue comparison gives

$$\|C_N(U_p)\|_{\mathrm{HS}}^2 \le 2N a_p.$$

Raw covariant differences are finite sums of isometrically transported curvatures. A nonconstant invariant polynomial for a simple Lie algebra has at least two factors. Its absolute value is bounded by $C(\sum_{p \in C_\alpha(x)} \sqrt{a_p})^m$, with $m \ge 2$. Cauchy–Schwarz and $0 \le a_p \le 2$ reduce this to a fixed multiple of $\sum_p a_p$. Therefore

$$|a^d O_{\alpha,a}(x)| \le C_\alpha \sum_{p \in C_\alpha(x)} a_p, \qquad |a^4 \sum_x f(x) O_{\alpha,a}(x)| \le C_\alpha a^{4-d} \|f\|_\infty \sum_p a_p. \tag{130}$$

Finite overlap proves the second inequality. The dimensional cost is explicit: a dimension-d source costs an additive Wilson shift of order a^{4-d}. For $d > 4$, a fixed physical analytic source disc does not follow from the marginal action-source bound.

Such a disc is not necessary for a composite field. For one standard Gaussian variable, $O =: X^4 := X^4 - 6X^2 + 3$ has every moment, but

$$\mathbb{E} e^{tO} = \infty \quad \text{for every } t > 0. \tag{131}$$

The positive quartic exponent dominates the Gaussian tail. Finite-order insertion transport and quantitative moment bounds are the appropriate extension, rather than demanding a two-sided exponential neighborhood for every higher composite.

33.2 Exact multitype transport and Gaussian Wick normalization

For a frozen transfer kernel K and unmarked input ρ, with nonzero output $\rho'(V) = \int K(V, \mathrm{d}U) \rho(U)$, put

$$(\mathcal{T} O)(V) = \frac{\int K(V, \mathrm{d}U) \rho(U) O(U)}{\rho'(V)}. \tag{132}$$

Take commuting labels z_i with $z_i^2 = 0$. Exponential and logarithm are finite polynomials in their nilpotent ideal, so $\mathcal{W}(z, V) = \log \mathcal{T} \exp(\sum_i z_i O_i)$ needs only finitely many moments. Its z_I coefficient is exactly

$$\kappa_{\mathcal{T}}(O_i : i \in I) = \sum_{\pi \in \mathcal{P}(I)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \mathcal{T}\left(\prod_{i \in B} O_i \right). \tag{133}$$

Gauge covariance of the frozen transfer preserves invariance. If a map or constraint is changed with the source, its derivatives and normalizer derivatives must instead be included; (132) does not erase them.

For two correctly normalized successive kernels,

$$e^{\mathcal{W}_{12}} = \mathcal{T}_2 e^{\mathcal{W}_1}. \tag{134}$$

For two labels this reads

$$\kappa_{12}(O_1, O_2) = \mathcal{T}_2 \kappa_1(O_1, O_2) + \kappa_2(\mathcal{T}_1 O_1, \mathcal{T}_1 O_2).$$

The general partition identity follows by expanding (134). Product insertions are part of the family and cannot be replaced by separate first-insertion expectations. In a reference replacement the numerator must be the carried physical numerator (127) on its history space.

For the Gaussian model let $X = HB + a$, with independent centered Gaussian a, B of covariances C_h, C_c, and $C = C_h + HC_cH^T$. The generating identity is

$$\mathbb{E}_a e^{t \cdot (HB + a) - t^T C t/2} = e^{t \cdot HB - t^T H C_c H^T t/2}. \tag{135}$$

Comparing any finite coefficient of the formal vector t proves

$$\mathbb{E}_a\big[: P(HB + a) :_C \big] = : P(HB) :_{HC_cH^T} \tag{136}$$

for every polynomial P. This is exact Gaussian mixing and subtraction, and is associative over hidden bands. Taking the curvature covariance $C_h = \sum_{j<n} x_j^{-1} \Gamma_j$ includes the running coefficients. Coincident contractions may diverge with the fine cutoff; the formula specifies their finite Wick subtraction. It is not an interacting distribution estimate.

33.3 Finite jet mixing and a scaling remainder

Fix a finite determining carrier and a finite-dimensional space of invariant weighted jets through degree D. Include the identity and the necessary tensor/reflection types. Choose actual analytic representatives P_α whose jets form a basis, with dual extraction functionals λ_β. The first-insertion map gives the exact finite decomposition

$$M_{\beta\alpha} = \lambda_\beta(\mathcal{T} P_\alpha), \qquad R_\alpha = \mathcal{T} P_\alpha - \sum_\beta M_{\beta\alpha} P_\beta, \qquad J_D R_\alpha = 0. \tag{137}$$

Jet extraction is bounded by Cauchy on the fixed complex chart. If a reference M_0 is invertible and $\|M_0^{-1}(M - M_0)\| \le \rho < 1$, then M^{-1} is holomorphic and bounded by $(1 - \rho)^{-1} \|M_0^{-1}\|$. The normalized transferred representatives

$$\widetilde{P}_\alpha = \sum_\gamma (\mathcal{T} P_\gamma)(M^{-1})_{\gamma\alpha}$$

have unit prescribed jets. Symmetries make M an intertwiner. Neither a triangular engineering filtration nor uniform invertibility follows merely from finite dimension; some insertions can have zero conditional expectation.

For the precise remainder gain, let z_i have positive integer weights w_i. If f is analytic and bounded by B on $|z_i| < R$, with every weighted Taylor coefficient of degree at most D zero, then for $|z_i| \le \rho R r^{w_i}$, $0 < r \le 1$, $\rho < 1$,

$$|f(z)| \le B \sum_{w\cdot\nu > D} \rho^{|\nu|} r^{w\cdot\nu} \le B(1-\rho)^{-m} r^{D+1}. \tag{138}$$

This is Cauchy's coefficient estimate followed by a geometric sum. A target operator of degree d leaves the gain r^{D+1-d} after its normalization; an integrated four-dimensional action leaves r^{D-3} after the root count. Actual windows must obey those weights on their entire analytic interpolation, and carrier-size costs must be included before claiming volume uniformity.

34 A clean charge-conjugation-odd curvature-cubic sector

For SU(N), $N \geq 3$, a specific dimension-six tensor sector extends the clean first-insertion construction without differentiating the sharp harmonic curvature lift. We prove the local three-curl Ward factorization and the resulting finite center/stable normalization. The fixed-clean paired-domain and real-g derivative inputs are those already used for the curvature bilinears. Full physical large-field transport, all-order bounds, and tensor rotation covariance are separate obligations.

34.1 The sector and its exact symmetries

For Hermitian Lie algebra variables charge conjugation is

$$\mathsf{C}A = -A^T, \qquad \mathsf{C}U = \overline{U}. \tag{146}$$

The Wilson action, Haar measure, principal regular logarithm, and path-log average commute with this map. In a standard Hermitian basis it is a signed coordinate map, preserving the original symmetric cutoff boxes. The independent current transforms by the same adjoint automorphism. Thus the equivariant frozen clean transfer preserves its odd sector. Projection of a local coefficient by $(O - sfCO)/2$ preserves support and changes its analytic norm by at most a fixed factor.

Write F_I, $I = 1, \ldots, 6$, for the oriented curvature components. The pointed local odd cubics are

$$P_{IJK}(F) = d^{abc} F_I^a F_J^b F_K^c, \qquad I \leq J \leq K, \tag{147}$$

where d^{abc} is the invariant symmetric trace tensor. There are $\binom{8}{3} = 56$ components. They are independent: take $F_I = t_I X$, with $X = \mathrm{diag}(1, 1, -2, 0, \ldots)$ and hence $\operatorname{tr} X^3 \neq 0$. A relation would be a zero cubic polynomial in six independent t_I.

Globally bounded lattice representatives need no logarithmic chart. For the six plaquettes based at one site, set

$$X_p(U) = \frac{U_p - U_p^{-1}}{2i} - \frac{1}{N}\operatorname{tr}\left(\frac{U_p - U_p^{-1}}{2i}\right) I. \tag{148}$$

The symmetrized traces of $X_I X_J X_K$ are gauge invariant and odd under (146). Their fixed normalization gives the leading jets (147). A physical density has mesh factor a^{-6}; its integrated lattice insertion has a^{-2} multiplying the plaquette polynomial.

Every odd gauge-invariant analytic member has zero joint link/current jets of ordinary degrees zero, one and two. The constant is fixed by charge conjugation and hence zero. A linear invariant on the simple adjoint representation vanishes. Each invariant bilinear form is a multiple of the Killing form and is even, including mixed link-current forms. For $\mathfrak{su}(N)$ the invariant cubic tensors are the commutator tensor and the symmetric trace tensor: equivalently, the conjugation-equivariant bilinear maps of traceless matrices are spanned by the traceless parts of XY and YX. The former cubic tensor is even and the latter odd. Thus the odd leading cubic has color tensor d^{abc}.

At least three adjoint factors are needed in an odd invariant. Curvature has dimension two, and differentiation increases dimension. There are therefore no lower-dimensional odd local counterterms. At dimension six the leading types are precisely (147); derivative bilinears are even. These are tensor fields. A Lorentz scalar made from the symmetric-color triple vanishes: transposing the three antisymmetric Lorentz matrices reverses their product with a minus sign, while d^{abc} is symmetric. The scalar angular theorem does not consequently provide their tensor covariance.

34.2 A local three-curl factorization

Lemma 34.1 (Cubic local Ward factorization)**.** *Let $O_X(A)$ be analytic and gauge invariant on a finite connected unit-lattice coarse-data carrier X, with its first two jets zero. Its symmetric cubic derivative T_X factors as*

$$T_X(A_1, A_2, A_3) = \widetilde{T}_X(dA_1, dA_2, dA_3). \tag{149}$$

The factorization costs only a polynomial in carrier diameter in the absolute coefficient norm, and preserves exponential locality after a fixed loss of exponent.

Proof. The infinitesimal gauge transformation is $\delta_\lambda A = d\lambda + O(A\lambda)$. In the order-two Ward identity the nonlinear part multiplies a lower jet and hence vanishes. Polarization gives

$$T_X(d\lambda, A_2, A_3) = 0 \tag{150}$$

separately in every slot.

Choose a root u and coordinate paths γ_x to the endpoints of each edge in X. The loop $\gamma_x + e - \gamma_y$ for $e = (x, y)$ has a cubical plaquette filling P_e. Let $R = 1 + \operatorname{diam} X$. Paths and fillings lie in a box of radius CR, and $\sum_p |P_e(p)| \le CR^2$. Put

$$\psi_x(A) = \sum_{b \in \gamma_x} A_b, \qquad (H\omega)_e = \sum_p P_e(p)\omega_p.$$

Discrete Stokes gives the exact identity

$$A = d\psi(A) + H(dA) \quad \text{on the edges of } X. \tag{151}$$

Use (150) in all three slots. Equation (149) follows with $\widetilde{T}_X(\omega_1, \omega_2, \omega_3) = T_X(H\omega_1, H\omega_2, H\omega_3)$.

For the norm, suppose O_X is bounded by B_X on a coarse-coordinate ball of radius r_X. Cauchy bounds each ordered cubic coefficient by $CB_X r_X^{-3}$. The box has at most CR^4 coarse edges and every filling at most CR^2 plaquettes. Consequently

$$\sum_{p_1,p_2,p_3} |\widetilde{T}_X(p_1,p_2,p_3)| \le CR^{18} B_X r_X^{-3}. \tag{152}$$

A polynomial dependence of r_X^{-1} on R changes only the power.

Expand into the three actual plaquette legs and attach each by a path to X. Do not file the whole filling surface as a carrier. Each plaquette lies within distance CR of the root, and R is bounded by a constant times the original connected tree length plus one. The new marked tree therefore has at most a fixed multiple of that length. A fixed reduction of the locality exponent pays this length increase and all polynomial factors in (152). □

For the actual activity this lemma is applied after its gauge-covariant local coarse-data reconstruction, whose analytic response is the original Lemma 4, (3.53), of [4] and (1.21)–(1.36) of [9]. These are the same reconstruction and Cauchy inputs used for bilinear extraction. Counting the uneliminated microscopic fine links in (152) would be invalid. The counting lattice here is the unit coarse-data lattice, so no inverse microscopic mesh appears. On tori apply the homotopy to contractible inner carriers; wrapping carriers retain their separate exponential tail bound.

The reconstructed coarse-data kernel need not be finite range. The tree-decay statement in (4.2)–(4.5) of [4], followed there by the explicit unit-lattice Ward identities, gives the needed weighted version. Here is the source-specific interface. Write H_j for the actual reconstruction into the physical pair. At the flat pair charge conjugation gives $DO_X(0) = D^2O_X(0) = 0$, so the chain rule reduces exactly to

$$D^3(O_X \circ H_j)(0)[e_1,e_2,e_3] = D^3O_X(0)[DH_j(0)e_1, DH_j(0)e_2, DH_j(0)e_3]. \tag{153}$$

The second and third derivatives of the nonlinear reconstruction have disappeared. The physical local derivative norm in (4.4)–(4.5) of [4], expressed through $(A, \nabla A, \Delta^* A)$ and the associated current, bounds each $DH_j(0)e$ on X by $Ce^{-c\,\mathrm{dist}(e,X)}$, uniformly in j. Cauchy on the fixed local pair core gives

$$|D^3(O_X \circ H_j)(0)[e_1,e_2,e_3]| \le CB_X R_0^{-3} \exp\left(-c\sum_{i=1}^{3} \mathrm{dist}(e_i,X)\right). \tag{154}$$

The unit-coarse edge sum costs a fixed polynomial in $|X|$. A tree joining $u \in X$ and the three legs has length at most $C|X| + C\sum_i \mathrm{dist}(e_i,X)$. Spending a fixed portion of the carrier exponential supplies the following weighted bound. No large global coarse-coordinate disc is needed.

Suppose a separately divergence-free kernel obeys

$$\sum_{e_1,e_2,e_3} e^{\kappa\tau(u,e_1,e_2,e_3)} |T(e_1,e_2,e_3)| \le B, \tag{155}$$

where τ is rooted tree length on unit coarse edges. Choose the same paths and fillings for all edges. Their coefficient sums are at most $C(1+\mathrm{dist}(u,e))^2$. If p_i occurs in P_{e_i}, then

$$\tau(u,p_1,p_2,p_3) \le C_0(1+\tau(u,e_1,e_2,e_3)).$$

For a sufficiently small fixed $\kappa' > 0$, the three polynomial filling factors and the new weight are bounded by the old exponential weight. The three-curl coefficient S consequently satisfies

$$\sum_{p_1,p_2,p_3} e^{\kappa'\tau(u,p_1,p_2,p_3)} |S(p_1,p_2,p_3)| \le CB. \tag{156}$$

Absolute convergence justifies every rearrangement. To use the Ward identity on the path potential $\psi(A)$, first truncate it outside radius R and then let $R \to \infty$; the exponential kernel tail dominates its polynomial growth. The original coarse derivative tree bound and unit-lattice shell sums supply (155) uniformly in microscopic depth. Anchoring at a point of X spends only a further fixed portion of its carrier exponent. Finite-window boundary terms retain their exponential remote tail. Thus no finite-range assumption on the actual reconstruction is being made.

34.3 Constant-curvature extraction and the own current

Set the three curvature legs of (149) to constant components. This defines a bounded local map $\mathcal{B}_3$ into the 56 coefficients, extracted before summing translated roots. Choose local completions P_α with $\mathcal{B}_3 P_\alpha = e_\alpha$.

Mixed cubic terms involving the independent current are retained. They are not declared irrelevant on an arbitrary independent-current ball. On an actual old inner window, with scale ratio r, the original reconstruction forms its own current. Its localized linear response annihilates affine constant-curvature data. Up to polynomial carrier factors, the verified readout is

$$A = r^2 A(F_0) + E_A, \qquad |E_A| + |J^{\mathrm{own}}| \le C(r^{2+\zeta}\alpha + r^4\alpha^2), \tag{157}$$

with the same $\zeta > 0$ as in the bilinear argument. It comes from the analytic own-current response, not differentiation of raw fine Hölder plaquettes.

If $\mathcal{B}_3 O_X = 0$, expand its cubic joint Taylor term. The pure A^3 term vanishes on the constant-curvature part. Its remaining terms have one E_A leg, and every $A^2 J$, AJ^2, or J^3 term has an own-current leg. Degree at least four supplies r^8. Hence on the actual inner core, after a fixed locality loss,

$$|O_X(\text{inner readout})| \le C B_X \left[(\alpha/R_0)^3 r^{6+\zeta} + (\alpha/R_0)^4 r^8\right]. \tag{158}$$

Polynomial carrier factors are paid before the old-carrier sum. Carriers longer than the inner window go into its separately controlled remote tail. No derivative of harmonic curvature beyond the existing Hölder comparison is used. This treats the complete zero-center remainder, including mixed-current terms; outer and removed pieces keep their own exponential response estimates.

34.4 The clean finite first-insertion cocycle

Use engineering weight six. A point insertion is rescaled by L^6; for an integrated density the L^4 root count and source factor L^2 give the same normalization. Write $\widehat{T}_g$ for the accordingly rescaled clean first-insertion transfer. Affine reproduction gives

$$\mathcal{B}_3 \widehat{T}_0 P_\alpha = e_\alpha, \qquad \mathcal{B}_3 \widehat{T}_0 R = 0 \quad (\mathcal{B}_3 R = 0). \tag{159}$$

All one-step constants may depend on the single fixed L. Equation (158), after the engineering and root powers, leaves the strict factor r^ζ. Store remainders with their birth norms and choose $0 < \sigma < \zeta$. Pure aging contracts by $L^{-\sigma}$; fresh stable formation has the original small cutoff factor. The same choices as in the bilinear proof give a stable norm $q < 1$.

The fixed-core real-g derivative lemma applies to every fixed analytic insertion. The connected polynomial Gaussian proof bounds its first two g derivatives uniformly. Simultaneous $(g, B) \mapsto (-g, -B)$ makes the transfer even. Therefore, with old clean parameters frozen,

$$\|\widehat{T}_g O - \widehat{T}_0 O\|_{\mathrm{loc}} \le C_L g^2 \|O\|_{\mathrm{loc}}. \tag{160}$$

The current cancellation and cutoff-boundary flatness are unchanged. No ordinary Gaussian completion for a cubic source is assumed. Directly, contraction of the leading Gaussian cubic to one color leg also vanishes because $\sum_a d^{aac} = 0$.

The resulting clean linear first-insertion state has the form

$$\begin{gathered} c_{j+1} = (I + A_j)c_j + B_j r_j, \qquad r_{j+1} = C_j c_j + D_j r_j, \\ c_j \in \mathbb{C}^{56}, \qquad \|A_j\| + \|B_j\| \le C/x_j, \qquad \|C_j\| \le C_0, \qquad \|D_j\| \le q < 1. \end{gathered} \tag{161}$$

There is no lower-dimensional odd unstable block. The finite graph calculation already proved for the bilinears applies verbatim:

$$M_j = I + A_j + B_j\Gamma_j, \qquad \Gamma_{j+1} = (C_j + D_j\Gamma_j)M_j^{-1}.$$

At a sufficiently weak floor the stable graph stays bounded, every M_j is invertible, and

$$\|Z_{n:0}\| + \|Z_{n:0}^{-1}\| \le 2\exp\left(C\sum_{j<n} x_j^{-1}\right), \qquad Z_{n:0} = M_{n-1}\cdots M_0. \tag{162}$$

The running tube makes this polynomial in ultraviolet depth or endpoint inverse-coupling ratio. Multiplying the finite-cutoff bare basis by $Z_{n:0}^{-1}$ fixes its terminal pointed cubic center to the identity. This gives a clean normalized first-insertion family on the actual fixed paired core.

35 Joint curvature insertions: the protected quadratic/cubic family

This note addresses a different problem from a one-insertion normalization matrix. Several renormalized curvature-cubic insertions generate even lower-degree coefficients, including a quadratic field term and a vacuum. Those terms cannot be discarded. The exact finite-source recursion below keeps them, and a completed source bridge makes their ultraviolet formation summable for separated supports.

The clean conclusion uses the fixed paired-core, localized old-member, and rooted connected-insertion estimates already stated in the clean notes. The physical conclusion additionally needs the original-map marked-history extension. Neither conclusion resolves the derivative-counterterm problem for a general higher-dimensional single insertion. All observable normalizations are fixed on the original Wilson sequence, before a source profile, separation grid, or tuple length is chosen.

35.4 4. A finite interacting filtration

The preceding Gaussian degree rule extends to a useful bound, though not to a claim that the complete interacting source state is polynomial. Define the even integer

$$e(I,q) = 2\left\lceil \frac{\max\{2(m_I-1),\, P_I - q\}}{2} \right\rceil . \tag{JC10}$$

For any fixed maximal source order and any fixed number of physical field derivatives, the rooted clean coefficient bound has the form

$$\|\mathsf{J}_q O_I\| \le C_I g^{e(I,q)} \prod_{i\in I} a_i + \text{retained cutoff-tail bound.} \tag{JC11}$$

The field/core powers have been included in the definition of the bounded local norms a_i; the claim concerns the coupling powers. Ordinary Taylor coefficients beyond any fixed required degree are kept as an analytic remainder on the smaller fixed core. No convergence of perturbation theory in g is asserted.

Here is the finite-jet argument. Temporarily expand only to the finite order needed in real current g and in the physical coarse data. A primitive source vertex has at least p_i field legs. A source-free nonlinear action vertex of total field valence $l_v \ge 3$ has the action factor g^{-2}, with one g for each hidden fluctuation leg. Background derivatives of the Gaussian covariance can be represented by the same vertices, with two hidden and the remaining external legs. Old bounded irrelevant vertices have

at least the same favorable powers. If there are V action vertices, E hidden pairings, and q external field legs, the current power is

$$2E - 2V \;\geq\; P_I - q + \sum_v (l_v - 2) \;\geq\; P_I - q. \tag{JC12}$$

Connectedness also gives $E \geq m_I + V - 1$, hence the lower bound $2(m_I - 1)$. The exact simultaneous change $(g, B) \mapsto (-g, -B)$ removes odd current powers. This proves (JC10). Nonlinear physical reconstruction adds external legs or higher source jets; it cannot invalidate either inequality.

The coefficient-vanishing argument is finite. The Taylor remainder is bounded by the actual fixed-core real-current derivative lemma and rooted Gaussian polynomial moments. It does not require termwise summation of an infinite Feynman series. The original frozen canonical cutoff differs from the full Gaussian by a tail; for fixed order its polynomial losses in g^{-1}, the source degree, and the coordinate count are retained in

$$C_I x^{b_I} e^{-cP(x)^2}. \tag{JC13}$$

A truncated Gaussian is not exactly Wick Gaussian. In particular, its higher connected moments of linear forms need not vanish. This is why (JC13) is kept separately rather than silently treating the cutoff Gaussian as the full one.

The filtration is stable under old mixed transport. Indeed, suppose a partition has r blocks B, with field degrees q_B. The inherited powers and the new connected joining give a lower bound

$$\sum_{B \in \pi} e(B, q_B) + \max\{2(r-1), \sum_B q_B - q\}. \tag{JC14}$$

It is at least $2(m_I - 1)$, by the first alternatives, and at least $P_I - q$, by the second alternatives. Its allowed parity is even. Thus it is at least (JC10). The one-block term retains the same conclusion after its Gaussian contractions. This is the actual triangular source recursion; no new assumption that old transported observables are current Wilson actions has been made.

For a Gaussian-allowed degree in (JC8), the term of order $g^{P_I - q}$ is exactly the Gaussian Wick coefficient. Additional source valence or nonlinear action valence costs a positive power; parity makes the first possible correction two powers higher. For other field degrees, nonlinear classical source terms may occur at the minimum $g^{2(m_I - 1)}$. They must be kept as analytic field functions, rather than mislabeled as a small quantum remainder.

35.5 5. Extending the fixed first normalization to smooth local profiles

This is the interface needed before multiplying distinct insertions. It is weaker than a complex product domain of arbitrary fine plaquette couplings. We prove it on a fixed finite smoothness class. A constant-profile cocycle and a one-step kernel moment alone do not automatically give a cutoff-uniform estimate on an arbitrarily prescribed low Holder class.

Let $\widehat{T}_j$ denote the engineering-rescaled linear marked transfer on one of the two protected sectors, including its actual stable complement. The constant-profile normalization gives its finite center cocycle M_j and bounded center/stable frames. In those frames the center propagates by matrices $M_j = I + O(x_j^{-1})$, the stable block contracts by $q < 1$, and the normalized propagators have at most a fixed polynomial loss in the endpoint inverse-coupling ratio. The original wall localization gives the one-step weighted kernel moment

$$\sum_v \|\widehat{T}_j(u, v)\| e^{\kappa \operatorname{dist}(u,v)/\ell_j} \leq C. \tag{JC15}$$

Here the zero-frequency center is a POINTED local density jet, before summing translated source roots and before quotienting by total derivatives. It is not defined by differentiating a global constant-source partition function. For example, the perturbative integral of $\mathrm{tr}(F \wedge F)$ vanishes in the periodic trivial sector, while its local curvature-quadratic density is nonzero. This direction is retained in the 21-dimensional pointed center. Section 11 makes the fixed reflection representation and the full R-origin first matrices explicit. The norm here is the actual local activity/center norm, with the old-member filing retained. An energy norm of a global Schur operator alone would not imply (JC15).

The translation representation must be specified at the FUNCTIONAL level. An external nonconstant background is not held fixed and treated as a translation-invariant numerical coefficient. At input root v, translate the entire pointed activity, its field/current arguments, and its origin records into coordinates relative to v. The resulting reference fiber $\mathcal{E}_j$ consists of analytic functions of those relative background variables, together with the finite center and the stable age state. Include bond/plaquette orientations and the finitely many within-block root offsets as fiber types. Output root u is represented in the same way on the next lattice.

Covariance of the original fixed blocking construction under block translations then gives an operator kernel $\mathcal{T}_j(r) : \mathcal{E}_j \to \mathcal{E}_{j+1}$, with $r = v - Lu$. Each operator still acts on the COMPLETE background functional. For a pure source mode, the exact identity is

$$\sum_v e^{ip\cdot v}\mathcal{T}_j(v - Lu)O = e^{iLp\cdot u}\widehat{T}_j(p)O, \qquad \widehat{T}_j(p) = \sum_r e^{ip\cdot r}\mathcal{T}_j(r). \tag{JC15b}$$

The sums and moment bounds are in the activity-operator norm. Writing $r = La + \delta$ retains the phase $e^{ip\cdot\delta}$ of every within-block offset. Thus the coarse momentum is Lp modulo 2π, and the aliases are part of the operator symbol, not discarded scalar modes. This is the precise Bloch interpretation used below. The weighted actual wall kernel gives its complex strip and its moment bound. If only a numerical kernel evaluated at a flat background were available, that would not establish (JC15b) on the full paired activity core.

For the zero-frequency norm statement one needs the stable foliation too, not just an invariant center graph. After the center graph has made the zero-frequency step upper triangular, write it as $\left(\begin{smallmatrix} M_j & B_j \\ 0 & D_j \end{smallmatrix}\right)$, with $\|D_j\| \le q < 1$ and $\|M_j^{-1}\| \le 1 + \epsilon$. On any finite interval ending at s, choose the stable graph $c = L_j r$ by $L_s = 0$ and

$$L_j = M_j^{-1}L_{j+1}D_j - M_j^{-1}B_j = -\sum_{k=j}^{s-1}(M_k \cdots M_j)^{-1}B_k(D_{k-1} \cdots D_j). \tag{JC15a}$$

The empty D product is the identity. Taking the weak ceiling so that $q(1 + \epsilon) < 1$ makes these graphs uniformly bounded. The resulting change of frame block-diagonalizes the zero-frequency cocycle. This auxiliary finite-interval stable frame changes no observable normalization.

The one-step symbol satisfies

$$\|\widehat{T}_j(p) - \widehat{T}_j(0)\| \le C\min\{|p|, 1\}, \qquad \|\widehat{T}_j(p)\| \le C_L. \tag{JC16}$$

At physical frequency k, put $p_j = \ell_j k$. First suppose $\ell_s|k| \le 1$. Conjugate the center at scale j by its remaining zero-frequency propagator to scale s; scale the stable norm by a sufficiently large fixed power of x_j/x_s. The zero-frequency step then has norm at most one, with a strictly contracting stable block, while the perturbation is at most $C(1 + s - j)^b\ell_j|k|$. The elementary product bound is uniform because

$$\sum_{j\le s}(1 + s - j)^b\ell_j|k| \le C_b\ell_s|k|. \tag{JC17}$$

The initial stable transient is exponentially dying and obeys the same bound after its polynomial normalization loss.

For $\ell_s|k| > 1$, stop this low-frequency argument at the last j_0 for which $\ell_{j_0}|k| \le 1$. Only $O(1+\log(\ell_s|k|))$ steps remain. Their norms are at most C_L each. The difference between the first normalization at j_0 and the one fixed at s is a polynomial in x_{j_0}/x_s, hence a polynomial in $1+\log(\ell_s|k|)$. Therefore, for some fixed finite B, the normalized complete Fourier-mode transfer is bounded by

$$C\,\mathrm{poly}(x_s)(1+\ell_s|k|)^B, \tag{JC17a}$$

uniformly in the number of microscopic steps. The exponent B depends on the fixed first family and block factor, not on a future tuple length. The weighted kernel in (JC15) also gives these estimates for $|\operatorname{Im} k| \le c/\ell_s$. After partitioning a compact profile into smooth own-ℓ_s patches, the same shifted-symbol estimate retains an exponential from each original source patch to its output carrier.

Choose one integer $s_0 > B+4$. A profile on a patch of diameter $C\ell$, with ℓ-scaled derivatives through order s_0 bounded by t, has Fourier coefficients bounded by $Ct\ell^4(1+\ell|k|)^{-s_0}$, with the usual torus-volume normalization. Summing (JC17a) proves the own-cell estimate

$$\|O_{i,\ell}(X)\|_{R_0} \le C\ell^{4-D_i}\,\mathrm{poly}(x_\ell)\sum_{|\gamma|\le s_0}\ell^{|\gamma|}\|\partial^\gamma f_i\|_\infty e^{-\kappa d(X,\operatorname{supp} f_i)/\ell}, \tag{JC18}$$

with the corresponding rooted carrier sum and smooth-patch decomposition understood. The fixed clean core has the expected $R_0^{p_i}$ zero-jet factor. A bare constant-profile matrix alone, without a local moment such as (JC15), would not prove (JC18). In the present construction (JC15) is supplied by the actual local reconstruction and generic wall/cluster localization; it is not inferred from a source Cauchy estimate in an unspecified Banach norm.

Equation (JC18) uses exactly the normalization selected before f_i or a tuple was given. The separation-scale analysis below introduces no new observable definition.

35.6 6. Separated coefficients: a fixed-order clean induction

Fix a finite number m of labels from the protected family. Suppose their supports have diameter at most Ch, are pairwise at distance at least $h>0$, and lie in a fixed multiple of an mh neighborhood. Put

$$t_i = \sum_{|\gamma|\le s_0} h^{|\gamma|}\|\partial^\gamma f_i\|_\infty. \tag{JC19}$$

Constants may depend on m and the fixed shape bound. The goal here is a uniform cutoff bound at every fixed m, not an optimal factorial constant.

The linear propagation used below is the already established one-root resolvent on the original paired domain. For a vacuum-subtracted gauge-invariant analytic member with its pointed counterterms retained, it gives a factor

$$C\,\mathrm{poly}(x_{\ell'}/x_\ell)(\ell'/\ell)^\lambda, \qquad \lambda = 4-\delta > 0, \tag{JC20}$$

from a birth scale ℓ' to ℓ. The stronger cubic readout can be used for an odd coefficient, but is not needed for mixed members. Genuine linear formations are summed in this resolvent before proper-subset joinings are counted. In particular there is no factor $C^{\#\text{scales}}$ left in (JC20).

35.6.1 6.1 What a genuine multi-label birth costs

At one scale $\ell \le h$, multiply the own-scale engineering weights of its input label subsets. Their product is always $\ell^{-\sigma_I}$. The connected cluster kernel and the previously retained source carriers form one connected carrier meeting every source support in I. If $|I| \ge 2$, that carrier has physical length at least h. Reserve its exponential before summing any internal roots. With spare wall decay, the local birth bound is consequently

$$C_I \ell^{-\sigma_I} \operatorname{poly}_I(x_\ell) e^{-c_I h/\ell} \prod_{i \in I} t_i, \tag{JC21}$$

in the rooted local norm. The proof is induction on $|I|$ in (JC5): a proper mixed child already carries its full source tree, while otherwise the new connected joining supplies the missing connection. The union of these trees connects all labels. An old one-block term is handled by (JC20), not by pretending that it is a new joining.

There are $O((h/\ell)^4)$ possible first-source roots at this scale for the stated shapes. The other source roots are summed against the connected kernel; they do not contribute independent ultraviolet volumes. Thus the integrated birth bound is (JC21) multiplied by $C_I(h/\ell)^4$.

35.6.2 6.2 The bridge pays both dimension and old-root counts

Set $r = \ell'/\ell \le 1$. Comparing an old density with the current density introduces the old-root factor r^{-4}, as well as the engineering factor $r^{-\sigma_I}$. The correct ratio is therefore

$$r^{\lambda - \sigma_I - 4} \operatorname{poly}(1 + |\log r|) e^{-ch/(r\ell)}. \tag{JC22}$$

The factor r^{-4} must not be omitted in a per-current-cell density norm. Equivalently, one may keep the integrated old-root norm throughout and never change its counting measure.

For $h/\ell \ge 1$, a fixed fraction of the exponential bounds every power in (JC22). On the geometric scales, for every fixed I,

$$\sum_{r \in \{L^{-n} : n \ge 0\}} r^{\lambda - \sigma_I - 4} (1 + |\log r|)^b e^{-ch/(r\ell)} \le C_{I,b} e^{-c'h/\ell}. \tag{JC23}$$

One proof puts $u = r^{-1}$, bounds the polynomial by a fixed power of u, and compares the geometric sum with $\int_1^\infty u^M e^{-c(h/\ell)u}\, du$. This also makes its finite dependence on I explicit. For cubic labels the needed power grows only linearly with $|I|$; the resulting constants can be bounded by a factorial of order $2|I| + O(1)$, apart from the independently stated cluster constants.

This is why a generated quadratic term does not destroy separated ultraviolet summability. It propagates with degree four, while its original two or more source anchors supply the missing small-scale suppression. Assigning it a false degree-six readout is unnecessary.

35.6.3 6.3 Closing the triangular induction

Expand the exact recursion at its proper-subset joinings, using the full linear resolvent between them. A graph with m_I primitive labels has at most $m_I - 1$ such joinings. This is a finite combinatorial statement, independent of the number of scales. At each joining use the connected kernel bound and reserve the source bridge. Sum the intervening linear ages using (JC23). A fixed locality reserve can be allocated to the complete finite joining graph before summing its internal roots.

This proves (JC21) for all mixed members, including all lower-degree field jets and the analytic remainder, with a possibly smaller fixed $c_I > 0$. It is a triangular induction, not a nonlinear smallness assumption on the highest source coefficient. Scalar outputs are bounded by the same rooted moment estimate and recorded once.

The total mixed scalar output below h is bounded by

$$\begin{aligned} C_I h^4 \prod_i t_i \sum_{\ell \le h} \ell^{-\sigma_I - 4} \operatorname{poly}_I(x_\ell) e^{-c_I h/\ell} \\ \le C_I h^{-\sigma_I} \operatorname{poly}_I(x_h) \prod_i t_i. \end{aligned} \tag{JC24}$$

The field-dependent outputs transported to h obey the same bound, with the additional favorable factor $(\ell/h)^\lambda$ inside the sum. The weak-tube relation $x_\ell \le x_h + C(1+\log(h/\ell))$ pays the older coupling powers by the same exponential. The lower cutoff has disappeared from the bound.

At $\ell \asymp h$, all retained subset packets have total rooted norms bounded by the right side of (JC24); first packets obey (JC18). Above this scale there are only their original finite sets of anchors. Resolve the linear one-root propagators first and sum their geometric decay. Each subsequent proper joining still increases label degree, so only finitely many joining vertices occur at fixed m. The unchanged local connected estimates therefore bound all later scalar emissions and retained field coefficients by

$$C_{I,T} h^{-\sigma_I} \operatorname{poly}_I(x_h) \prod_i t_i \tag{JC25}$$

on a fixed physical reference torus. There is no ultraviolet translation count in this post-h estimate.

Finally, the exact logarithm of the retained integral is the sum, over partitions of I, of cumulants of the retained zero-flat subset functions. For any positive retained source-zero law, each such cumulant is bounded by the product of their sup norms times a finite partition constant. Since $\sum_{B\in\pi} \sigma_B = \sigma_I$, every partition obeys (JC25). The retained field integral is included; it has not been replaced by its flat value.

Clean fixed-order conclusion. Under the stated original clean paired-core, rooted linear-resolvent, and connected cluster inputs, the already normalized 21 quadratic and 56 odd-cubic insertions have joint separated moments and cumulants bounded uniformly in the fine cutoff by (JC25), at every fixed order. For cubic labels only, the natural physical factor is h^{-2m}, up to the displayed coupling polynomials. The same first-insertion definitions are used for every tuple and every order.

Uniform boundedness gives common subsequential distribution limits on separated configuration domains by the usual diagonal compactness argument. It does not by itself prove uniqueness of the continuum limit along the whole bare sequence, or tensor rotation covariance.

35.6.4 6.4 A once-per-label engineering payment

There is a stronger way to organize the preceding clean proof when the dependence on m matters. Repeating the crude integral estimate (JC23) at every joining needlessly produces a new factorial cost. Retain the scale-labelled source tree instead.

For each original support K_i, put $N_i = \{x : \operatorname{dist}(x, K_i) < h/4\}$. These neighborhoods are disjoint. A cubic label remains UNPAID while its branch has only that label and lies below the separation scale. Its branch is then governed by the already normalized linear cubic transfer, not a small ordinary cubic source disc. If it first joins another label at scale $\ell_i \le h$, the retained birth tree has a path from K_i to

another original support, all of whose edges were created at scales $\ell_e \le \ell_i$. Follow that path only until it first leaves N_i. Its physical length is at least $h/4$.

The selected portions for different labels lie in disjoint neighborhoods. Splitting an edge into its portions inside these neighborhoods, when necessary, gives the literal inequality

$$\sum_{i \text{ first joined below } h} \frac{h}{4\ell_i} \le \sum_e \frac{\text{length}(e)}{\ell_e} =: \mathfrak{L}. \tag{JC25a}$$

The paths are selected in the retained source birth forest, before any geometrical reserve is spent. A future joining does not erase or reassign this earlier payment. A label that has not joined below h is first transported linearly to scale h, where its engineering factor is already h^{-2}.

With $t_i^{\text{eng}} = h/\ell_i \ge 1$, (JC25a) implies

$$\prod_{i \text{ newly paid}} (t_i^{\text{eng}})^2 \, e^{-\kappa\mathfrak{L}} \le C^{|I|} e^{-\kappa\mathfrak{L}/2}. \tag{JC25b}$$

Indeed $t^2 \le Ce^{\kappa t/8}$ for $t \ge 1$, and the sum of those exponents is at most $\kappa\mathfrak{L}/2$. A fixed power of $1 + \log t$, arising from one label's first normalization or own-core conversion, is paid by another fixed fraction in exactly the same manner. Each such power is fixed by the chosen observable family, not by the eventual tuple length.

Thus each cubic ℓ_i^{-2} becomes Ch^{-2} once. Mark that label PAID and retain this flag in every descendant. Products in (JC5) involve disjoint label subsets, so their paid flags unite without ambiguity. A generated quadratic field term keeps all these flags; its smaller field degree does not undo their payment. For the quadratic family there is no negative engineering factor to pay.

After this conversion, keep the INTEGRATED old-root norm of that multi-label component. It is not converted back into a current-cell density at each later merger. If a density comparison is needed once, the ratio in (JC22) is only $r^{\lambda-4}$, independent of $|I|$, instead of $r^{\lambda-\sigma_I-4}$. A separate fixed portion of the source bridge pays that conversion.

A first multi-label component made entirely from unconverted branches has one first-root count $O((h/\ell)^4)$. Since that first joining contains at least two new labels, assign at most two additional powers of h/ℓ to each of two of their unused neighborhood portions. The same elementary inequality as in (JC25b), with power four in place of two, pays this count with a fixed constant per newly paid label. A joining with an already integrated multi-label component uses its existing integrated root norm and the local row sum for a new branch; it introduces no second fine-volume count. Thus root conversion is charged once as well.

These choices leave fixed geometric/resolvent constants per genuine joining, rather than a new $(2m)!$ at each joining. The remaining all-order constants are those of the marked cluster coefficients, Gaussian moments, and the physical R transfer. The clean engineering argument itself therefore does not force a growing Gevrey index.

This is a proof with a reserved tree cost. It is not an inference from a bound on an already integrated scalar island mass, or from a carrier exponential that has already been spent on another estimate. In a live region the corresponding complete-component or age cost must first be supplied by the original unnormalized R account; a relative tree with a free crossing cannot be substituted into (JC25a).

35.7 7. What coincidence would require

The power ω_I in (JC3) is the superficial local collision degree. A local source counterterm of the form

$$\int dy \prod_{i\in I} \partial^{\beta_i} f_i(y)\, \mathcal{O}_D(y) \tag{JC26}$$

can be divergent only within the dimension filtration

$$D + \sum_i |\beta_i| \le \omega_I. \tag{JC27}$$

This is dimension counting, not a claim that every allowed coefficient is nonzero. It follows also from (JC21): a field output of dimension D and a spatial Taylor order k changes the scale power of the local density to $\ell^{D+k-\omega_I}$.

For two cubic labels, $\omega_I = 8$. Besides the identity and F^2, the filtration may contain derivative descendants of F^2, dimension-six even cubic types, and dimension-eight quartics, subject to the actual symmetries. The three Gaussian terms in (JC9) are only the first explicit members; they are not the complete interacting contact list.

For pairwise disjoint supports, every (JC26) involving at least two labels vanishes exactly. This explains why the separated induction can carry the complete mixed analytic member without solving a new coincident-insertion renormalization problem. It does not turn derivative operators into contacts. For example, $\Box\,\mathrm{tr}(F^2)$ is a genuine dimension-six descendant with nonzero separated correlations, although its constant-curvature value is zero.

Consequently a general single curvature polynomial of degree $p \ge 4$ is not controlled merely by subtracting all constant-curvature polynomials of degree at most p. Its dimension-$2p$ first-insertion space must also include lower-field-degree momentum/current jets: at dimension eight, for instance, the quadratic sector through momentum degree four and the cubic sector through momentum degree two. The protected odd cubic sector avoids that obstruction because every lower-degree color tensor is charge-conjugation even. The present joint result therefore applies to the protected $\mathcal{P}_2 \oplus \mathcal{P}_3^-$ family, or to any other finite family for which the required complete FIRST-insertion dimension filtration has separately been constructed.

35.9 9. Quantitative clean coefficients on one fixed Gevrey scale

The clean separated induction can be made uniform in the label number without changing the weak floor. This part uses real Gaussian/cold growth bounds and the generic local marked-cluster allowance; it does NOT require current-coupling Taylor estimates through an order that grows with the tuple length. The finite-jet coupling grades in Section 4 are a separate refinement.

35.9.1 9.1 The coefficient algebra and its polynomial reserve

For finitely many squarefree labels set

$$\|F\|_{\mathfrak{A}(\mathbf{r})} = \sum_I \frac{r_I}{\sqrt{|I|!}} \|F_I\|, \qquad r_I = \prod_{i\in I} r_i. \tag{JC29}$$

The coefficient norm may be an analytic local activity norm, a field sup norm, or an integrated rooted norm with its reserved carrier weights. The scalar factorial weight is submultiplicative because $(|A| + |B|)! \geq |A|!|B|!$. The usual carrier convolution norm can be rescaled by one fixed constant to absorb its fixed product constant. Field and wall Cauchy estimates act coefficientwise and preserve (JC29).

Let $E \geq 0$ be the energy in one joint Gaussian reference integrand. A normalized first insertion from

the protected family obeys, on the actual fixed core and its fluctuation box,

$$|Q_i| \le a_i(1+E)^{3/2}. \tag{JC30}$$

For a cubic this follows from its zero pair jets through degree two, the actual Lipschitz analytic substitution, and Taylor's integral remainder on a spare fixed core. A quadratic has the better degree-two bound and is included in (JC30). The fixed normalization matrix is included in a_i. This estimate is made on a genuine admitted regular representative; it is not an extension into a wild field.

If $R = \sum_i r_i a_i$, the exact squarefree exponential and the elementary-symmetric-sum inequality give

$$\left\| e^{\sum_i z_i Q_i} - 1 \right\|_{\mathfrak{A}} \le \sum_{k\ge 1} \frac{[R(1+E)^{3/2}]^k}{(k!)^{3/2}}. \tag{JC31}$$

Choose a fixed quadratic reserve $0 < \gamma < \gamma_0$, where γ_0 is the spare allowance in the original conditioned Gaussian integral. For

$$K_\gamma = e^\gamma \max\{1, (3/(2\gamma))^{3/2}\},$$

maximization in $1+E$ and $k! \ge (k/e)^k$ show

$$\sup_{E\ge 0} e^{-\gamma E} \frac{(1+E)^{3k/2}}{(k!)^{3/2}} \le K_\gamma^k.$$

Consequently, for $K_\gamma R \le 1/2$,

$$\left\| e^{\sum_i z_i Q_i} - 1 \right\|_{\mathfrak{A}} \le 2K_\gamma R\, e^{\gamma E}. \tag{JC32}$$

All labels may share the same variables. There is one quadratic reserve and one colder integral, not a product of separately integrated polynomial factors. The constant is independent of the number of labels. More explicitly, for a positive reference measure μ,

$$\int \left\| e^{\sum_i z_i Q_i} - 1 \right\|_{\mathfrak{A}} d\mu \le 2K_\gamma R \int e^{\gamma E} d\mu. \tag{JC32a}$$

The integral on the right is the one weakened joint majorant after that fraction of action has been spent. Later steps retain only its REMAINING reserve. The old unspent measure is not restored, and (JC32a) is not multiplied over factors sharing the same energy. The Gaussian determinant cost is paid once in the actual integrated dimension.

The generic clean Mayer proof needs exactly this kind of small marked factor with a fixed quadratic-growth tilt. Its absolute expansions, Gaussian conditioning, local wall Cauchy estimates, and final polymer resummation use only the triangle inequality, coefficient-norm submultiplicativity, and that tilt. Replacing the scalar coefficient norm by (JC29) changes none of their constants. A primitive marked factor already has the small factor $2K_\gamma R$ in (JC32), so no scalar source-Cauchy disc is needed to manufacture that factor. This is a direct estimate of the Mayer factor itself.

35.9.2 9.2 Applying the reserve only after engineering payment

For $h \le 1$, define the fixed first-profile loads

$$\Lambda_i(h) = Ch^{4-D_i}(1+x_h)^B \sum_{|\beta|\le s_0} h^{|\beta|} \|\partial^\beta f_i\|_\infty. \tag{JC33}$$

Here B and s_0 depend only on the protected family and the fixed background construction. They do not depend on the tuple length. The constant includes the once-per-label tree payment and the first-profile transfer.

Assume

$$\sum_i r_i \Lambda_i(h) \le R_*. \tag{JC34}$$

At a finer mesh, the literal primitive cubic load $r_i \ell^{-2}$ may be arbitrarily large. We therefore do not apply (JC32) to that unconverted fine-scale load. Expand finite source coefficients first, retaining the complete scale-labelled source trees. A one-label component is propagated by the fixed normalized linear cocycle. At its first genuine joining, apply (JC25a)–(JC25b), with the additional first-root payment just described, and mark its labels paid. A label still alone is first transported to h.

In the resulting positive termwise majorant, every primitive marked leaf now has weight at most $r_i \Lambda_i(h)$. Each source label occurs once in a squarefree coefficient, and its fixed first-normalization polynomial is paid once. The source-tree exponential has been weakened by one fixed fraction, not by a new fraction at every scale. Integrated old-root norms are retained after the first joining. Every completed mixed analytic member remains in the state; no field-degree-dependent renormalization is repeated on it.

The marked Gaussian/cluster factors in this NEW majorant satisfy (JC32) with $R \le R_*$. This is a comparison of absolute, geometrically completed source graphs. It is not an assertion that the original integral has a small nonzero cubic source potential at the fine cutoff. The remaining carrier exponential and the original spare Gaussian reserve still pay the unmarked entropy and locality sums.

35.9.3 9.3 A uniform local closure

The linear one-root propagation has already been resolved. Let C_{lin} be its integrated norm and let C_{join} be the fixed analytic constant of the generic connected-formation map on its small activity ball. These constants belong to the original source-zero construction and to the fixed second/third field readout; they are independent of the coefficient algebra dimension. The Banach algebra bound controls the sum of proper joinings of every order by

$$C_{\mathrm{join}} \frac{\mathfrak{J}^2}{1 - C_{\mathrm{join}} \mathfrak{J}},$$

where $\mathfrak{J}$ is the total paid regular marked norm. The one-label forcing at scale h and the geometrically completed forcing below h are bounded by $C R_*$. Thus the finite-prefix induction has the scalar majorant

$$\mathfrak{J} \le C_{\mathrm{lin}} R_* + C_{\mathrm{lin}} C_{\mathrm{join}} \frac{\mathfrak{J}^2}{1 - C_{\mathrm{join}} \mathfrak{J}}. \tag{JC35}$$

This product bound is applied only to genuine regular analytic activities in their common admitted domains. It is not a product of live cold masses.

Choose R_* once small enough that the right side maps $\mathfrak{J} \le 2 C_{\mathrm{lin}} R_*$ strictly into that ball. The usual first-exit argument proves the bound for every finite number of scales and every finite number of labels. Equivalently, the analytic majorant map is a contraction on that ball after decreasing R_* once. Neither the weak floor nor the first normalization matrix changes with the label number.

Only fixed field Cauchy orders are used here. A complete vacuum-subtracted even coefficient has zero first pair jet, so the ordinary quadratic readout controls its age. The odd first family uses its cubic center and its fixed third-order readout. The higher mixed analytic function is carried without extracting a growing list of higher field jets. Thus there is no hidden $\alpha^{-c|I|}$ loss at every step of (JC35).

Scalar outputs obey the same algebra norm. Their remaining first-root sum is (JC26) of the companion history note, namely $\sum_{\ell \le h} (h/\ell)^4 e^{-ch/\ell} \le C$, after the engineering powers have been paid. Output

constants are recorded once. Divergent one-label scalar terms are canceled algebraically by the fixed one-point renormalization before this estimate; genuinely local mixed contact terms vanish on the separated profiles.

35.9.4 9.4 Coefficient extraction and terminal retained fields

On a fixed physical terminal torus, the paid vacuum norm and the retained-field source norm are bounded by fixed constants under (JC34). Their exponential is an exponential in the finite Banach algebra (JC29). Integrate the retained actual source-zero field law, or use the clean source-zero positive mass lower bound when writing the unnormalized formulation. This proves a bound

$$\|Z^{\mathrm{sep}}/Z(0)\|_{\mathfrak{A}(\mathbf{r})} \le C_T \tag{JC36}$$

for the actual separated source coefficients, after the fixed one-point subtractions. The complete retained integral remains inside this formula.

For a desired label set of size m, choose $r_i = R_*/(m\Lambda_i)$. Extracting its coefficient and using $m^m \le e^m m!$ gives

$$|S_{m,a}(f_1,\dots,f_m)| \le C_T C^m (m!)^{3/2} \prod_{i=1}^{m} \Lambda_i(h). \tag{JC37}$$

The constants C, B, s_0, R_* and the exponent $3/2$ are fixed for this protected family. Formula (JC37) concerns the clean separated hierarchy and supplies the incoming norm (CH27) of the companion cubic-history theorem. A larger fixed Gevrey exponent is also harmless.

Connected coefficients have the same factorial exponent. Indeed the normalized constant term in (JC36) is one. Shrink all r_i by the fixed factor $[4(C_T+1)]^{-1}$; the norm of its nonconstant part becomes at most $1/4$. The finite-algebra logarithm is then bounded by its geometric majorant, so coefficient extraction gives $C_{1,T}C_{2,T}^m (m!)^{3/2} \prod_i \Lambda_i$. This is a formal coefficient statement, not an ordinary complex zero-free source disc.

The full physical version uses the history theorem's SAME factorially weighted algebra, its first full normalization, and its complete source-component engineering bank. With those source-free preparation and marked-history inputs, (JC37) passes through the original R sum on this one fixed Wilson sequence. A volume-uniform physical cell/chessboard estimate and the tensor rotation law remain distinct requirements. Neither follows merely by taking roots of the fixed-torus factorial bound.

35.10 10. Compact completions and a local replicated-cell majorant

The global norm in (JC29) is not the norm needed for a chessboard replication. A separate local coefficient norm is available on the CLEAN fixed core. Its physical R transfer requires the represented-state qualification below.

35.10.1 10.1 A completion is an exact split, not another observable

Fix the microscopic operators (JC2), their first normalization matrices, and their one-point subtractions once. Let $\mathcal{Z}_a(z)$ be their exact finite squarefree source integral on the original Wilson sequence. Its coefficients define the physical products under discussion.

Choose once, on every actual RG carrier, a source-independent linear completion map from an extracted curvature kernel to a globally defined compact-link function. For a quadratic kernel use two traceless

plaquette variables $X_p(U)$, parallel transported along a fixed carrier path; for a cubic kernel use three such variables and the invariant color tensor. On real compact links the transports are isometric, and

$$\|\mathcal{C}_q(K)(U)\| \le C_q \|K\|_{\ell^1}, \qquad q = 2, 3. \tag{JC38}$$

The corresponding holomorphic bound on a prepared complex regular domain spends the already available path-length locality reserve. The two- and three-curl constructions give precisely the kernel norms required in (JC38). A third-order extraction is applied only after all lower field jets have been matched and removed; this is automatic for the first C-odd family. For an even mixed member, subtract the COMPLETE quadratic completion first, including its higher ordinary field terms. The third jet of a function with an unsubtracted quadratic jet need not satisfy the homogeneous three-leg Ward identity.

For an actual regular source coefficient, the operation is the identity

$$O_I = \mathcal{C}_q(\mathsf{P}_q O_I) + [O_I - \mathcal{C}_q(\mathsf{P}_q O_I)]. \tag{JC39}$$

The physical own-current jet is formed before P_q. The bracket is retained on its original paired domain, with the same source roots and all its own counterterms. It is not dropped because its selected jet vanishes.

On a live branch, (JC39) is not used to extend an arbitrary analytic remainder to a wild field. The original source-containing unnormalized kernel is kept. If a universal completion is extracted outside that kernel, its inverse factor is inserted inside the SAME kernel. In the finite algebra this is the literal identity

$$e^{-z_I C}\,\{e^{z_I C} K(z)\} = K(z), \tag{JC40}$$

with the appropriate complete coefficient function C. The original completion/subtraction filing determines the regions on which the two factors are placed. No sourced island inverse is needed.

Induction over the exact transfers and (JC39)–(JC40) proves that the resulting source integral is still $\mathcal{Z}_a(z)$, coefficient by coefficient. In particular all of its physical derivatives are unchanged. The completion map depends on the fixed carrier convention, not on a later separation scale, tuple length, or observation grid. Keeping an exact Gaussian Schur state for a calculation is another bookkeeping of the same jets; a compact completion is never multiplied into that state as a second copy of the source.

For bounds which do not require a sharp Gaussian asymptotic, this representation permits one to retain original compact source atoms and admitted-core analytic remainders rather than an unbounded polynomial surrogate on an inadmissible wild chart. It is essential to keep the compensating terms in (JC39)–(JC40).

35.10.2 10.2 The clean local coefficient norm

Place at most a fixed number of source profiles in each cell of an h-scale grid, with their actual supports separated by the required fixed multiple of h. Let q_i be the cell containing source i. After its engineering payment use the load Λ_i from (JC33). The local smallness condition is

$$\sup_u \sum_i r_i \Lambda_i e^{-\kappa d_h(u,q_i)} \le r_0. \tag{JC41}$$

For uniformly replicated profiles it holds with one fixed $r_i\Lambda_i \le c r_0$, independently of the number of copies. It is not the global condition $\sum_i r_i\Lambda_i \le r_0$.

Use the ordinary squarefree coefficient norm, without a factorial divisor, inside the LOCAL polymer norm. Schematically it is

$$\|\mathcal{E}\|_{h,\kappa,\mathbf{r}} = \sup_{u} \sum_{X \ni u} \sum_{I \neq \varnothing} r_I e^{\kappa \tau_h(X,\{q_i : i \in I\})} \|E(X,I)\|_{R_0}. \tag{JC42}$$

All source origins and the root convention are retained. As in the original polymer norm, equivalent pointing conventions only change one fixed constant. The output field variables in $\|\cdot\|_{R_0}$ remain arbitrary in the common admitted clean core.

On that physical core, a resolved first-insertion representative is bounded by its stored analytic sup norm. Thus after payment its marked Mayer atom has a bounded load, not merely the global polynomial-energy bound (JC30). The zero through second jet of a cubic is still needed in its one-label normalization and age readout; it is not needed to replace this stored sup by a growing total-energy polynomial.

The literal small-potential proof can therefore be applied to an ordinary Banach coefficient algebra. Below h, first expand source coefficients and resolve their one-label branches, then perform the once-per-label payment. The resulting positive completed-source-graph majorant has the same local bounds as a uniformly small family of bounded marked atoms satisfying (JC41), with a fixed weaker locality exponent. This comparison is justified at the homogeneous coefficient level: the actual one-step cluster map is analytic in a bounded local potential on its fixed core, so its order-r coefficient is multilinear in the r input norms and has the original connected wall kernel. Engineering payment replaces those input norms by Λ_i before their positive series is summed. No nonzero fine-scale source is introduced into the actual measure.

The generic local Mayer/polymer majorant now closes in (JC42) for a fixed small r_0, with

$$\|\mathcal{E}\|_{h,\kappa',\mathbf{r}} \leq C r_0. \tag{JC43}$$

It uses one nilpotent label per separated cell; it makes no assertion about arbitrarily many copies of one source at the same cell. Those coincident copies would require the missing contact analysis and would invalidate the engineering comparison.

For completeness, this is a finite-prefix bootstrap, not an assumption on the incoming mixed coefficients. Let A_n be their paid regular norm after the full one-label and ordinary zero-flat linear resolvents have been taken. The once-resolved primitive forcing has norm at most $C_0 r_0$. Proper connected joinings have at least two incoming regular factors and the actual analytic formation estimate gives

$$A_n \leq C_0 r_0 + C_1 \frac{(r_0 + A_n)^2}{1 - C_1(r_0 + A_n)}. \tag{JC43a}$$

Choose $K > 2C_0$, then choose r_0 so that the right side is below $K r_0$ at $A_n = K r_0$. The positive finite-prefix series starts at zero as all fugacities are scaled radially from zero. First exit proves $A_n \leq K r_0$, uniformly in the prefix and the number of labels. The input to this inequality is the coefficientwise formation bound on a COMMON admitted core; an unresummed factor greater than one at each fine scale would not give (JC43a).

The specific map-locality input in this argument is the original FULL-CORE operator-valued local-potential theorem, including its connected wall kernel and its coefficient-algebra stability. A numerical covariance estimate at the flat background or a scalar constant-profile normalization is not a substitute. Under that stated input, (JC41)–(JC43) are consequences of its same positive majorant and the explicit source-tree payment, with no new perturbation parameter or label-number-dependent floor.

35.10.3 10.3 What the local norm gives, and what its R transfer must preserve

The total paid scalar vacuum and retained field source norm in a region of N_h cells are at most Cr_0N_h. Exponentiation and the positive terminal source-zero integral give a coefficient-norm bound $\exp(CN_h)$, including the terminal stability factor. If M replicated profiles fill a fixed positive fraction of those cells, $N_h \le C_0M$, extraction with the fixed radii allowed by (JC41) gives

$$|S_{M,a}(f_1,\dots,f_M)| \le C_1^M \prod_{i=1}^{M} \Lambda_i. \tag{JC44}$$

This is a clean replicated-cell estimate. It comes from a local coefficient majorant, not from taking an M-th root of (JC37).

For a physical version of (JC44), the same LOCAL norm must survive the original R sum. The represented state must keep each original compact source atom and every compensating completion from (JC40), and it may use a regular remainder only where the actual prepared background lies in that remainder's stored domain. After the same once-per-label engineering payment, compact atoms obey a pointwise constant bound; admitted regular remainders obey their stored sup bounds. Products can then be bounded BEFORE integrating their common live kernel. A local fugacity contributes only a fixed factor per actual source/volume record, which can be paid from the own-layer volume reserve.

If instead an implementation retains unbounded auxiliary Gaussian source polynomials, it needs the stronger local energy-slot statement: one slot per separated cell, bounded column overlap in the common quadratic energy, and a joint integral. The global estimate (JC32) alone does not prove it. The physical theorem therefore requires verification of one of these two precise represented-state alternatives. Replacing an analytic regular function by an arbitrary bounded compact extension would not be such a verification and would change the physical derivatives.

The companion cubic-history note gives that verification for the compact/stored-core representation in (CH38)–(CH39d). Its additional norm is a FIRST complete-healing/terminal-exit flux of an unfolded joint excursion. The nonlinear term is a product of bounded incoming source loads before the one cold integral, and there is no square of an integrated live flux. The explicit two-norm first-exit argument there is needed in addition to the list of represented states above. All action, carrier and engineering fractions already used are removed from the common positive majorant before its remaining record-volume reserve pays the local fugacity.

36.1 1. A uniform factorially weighted source algebra

Fix $p \geq 2$, the maximum ordinary polynomial degree in the canonical Gaussian coordinates, and set

$$\sigma \geq p/2 - 1. \tag{CH1}$$

For n distinct source labels define

$$\mathfrak{A}_{n,\sigma} = \mathbb{C}[z_1, \dots, z_n]/(z_1^2, \dots, z_n^2), \qquad \|F\|_{\sigma,\boldsymbol{r}} = \sum_{S \subset [n]} \frac{r_S}{(|S|!)^\sigma} \|F_S\|, \quad r_S = \prod_{i \in S} r_i. \tag{CH2}$$

The convention is $0! = 1$. Its constant term is the ordinary source-free coefficient.

This norm is submultiplicative. Indeed, for disjoint sets A, B,

$$\frac{1}{(|A| + |B|)!^\sigma} \leq \frac{1}{|A|!^\sigma |B|!^\sigma}, \tag{CH3}$$

and the coefficient of z_S in a product sums exactly over $A \dot\cup B = S$. Thus every algebraic operation in the local Mayer/cluster proof that uses only a submultiplicative coefficient norm has constants independent of n.

The same assertion holds with coefficient functions holomorphic in the frozen localization parameters or in a mask parameter $|u| < 2$, taking their sup norm first. Cauchy in a FIELD or WALL parameter does not change S, so it commutes with this weight. No ordinary holomorphic source disc is asserted.

36.1.1 1.1 The pointwise cold-moment lemma

Let $E \geq 0$ be the nonnegative energy in ONE real joint reference integrand. Suppose the source factors satisfy

$$|Q_i| \leq a_i (1 + E)^{p/2}, \qquad R = \sum_i r_i a_i. \tag{CH4}$$

The factors may share all their variables. Put $V = \sum_i z_i Q_i$. Because the labels are squarefree,

$$e^V = \prod_i (1 + z_i Q_i).$$

The elementary-symmetric-function inequality gives

$$\|e^V - 1\|_{\sigma,\boldsymbol{r}} \leq \sum_{k \geq 1} \frac{[R(1 + E)^{p/2}]^k}{(k!)^{\sigma+1}}. \tag{CH5}$$

There is no omitted factorial: the additional $k!$ comes from bounding the elementary symmetric sum by the k-th power of its first sum divided by $k!$.

For every fixed $\theta > 0$,

$$\sup_{E\geq 0} e^{-\theta E}\frac{(1+E)^{pk/2}}{(k!)^{p/2}} \leq C_{\theta,p}^{k}. \tag{CH6}$$

For example, write $y = 1 + E$, maximize $y^{pk/2}e^{-\theta y}$, and use $k! \geq (k/e)^k$. The finitely many cases where the maximizing y is below one are covered by the same constant. Equations (CH1), (CH5) and (CH6) prove

$$\boxed{\sup_{E\geq 0} e^{-\theta E}\|e^{V}-1\|_{\sigma,\boldsymbol{r}} \leq \frac{C_{\theta,p}R}{1-C_{\theta,p}R}, \qquad C_{\theta,p}R < 1.} \tag{CH7}$$

This is a fixed quadratic-growth allowance for the Mayer factor, uniform in the number of labels. It is stronger than an estimate at each separately fixed source degree.

For a real positive reference measure $d\mu$, it follows directly that

$$\int \|e^{V}-1\|_{\sigma,\boldsymbol{r}}\, d\mu \leq \frac{C_{\theta,p}R}{1-C_{\theta,p}R}\int e^{\theta E}\, d\mu. \tag{CH8}$$

The right side uses ONE colder integral. It is never replaced by a product of the integrals of the individual Q_i.

In a Gaussian block, $\int e^{\theta E}d\mu$ has the usual $\exp(C_\theta N)$ cost in its actual integrated dimension N. In an original live island, the same colder integral uses a fixed fraction of its actual Wilson/quadratic action. Those volume factors are the original own-layer charges, not one charge for every source degree. The fixed bank reserve is chosen once to pay them.

For cubic degree $p = 3$, $\sigma = 1/2$ suffices. Choosing $r_i = R_0/(na_i)$, with fixed sufficiently small R_0, and extracting the coefficient of $z_1 \cdots z_n$ gives

$$\left|\int Q_1 \cdots Q_n\, d\mu\right| \leq C^n (n!)^{3/2}\prod_i a_i \int e^{\theta E}\, d\mu. \tag{CH9}$$

Here $n^n \leq e^n n!$ absorbs the radius allocation. For general degree p, the exponent is $p/2$, or any larger fixed exponent allowed by (CH1).

36.1.2 1.2 Why an ordinary source disc is the wrong target

Already for one real standard Gaussian B, $\int e^{zB^3}d\mu(B)$ diverges for every nonzero real z. Its even moments have order $C^n(n!)^{3/2}$, not $C^n n!$. Thus ordinary uniform holomorphy of a cubic source is not a consequence of cold Gaussian domination and is not needed for the corrected OS factorial-growth condition.

The finite-cutoff compact-group generating function can be entire while its source radius in a uniform analytic estimate tends to zero. Formula (CH7) concerns its coefficient majorant and does not confuse these two statements.

36.3 3. One physical cubic insertion through R histories

This section concerns one insertion, including all 56 components linearly. It constructs the rare correction to the first-insertion center under the original prepared domain and first-healing operator bounds.

36.3.1 3.1 The admissible state space and its linear norm

The one-source state has three kinds of entries:

1. The 56 fixed globally defined local polynomial center completions, with their physical ℓ^{-2} factor. Their values on an arbitrary real core satisfy (CH10), linearly in the coefficient vector.
2. Stored regular remainders on their ORIGINAL paired analytic domains. At a new wild encounter, the source-free preparation theorem places their old readouts in those domains, so their stored sup norm is a legitimate factor in the common cold integral.
3. Original unnormalized live kernels, with their actual retained variables, original entry label, and a cold-integrand majorant. The reserved rare factor from that entry is recorded separately from their subsequent propagation norm. It is never granted again at healing.

An arbitrary holomorphic function on a regular core is not declared to be an insertion on a wild core. The operator norm below is on this admissible represented state space. Every operation is linear in that one physical source state.

The source-free conventions can be chosen equivariant under charge conjugation. If an individual deterministic filing is not equivariant, pair its history with the charge-conjugate history before extracting a completed coefficient. The paired physical marked coefficient is C odd. Projection onto that odd part does not increase the support or the absolute norm by more than a fixed factor.

36.3.2 3.2 Extraction occurs only at first complete healing

At the FIRST complete marked healing, the output is an actual gauge-invariant C-odd scalar member on its original analytic pair domain. Its Taylor jets of ordinary degrees zero, one and two vanish. A partial dropout or an open boundary operator is not separately declared to be such a member.

Let its birth core be α_j, and let $\mathcal{H}_j$ be the actual source-free coarse-data reconstruction. At the flat point the exact chain rule is

$$D^3(O \circ \mathcal{H}_j)(0) = D^3O(0)\,[D\mathcal{H}_j(0)]^{\otimes 3}. \tag{CH13}$$

There are no nonlinear-lift derivative terms because the lower jets vanish. The original localized linear response kernel and Cauchy on this LOCAL pair core give

$$\|D^3(O \circ \mathcal{H}_j)(0)\|_{\text{rooted tree}} \le C\alpha_j^{-3}\|O\|_{\text{loc}}. \tag{CH14}$$

The weighted three-curl homotopy in the cubic note therefore gives the complete local three-curvature kernel with the same bound, after a fixed locality loss.

This proves the local third-jet/three-curvature-kernel extraction and its spatial moment bounds on the shrinking R core. It does not borrow the fixed clean radius R_0. Its cost α_j^{-3} is a fixed power of x_j and its logarithms.

There is one further symmetry qualification. A fresh R member has the covariance of its original MR_j-block partition, as in III (2.32), not yet every current-lattice translation. Its LOCAL cubic kernel can already be split off at healing using its exact compact completion. However, its contribution to the universal finite 56-dimensional center is filed only at the first compatible promotion, just as in the original III pp. 270–271 promotion of R origins. The tagged cubic kernel and its exact remainder are retained in the intervening interval.

That waiting interval has length $h_j \le C(1+\log\log x_j)$. Before promotion the zero through second jets give $Cb_j(\alpha_k/\alpha_j)^3 r^6$, without claiming a strict extra age power. Root density and cubic engineering cancel r^6, and the remaining radius ratios and waiting count cost only a fixed power of x_j and its logarithms. The R rare envelope pays that fixed cost.

At promotion subtract the corresponding equivariant 56-component cubic completion. This is observable bookkeeping; it changes neither the source-zero coupling nor the field tests. The R-origin flag remains on the remainder. The local compact split at healing and this finite-center split are algebraic identities, not two copies of the source.

If the old birth has norm b_t on α_t, the actual own-current readout after this extraction obeys

$$Cb_t\left[(\alpha_j/\alpha_t)^3 r^{6+\zeta} + (\alpha_j/\alpha_t)^4 r^8\right], \qquad r = \ell_t/\ell_j. \tag{CH15}$$

For an integrated insertion the root-density factor r^{-4} and engineering factor r^{-2} cancel r^6. The remainder has strict age decay, after the polynomial radius ratios are paid. The physical own-current composition is performed before taking the center, exactly as in the clean cubic construction.

36.3.3 3.3 The one-root live-age account

Here t is the actual NEW wild-entry scale of a regular/center insertion, and $a = \ell_t$ is its stored-coordinate scale at that entry. It is not the bare source scale if the source spent intervening years in a regular center. Those years have already been transported by the clean or completed center matrix. A center-subtracted old remainder reaches the entry with its strict regular age factor (CH15) already included in its incoming norm.

An insertion entering at $a = \ell_t$ and first completely healed at ℓ_s has the remaining engineering excess

$$(\ell_s/a)^2 = L^{2(s-t)}. \tag{CH16}$$

Assign it once to that source's actual live epoch. Let B_{hist} include the original one-lineage entropy, possible old-root count, and fixed chart charges. Choose ONE fixed cubic reserve

$$B_3 > B_{\mathrm{hist}} + 2\log L + b \tag{CH17}$$

with $b > 0$. The age bound for the whole unnormalized lineage then pays (CH16), all polynomial coupling and coordinate costs, and leaves exponential age decay. No intermediate partial healing emits an additional independent marked copy.

Define $\mathcal{E}_{j,t}$ as the complete excursion operator from a regular/center state at this NEW entry t, through its original unnormalized live lineage, to its first complete healing at s, followed by its compatible finite-center promotion at $j = s + h_s$. The row includes the ONE rare factor from the actual entry record. An already live kernel retains its original t through mergers, partial dropouts and further propagation; it is not a new input with another rarity credit. The first-healing excursion row satisfies

$$\|\mathcal{E}_{j,t}\|_{\mathrm{regular/center}\to\mathrm{cubic}} \le CF_3(x_j)e^{-b(j-t)}(1+j-t)^q, \quad F_3(x) = (1+x)^Q e^{-cP_0(x)}. \tag{CH18}$$

This is a bound times the incoming state norm in subsection 3.1 at entry, not merely a bound for one selected basis element. Center completions use (CH10), and stored regular entries use their actual-domain sup. Inside the excursion, still-live entries propagate by their defining joint cold majorant, with dwell decay and no additional entry factor. Linearity preserves these bounds.

To obtain (CH18), first keep the entry envelope at x_t, the factor α_s^{-3}, and the polynomial in x_t/x_j. The two-sided running tube bounds x_t by $x_j + C(1+j-t)$ above and by $x_j + c(j-t) - C$ below. Replace F_3 by a monotone tail envelope with a slightly smaller exponent constant. Polynomial factors in the age are paid by (CH17). There is NO live-age charge during the regular promotion wait: the factor missing from $e^{-b(j-t)}$ is at most $e^{bh_s} = (\log x_s)^{O(1)}$, and the waiting readout and number of possible healing times have only fixed polynomial/logarithmic costs. They are absorbed by the weakened F_3. This gives (CH18) without assuming pointwise monotonicity of the actual run.

After COMPLETE healing, a subsequent regular-to-wild encounter starts a new excursion and has its own actual entry record. Center-subtracted regular aging between such excursions has (CH15). The resulting convolutions preserve the displayed exponential row bound, with a smaller b, by the same one-root resolvent as in history sections 16 and 22. If the marked component has not completely healed, a new geometric merger stays inside the same excursion instead. A terminal surviving lineage is an exit flux, not a regular coefficient to which (CH14) has been applied.

36.3.4 3.4 The full 56-dimensional normalization

Keep the clean stable age state and the just described R/live memory. Its first-healing rows are (CH18); there is no assumed product of integrated live norms. The clean center response is $I + O(x_j^{-1})$,

and its stable resolvent has fixed norm and exponential age decay. The finitely delayed R kernels remain in their separate tagged age state until promotion; they are not mistaken for a fully translation covariant finite center at their birth.

Here is a direct finite-dimensional way to close the memory. Let V_j be the center fundamental matrix for the 56 physical starting completions. Provisionally assume its earlier ratios obey

$$\|V_t V_j^{-1}\| \le e^{\varepsilon(j-t)} \quad (t \le j), \qquad 0 < \varepsilon < b. \tag{CH19}$$

The clean stable resolvent bounds its stable columns relative to V_j. Equation (CH18) and (CH19) bound all additional center feedback by

$$C F_3(x_j) \sum_{a \ge 0} e^{-(b-\varepsilon)a} (1+a)^q.$$

Thus the actual successive PROMOTED center matrix satisfies

$$M_j := V_{j+1} V_j^{-1} = I + O(x_j^{-1}) + O(F_3(x_j)). \tag{CH20}$$

Choose the original weak floor once so that the right-side perturbation and the norm of its inverse are small enough to reproduce (CH19). Induction then proves invertibility at every step and closes the provisional ratio bound. The stable graph is bounded by its same resolvent.

Consequently

$$\|Z_{n:0}\| + \|Z_{n:0}^{-1}\| \le C \exp\left[C \sum_{j<n} \{x_j^{-1} + F_3(x_j)\} \right]. \tag{CH21}$$

The second sum is uniformly small at a weak floor; the first has the usual logarithmic running bound. This is one fixed normalization of the 56 physical insertions on the given Wilson blocking sequence. It is independent of source profiles, the number of inserted fields, and any later observation grid. At a finite stopping scale the still-unpromoted R kernels remain explicit parts of the insertion state. They are neither discarded nor absorbed by redefining the finite matrix in that volume.

The same row bound holds in a fixed exponentially weighted coarse spatial-kernel norm. In particular, with dimensionless coarse momentum p,

$$\|\mathcal{E}_{j,t}(p) - \mathcal{E}_{j,t}(0)\| \le C|p| F_3(x_j) e^{-b(j-t)} (1+j-t)^q, \tag{CH22}$$

and any fixed number of momentum derivatives is available by spending spare locality. This is the R input to a smooth-profile first-insertion transfer; a constant-profile center alone would not prove such a spatial statement.

36.4 4. A separation-scale engineering bank

Let the n physical source supports have mutual distance at least $h > 0$. Decompose a source into its actual fine-cell origins before summing those origins. The geometry in this section is applied to each such term. It does not replace a source-support count by an unmarked fine physical volume.

36.4.1 4.1 Full carriers, including live interiors

The original wall construction retains every source origin in a marked carrier. A completed regular carrier has an ordinary connecting tree. A live boundary carrier has a relative tree with its islands

contracted. Expand that tree inside each ACTUAL connected island using a spanning tree of its cells. Its added length is at most a fixed constant times the actual island cell count in UNIT cells of that stored layer. A count only of larger side-length blocks would not suffice unless the corresponding volume factor were reinstated.

Thus a relative-tree cost plus its own-layer live-volume cost controls a full connecting carrier:

$$d_{\mathrm{full}} \le C\{d_{\mathrm{rel}} + N_{\mathrm{live}}\}. \tag{CH23}$$

All determining collars and old source-attaching paths are included. This statement would be false if the source anchors were dropped from the operator carrier, or if a hole interior were treated as free without its own volume record.

The source-free completed-R expansion supplies spare deterministic costs $\exp[-\kappa d_{\mathrm{rel}} - \kappa N_{\mathrm{live}}]$ in the same JOINT cold majorant. These factors can be spent on (CH23) without dividing the component into independent rare events.

36.4.2 4.2 A label is converted only once

Call a label unconverted while it belongs to a one-label branch below the separation scale. A regular one-label branch uses the physical cubic center/stable transport, including (CH15). A live one-label branch uses the one-mark age account (CH17). At a first multi-label join below h, convert the remaining engineering ℓ^{-2} factors to h^{-2}. A join is declared as soon as two source branches share ONE complete live component, even if that component has not healed. Its labels are not kept artificially unjoined until a regular analytic output is emitted. A converted label retains this record thereafter; its inverse-mesh factor is not reintroduced at later joins. An unjoined label is transported linearly to h and converted there.

For precision, let $\ell_i \le h$ be the first joining scale of a previously unconverted label i. In the retained scale-labelled source tree choose a path from its source origin until first exit from the $h/4$ neighborhood of that source support. Its edge scales obey $\ell_e \le \ell_i$, by the causal ordering of births. The source neighborhoods are disjoint. Therefore the selected path portions can be charged disjointly and

$$\sum_i \frac{ch}{\ell_i} \le \sum_e \frac{\mathrm{length}(e)}{\ell_e} =: \mathcal{T}. \tag{CH24}$$

Fixed collars only alter $c > 0$. Joins with ℓ_i comparable to h cost a fixed constant per label and can be handled separately. For live records, the filled trees in (CH23) supply the same scale-labelled path cost.

It follows that

$$\boxed{\prod_i (h/\ell_i)^2\, e^{-\kappa\mathcal{T}} \le C^n e^{-\kappa\mathcal{T}/2}.} \tag{CH25}$$

Indeed $t^2 e^{-ct}$ is bounded. Fixed coupling-normalization factors $(1 + \log(h/\ell_i))^B$ are paid by the same per-label estimate after reducing the spare fraction once.

The charge is made on these disjoint path portions in the EXPANDED term, not each time an inherited scalar norm is read. This is why one does not accumulate a new $(2|S|)!$ at every joining vertex. Every label spends its engineering credit once. The unused part of the tree/volume exponent still pays the original locality and entropy sums.

A component containing only one unconverted live label has at least one actual live record at each layer. Its fixed additional age charge is therefore paid by that component's own-layer volume. When

several labels meet, the single joint component uses (CH23)–(CH25), not several copies of its rarity. At scales at least h there is no further inverse-mesh conversion: all retained physical prefactors are bounded by h^{-2} per cubic label.

36.4.3 4.3 Root-density and scalar-output counts

An additional fine-root density remains when a connected scalar coefficient is emitted. It is separate from (CH25). A mixed scalar carrier meets at least two separated supports; a spare part of its full tree then supplies $e^{-ch/\ell}$. Choosing a root in one $O(h)$-diameter source support gives at most $C(h/\ell)^4$ positions. Hence

$$\sum_{\ell \le h} (h/\ell)^4 e^{-ch/\ell} \le C. \tag{CH26}$$

This constant is independent of the label number because the engineering factors have already been converted by (CH25). There is no additional $\ell^{-2|S|}$ inside (CH26).

A one-label scalar vacuum is zero in this C-odd family. Every scalar output is recorded once, with all of its source origins. Its descendants are not generated again at every subsequent scale. Multiple connected components are combined by the ordinary exponential in the coefficient algebra, whose norm is submultiplicative.

36.5 5. The quantitative mixed-history interface

The following is the precise interface supplied by the preceding estimates. It distinguishes the new quantitative history statement from the clean formation theorem still needed to feed it.

Assume the CLEAN separated cubic construction, using the single full normalization (CH21), supplies its actual regular packets in $\mathfrak{A}_{n,\sigma}$ with a uniform small incoming local norm whenever

$$\sum_{i=1}^{n} r_i \Lambda_i \le \eta, \qquad \Lambda_i = C h^{-2} (1 + x_h)^B \|f_i\|_{C^s}. \tag{CH27}$$

Here s, B, σ and $\eta > 0$ are fixed independently of n and the cutoff. A scale-dependent one-label branch retains its engineering factor until the exact conversion in section 4; (CH27) is not asserted to be a small primitive source potential at every finer mesh. Any separately required contact/mixing operations must be bounded in this same fixed factorial class.

Then the ORIGINAL mixed-history completion preserves this coefficient allowance with constants independent of n. Its proof consists of the following operations.

- Wall localization and regular Mayer formation use the submultiplicative algebra (CH2), with their fixed field-domain margins.
- First-label live entries use the operator norm of section 3 and its one fixed age charge.
- Multi-label live entries use the single cold estimate (CH7)–(CH8), with the complete old variables retained.
- Engineering and coupling factors are paid once by the source-tree allocation (CH25).
- Scalar root sums use (CH26), and source-free scalar vacua are completed by the original per-history identity before absolute values are taken.

None of these steps changes the weak floor with n. If a finite-window mask is also included, its ordinary $H^\infty(|u| < 2)$ factor and its ONE distinguished derivative owner are treated exactly as in

history section 22. The finite-algebra source norm changes, but no additional live rarity or live-age factor is introduced for each source subset.

On a fixed physical terminal torus, the resulting coefficient bound has the form

$$|S_{n,a}(f_1,\dots,f_n)| \le C_T\, C^n (n!)^{\sigma+1} \prod_{i=1}^{n} \Lambda_i, \tag{CH28}$$

after choosing $r_i = \eta/(n\Lambda_i)$. The constant C_T is the fixed source-zero terminal stability constant. The number of source species is fixed and can be included in C. Since $(1+x_h)^B \le C_\epsilon h^{-\epsilon}$ for $0 < h \le 1$, this has a finite factorial exponent and a separation exponent linear in n.

The assertion in (CH28) is conditional on the CLEAN all-order norm (CH27). A collection of estimates with unrelated constants C_n, even if valid for every fixed n, does not supply that hypothesis. The one-step cold and the mixed-history operations above do not require increasing its factorial exponent. This identifies where quantitative clean constants still have to be checked.

36.5.1 5.1 The complementary coarse meshes cost only a factorial power

There is an elementary all-cutoff complement to the fine analysis. On a source cube of diameter $O(h)$, compactness of the plaquette polynomial and (CH21) give

$$|O_a(f)| \le C a^{-2}(1+h/a)^4 (1+x_a)^B \|f\|_\infty. \tag{CH29}$$

If a prospective n-profile grid requires $a \le ch/n$, all remaining meshes satisfy

$$a^{-2}(1+h/a)^4 \le C h^{-2}(1+n)^6.$$

On the same bare sequence, $x_a \le C(1+\log(1/a))$, so for these meshes

$$(1+x_a)^B \le C_B\{1+\log(1/h)+\log(1+n)\}^B \le C_B h^{-1}(1+n) \quad (0<h\le 1). \tag{CH30}$$

Thus the product of n such physical insertions, pointwise and under any probability law, is at most

$$C^n (n!)^7 h^{-3n} \prod_i \|f_i\|_\infty. \tag{CH31}$$

The exponent seven is deliberately coarse. The normalization matrix is the same one in (CH21), not an order-dependent retuning. For the C-odd physical law the one-point expectation is zero; an additional physical centering would only multiply this bound by 2^n.

Consequently a fine-mesh separated estimate with a fixed factorial exponent can be supplemented uniformly at all coarse meshes. This does not resolve the volume interface below, but removes a possible cutoff-order gap in a later flat-test extension.

36.6 6. Contacts and the remaining volume interface

For n dimension-six insertions, the superficial local vacuum contact order is $2n+4$. A Taylor subtraction at that order has a number of spatial derivatives linear in n, as required by the flat-test OS framework. Exponential carrier moments bound a derivative of total order q by $C^q q!$. Thus a SINGLE such local projection has at worst $C^n(n!)^2$ additional coefficient cost when $q \le 2n+4$, apart from its explicit cutoff-dependent engineering power.

This observation is not an all-order mixing theorem. To retain a fixed factorial exponent, scalar contact coefficients must be extracted and recorded once when born, and a recursively propagated field-mixing family must have its own invariant norm. Reapplying a degree-dependent factorial loss at every RG step would not be legitimate. The sharp-background higher-derivative obstruction in the separate higher-jets note is not removed by (CH7).

For mutually separated supports, genuine local contact coefficients have zero mixed derivative. The bound (CH28) can therefore be proved on that algebra without assigning values on the coincidence diagonals. The flat-test extension then concerns those separated values; it is not a proof of a new local operator-product correspondence.

There is also a distinct volume issue. The fixed-torus factor C_T in (CH28) does not, by itself, give an infinite-volume tempered hierarchy. A physical composite cell seminorm or another local relative partition estimate is needed to replace that factor uniformly in the larger volume. The ordinary holomorphic scalar-action source argument cannot simply be reused for a cubic source, in view of subsection 1.2.

A potentially sufficient stronger input is a uniform replicated-cell coefficient bound: one cubic label per separated physical cell, with a local energy/source budget and growth C^N for N such cells. Its physical chessboard N-th root would be volume independent. Establishing that bound requires retaining the local distribution of the polynomial energy marks; the global estimate (CH9), with its $(N!)^{3/2}$, would not prove it. This is a precise remaining interface, not a consequence hidden in the term "Gevrey."

36.7 7. A joint replicated-cell polynomial lemma

The following estimate is a useful alternative when Gaussian source polynomials have been retained as explicit analysis variables. It also identifies precisely why one must retain the local distribution of the marks.

Let B be a centered finite-dimensional real Gaussian vector with covariance K. Let $A_i \geq 0$, $1 \leq i \leq M$, and assume

$$\|K\| \leq k_0, \qquad A := \sum_i A_i \leq w_0 I, \qquad \mathrm{Tr}(KA) \leq t_0 M. \tag{CH32}$$

No independence between the different $A_i^{1/2}B$ is assumed. For $0 < \theta < (2k_0 w_0)^{-1}$,

$$\begin{aligned} \mathbb{E}e^{\theta B^T A B} &= \det(I - 2\theta K^{1/2} A K^{1/2})^{-1/2} \\ &\leq \exp\left[\frac{\theta t_0 M}{1 - 2\theta k_0 w_0}\right]. \end{aligned} \tag{CH33}$$

The identity follows by diagonalizing the one JOINT quadratic form. For the inequality use $-\log(1-u) \leq u/(1-u_{\max})$ on all its eigenvalues.

For every fixed $p \geq 2$ and $0 < r \leq 1$, elementary one-variable maximization gives

$$\log\{1 + r(1+t)^{p/2}\} \leq C_p r + C_p r^{2/p} t, \qquad t \geq 0. \tag{CH34}$$

Indeed $(1+t)^{p/2} \leq C_p(1+t^{p/2})$, and $\sup_{u>0} u^{-1}\log(1+u^{p/2}) < \infty$. Taking $E_i = B^T A_i B$, and choosing r once so that $2C_p r^{2/p} k_0 w_0 < 1/2$, proves

$$\mathbb{E}\prod_{i=1}^{M}\{1 + r(1+E_i)^{p/2}\} \leq e^{C_{p,r}M}. \tag{CH35}$$

For a nonzero Gaussian mean m, the same calculation has the additional factor $\exp(C_p r^{2/p} m^T A m)$. It is still exponential in M whenever that sum of local mean energies is $O(M)$, or after its known mean load is included in each source normalization.

A concrete sufficient condition is $E_i = \sum_y w_{iy}|B_y|^2$, with nonnegative weights and both $\sup_i \sum_y w_{iy}$ and $\sup_y \sum_i w_{iy}$ bounded. The first bound controls the trace in (CH32); the second controls its operator norm. Exponentially localized source kernels at separated cell centers have these two bounds by the elementary packing sum. One must use curvature/readout or actual cold-energy coordinates: massless gauge potentials do not have a uniformly bounded covariance operator merely because their kernels are local in a formal expression.

In the ordinary squarefree coefficient norm,

$$\left\| \prod_i (1 + z_i Q_i) \right\|_{\boldsymbol{r}} = \prod_i (1 + r_i |Q_i|). \tag{CH36}$$

Thus (CH35) bounds all those coefficients simultaneously, with a constant to the power M, when $|Q_i| \le \Lambda_i (1 + E_i)^{p/2}$ and $r_i \Lambda_i = r$. It does not assert the existence of $\mathbb{E} e^{z Q_i}$ at a nonzero ordinary cubic source.

For a live cold integral the pointwise part of (CH34) is still valid. If $\sum_i E_i \le w_0 E_{\text{live}}$, one fixed fraction of that SAME retained action pays all the marks. Its change of Gaussian normalization costs $\exp(C\theta N_{\text{actual}})$, counted on the actual retained coordinates as in history (13.12). No product of integrated island masses is used. The following section uses a bounded-atom representation and therefore does not need to infer (CH32) from the earlier global Gevrey norm.

36.9 9. The terminal weak-coupling cost is polynomial per cell

Stop this replicated-cell calculation at the observation scale h, with inverse coupling $x = x_h$, on the SAME original bare sequence. All source-zero RG steps after that stopping point are simply left inside the retained physical integral. Let N_h be its terminal cell count.

The upper absolute R bank is uniform for all endpoints above the fixed weak floor. In particular, the remaining compact/Gaussian normalization powers cost at most $C(\log x_j)N_j$ on their ACTUAL stored coordinates; these are precisely the normalization costs in R-II p. 380 and (1.79), already paid in history section 13.4. Original creation and waiting reserves keep a fixed spare fraction after this payment. The completed regular source-free norm costs e^{CN_h}. Hence (CH39) and the common-vacuum identity give

$$\|\mathcal{Z}_a(z)\|_{\boldsymbol{r}} \le \exp[V_N + CN_h]. \tag{CH40}$$

Here the local source budget is fixed small; its $C\eta N_h$ is included in CN_h.

For the lower bound use the completed positive branch with no surviving terminal island. It still includes healed fine-scale R histories and therefore has this SAME V_N. Restrict every terminal link to $U(b) = \exp(ix^{-1/2}B(b))$, $|B(b)| \le c$, with fixed small c. This set lies inside the original shrinking core $\alpha_j = g_j(\log x_j)^q$. Its leading Wilson action is at most CN_h; its regular effective remainder is bounded by CN_h. Haar measure on every independent link has mass at least $c_1 x^{-\dim G/2}$, and the number of those links is at most a fixed multiple of N_h. Consequently

$$Z_a(0) \ge \exp[V_N - C(1 + \log x_h)N_h]. \tag{CH41}$$

This proves the required terminal dependence, rather than leaving it inside an unspecified $C(x_h)$ which might grow too fast.

Combining (CH40)–(CH41) gives

$$\|\mathcal{Z}_a(z)/Z_a(0)\|_{\boldsymbol{r}} \le (Cx_h^C)^{N_h}. \tag{CH42}$$

If M occupied replicated cells fill a fixed positive fraction of this torus, $N_h \le C_0 M$. Choose the fixed radii $r_i = c\eta/\Lambda_i$ allowed by (CH37). Coefficient extraction proves

$$|S_{M,a}(f_1, \dots, f_M)| \le C^M \prod_{i=1}^{M} \{x_h^C \Lambda_i\}. \tag{CH43}$$

Thus the additional terminal factor is only one fixed power of x_h per physical insertion. Since $x_h = O(1 + \log(1/h))$ on the fixed run, it is compatible with a fixed tempered short-distance power.

The first normalization matrices in this argument must be fixed from the universal local coefficient families, before the torus or observation grid is replicated. A wrapping carrier at scale $\ell \le h$ has the additional locality factor $e^{-cT/\ell}$, where T is the physical torus side. Polynomial running ratios are paid by a fixed fraction of this factor. Its sum over $\ell \le h$ is bounded uniformly in the bare cutoff. Such wrapping contributions stay in the retained source activities or a bounded wrapping cocycle; they do not redefine the physical field separately in each volume.

Using (CH43) in a physical reflected cell seminorm also requires the fixed finite observable family to transform covariantly under the lattice reflections used in the chessboard. One must verify that covariance for the selected universal first normalization, rather than bound a possible conjugation by its cutoff-dependent condition number. The replicated bound above does not silently assume a general tensor rotation theorem.

37 Separated source marks and the historical normalization cost

This section proves the geometric and numerical estimate that removes an apparent superfactorial cost of historical normalization. It applies to separated product tests and retains the actual scale of every stored record. It does not replace the dimension-filtered construction for higher composites.

37.1 The length of a carrier containing separated marks

Lemma 37.1 (Separated-anchor length)**.** *Let $y_1, \dots, y_m$, $m \geq 2$, have pairwise distances at least h in Euclidean space or a flat torus. On the torus assume that h is smaller than a fixed fraction of every period. Every connected rectifiable graph T containing these anchors satisfies*

$$\operatorname{length}(T) \geq mh/4. \tag{163}$$

Proof. The open balls $B(y_i, h/4)$ are pairwise disjoint. A path in T from y_i to any other anchor exits $B(y_i, h/4)$, contributing at least $h/4$ of length inside that ball. Sum over the disjoint balls. The argument uses the ambient torus metric and allows winding graphs. □

In the small-field localization, $d_j(X)$ is the length of the shortest continuous tree meeting every M-cube of X, divided by M in scale-j coordinates; see the definition following (0.24) in [4]. Here M is fixed. The increasing logarithmic buffer R_j is not the unit of this distance.

Let ℓ_j be the physical scale. If an anchor is filed by its containing $M\ell_j$ cube, join it to the localizing tree inside that cube at cost at most $CM\ell_j$. Lemma 37.1 gives

$$d_j(X_{\mathrm{full}}) \geq c\, mh/\ell_j - Cm. \tag{164}$$

The constants include M and the filing collar. The full carrier must retain every physical source anchor. An anchor kept only as an unnamed parameter cannot be used in this estimate.

37.2 A spare exponential pays the polynomial normalization

Assume the bounded-drift comparison along the weak trajectory, in the form

$$x_j \leq A + B \log(H/\ell_j), \qquad 0 < \ell_j \leq H, \tag{165}$$

with terminal physical scale H and fixed A, B.

Lemma 37.2 (Scale supremum)**.** *For fixed $\gamma, D \geq 0$ and $a > 0$, and $0 < h \leq H$,*

$$\sup_j x_j^{\gamma} \ell_j^{-D} e^{-ah/\ell_j} \leq C h^{-D} \bigl(1 + |\log(h/H)|\bigr)^{\gamma}. \tag{166}$$

The constant depends on A, B, γ, D, a, H, not on the ultraviolet length.

Proof. Put $t = h/\ell_j$. Equation (165) bounds x_j by a constant times $(1 + |\log(h/H)|)(1 + \log_+ t)$. Use $\ell_j^{-D} = h^{-D} t^D$ and

$$\sup_{t>0} t^D (1 + \log_+ t)^{\gamma} e^{-at} < \infty.$$

□

Combining (164) and Lemma 37.2, a fixed portion of the localization exponential gives

$$x_j^{\gamma m}\ell_j^{-Dm}e^{-\kappa d_j(X_{\mathrm{full}})/2}\le\left[Ch^{-D}(1+|\log(h/H)|)^{\gamma}\right]^m. \tag{167}$$

Another fixed portion remains for carrier entropy and the output marked-tree bound. The constants do not depend on m or the ultraviolet length. A root-position count ℓ_j^{-4} can be included by increasing D by 2, since $m\ge 2$, with a harmless constant depending on H. It can also be paid with another fixed portion of the exponential.

Optimizing $x^{\gamma m}$ against just $e^{-c(\log x)^p}$, $p>1$, would cost $\exp(Cm^{p/(p-1)})$, which is not a fixed factorial power. Separation supplies the additional geometric cost before the ultraviolet supremum. Also any fixed power of $1+|\log(h/H)|$ is bounded by a fixed extra power of h^{-1}. Thus (167) has the form $C^m h^{-bm}$.

37.3 One-mark components and propagation

A completed one-mark marginal large-field vertex, with normalization at most x_j^{γ}, has bound

$$x_j^{\gamma}e^{-cP_0(x_j)},\qquad P_0(x)=(\log x)^p, quad p>1. \tag{168}$$

Its supremum and actual scale sum are finite. The two-sided running tube gives $x_j\ge x_*+c_-(n-j)-C_0$ with $c_->0$. Eventual monotonicity of the envelope, with the weak floor beyond its initial interval, gives

$$\sum_j x_j^{\gamma}e^{-cP_0(x_j)}\le C\int_{x_*/2}^{\infty}(1+u)^{\gamma}e^{-c'(\log(1+u))^p}\,du. \tag{169}$$

This is a fixed constant per one-mark component, not the bound obtained by replacing γ by γm.

Zero-vacuum field-dependent propagation is rooted propagation: its fine-to-coarse readout has an r^4 factor without a translation density at the prescribed root. The regular/live Volterra bound sums repeated encounters. A normalization paid at an actual record is not paid again when that source is carried to the next scale.

For a stable old clean family this follows quantitatively from $x_t\le x_j+C(j-t)+C$ for $t\le j$. For every $\sigma>0$,

$$\sum_{t\le j}x_t^{\gamma}L^{-\sigma(j-t)}\le C'x_j^{\gamma}. \tag{170}$$

The complete old readout at one layer costs that layer's polynomial once, not a product over all birth origins. In the finite normalization cocycle, segment factors are bounded by $\exp(C\sum_s x_s^{-1})$ over disjoint scale intervals. The intervals along one insertion line do not overlap. This remains valid with the bounded nonmonotone coupling correction.

37.5 The marked-forest estimate and its hypotheses

Proposition 37.3 (Separated normalization payment)**.** *Consider a localized squarefree-source coefficient expansion for a fixed finite operator list, with the following properties.*

1. *Products partition labels among disjoint children; labels are never duplicated.*
2. *Entry/readout records have factors at most x_j^{γ}, with fixed γ, and subsequent propagation retains their paid tags.*
3. *Completed multi-mark carriers retain all anchors and a fixed surplus of the full exponential tree bound.*
4. *One-mark marginal vertices satisfy (168). Unchanged label groups have the rooted propagation estimate after entry normalization; unmarked histories obey their original locality and cold-action budgets.*
5. *A fixed number of polynomial fields per operator is integrated with the real joint reference Gaussian or cold action, and connected formation retains its source roots.*

Then the normalization and history payments contribute at most

$$C^m (m!)^q h^{-bm} \prod_{i=1}^{m} \|f_i\|_{C^s_{\mathrm{scaled}}}, \tag{171}$$

with fixed q, b, s depending on the operator list and localized class, not on m or the ultraviolet depth.

Proof. Use (167) for completed multi-mark entries and (169) for one-mark entries. Collapse maximal stretches carrying one unchanged nonempty label set into its rooted propagator. Every remaining branch point joins at least two disjoint nonempty sets. There are at most $m-1$ branch points and m

marked leaves. Rooted ordered tree shapes number at most C^m; leaf labels and derivative partitions add only a fixed factorial power.

If total polynomial degree is at most $D_0 m$, Gaussian moments cost at most

$$C^{D_0 m}\sqrt{(D_0 m)!} \le C_{D_0}^m (m!)^{D_0/2}. \tag{172}$$

A polynomial in the number N of actual stored coordinates satisfies

$$N^m \le m!\, \varepsilon^{-m} e^{\varepsilon N}. \tag{173}$$

Its exponential uses one fixed additional record charge, not a charge proportional to m. The remaining unmarked sums and propagation stretches are bounded by the assumed budgets. Multiply these estimates. □

This is a payment theorem for the stated localized class. Scalar island masses alone do not imply the marked-carrier or one-time-record hypotheses.

37.6 The remaining higher-dimensional distinction

For the 21 curvature-bilinear types the integrated engineering dimension is four, so (168) has no power of ℓ_j^{-1}. Clean finite center/stable normalization gives polynomial coupling factors. Large-field descendants keep their own radii and paired counterterms.

For dimension $d > 4$ a raw one-mark insertion instead has factor ℓ_j^{4-d}. Since x_j grows only logarithmically in ℓ_j^{-1},

$$\ell_j^{4-d} e^{-c(\log x_j)^p} \longrightarrow \infty \qquad (\ell_j \longrightarrow 0). \tag{174}$$

The one-mark reserve and separation do not fix this divergence. One must extract lower-dimensional mixing and normalize the dimension-d block. A weighted Taylor remainder above degree $D \ge d$ has an old-readout gain r^{D+1-d}, which pays the fine-origin sum. The actual all-depth filtered insertion cocycle with that property still has to be constructed for the full operator family.

The separated geometry therefore removes the identified factorial-growth obstruction without promoting scalar-source success or clean 21-dimensional normalization to the full Yang–Mills observable algebra.

38 Sharp conditioning, higher composites, and a Gaussian repair

The fixed-core first-insertion argument for curvature bilinears does not automatically extend to every engineering dimension. A sharp average can make the conditional mean of a high derivative singular on a block face. This already occurs in a source-free Gaussian model. We give an exact counterexample, explain its relation to the flat Wilson conditioning, and prove a separated-correlation Gaussian repair. The additional interacting estimate is kept explicit.

38.1 An exact conditional counterexample

Let ϕ be the stationary centered real Gaussian field on the line with

$$C(x-y) = \frac{e^{-m|x-y|}}{2m}, \qquad m > 0. \tag{175}$$

Thus $(-\partial_x^2 + m^2)C = \delta$. Define

$$Y = \int_0^1 \phi(x)\,dx, \qquad v = \mathbb{E}Y^2 > 0, \qquad h(x) = \mathbb{E}[\phi(x)Y] = (C * 1_{[0,1]})(x).$$

The locally integrable covariance makes Y a legitimate Gaussian variable. In the distributional sense,

$$(-\partial^2 + m^2)h = 1_{[0,1]}, \qquad h''' = m^2 h' - \delta_0 + \delta_1. \tag{176}$$

Choose a real even smooth mollifier ρ supported in $(-1,1)$, of integral one, and set $\rho_\varepsilon(x) = \varepsilon^{-1}\rho(x/\varepsilon)$. For $\varepsilon > 0$ let

$$X_\varepsilon(x) = \partial^3(\phi * \rho_\varepsilon)(x), \qquad O_\varepsilon(x) = X_\varepsilon(x)^2 - \mathbb{E}X_\varepsilon(x)^2. \tag{177}$$

These are finite-variance Gaussian polynomials. The second expression contains the full source-free Wick subtraction.

Proposition 38.1 (Divergent conditional quadratic coefficient)**.** *For every x and $\varepsilon > 0$,*

$$\mathbb{E}[O_\varepsilon(x) \mid Y] = \big(h''' * \rho_\varepsilon(x)\big)^2 \frac{Y^2 - v}{v^2}. \tag{178}$$

If $f \in C_c^\infty((-1/4, 1/4))$ and $f(0) > 0$, then

$$a_\varepsilon(f) := \int f(x)\big(h''' * \rho_\varepsilon(x)\big)^2\,dx = \frac{f(0)}{\varepsilon}\int \rho(u)^2\,du + O(1), \tag{179}$$

$$\big\|\mathbb{E}[O_\varepsilon(f) \mid Y]\big\|_{L^2} = \frac{\sqrt{2}\,|a_\varepsilon(f)|}{v}. \tag{180}$$

Subtracting the conditional value at $Y = 0$ still leaves an unbounded coefficient on every fixed nonzero coarse-field ball.

Proof. Regression gives conditional mean $(h''' * \rho_\varepsilon(x))Y/v$ and variance equal to the unconditional variance minus $(h''' * \rho_\varepsilon(x))^2/v$. Wick subtraction proves (178).

For $\varepsilon < 1/8$, the delta at 1 in (176) does not contribute on $\operatorname{supp} f$. The function $m^2 h' * \rho_\varepsilon$ is uniformly bounded there, whereas

$$\int f(x)\rho_\varepsilon(x)^2\,dx = \varepsilon^{-1} f(0)\int \rho(u)^2\,du + O(1).$$

The cross term and the square of the bounded part are $O(1)$. This proves (179). The identity $\mathbb{E}(Y^2 - v)^2 = 2v^2$ proves (180). Flat-value subtraction leaves $a_\varepsilon(f)Y^2/v^2$. □

The test f was already smooth, and the conditional expression is an entire quadratic polynomial in Y. Wick subtraction, smooth test smearing, and analyticity in the retained variable therefore do not give the proposed uniform first-insertion norm. Bounds for the covariance or the first derivative of its harmonic mean do not imply bounds for arbitrary higher composites. This does not assert divergence of the completed Gaussian off-diagonal correlation.

38.2 The sharp transverse average at flat background

In the flat free vector problem take a transverse color component A_μ depending only on x_ν, with $\nu \neq \mu$, and constant in tangential directions. The gauge-fixing correction vanishes since $\partial_\mu A_\mu = 0$. The extra line average in direction μ is trivial, but the actual transverse box average remains sharp.

For a coarse multiplier λ, the constrained mean solves

$$-\partial_\nu^2 A_\mu = Q_\mu^* \lambda. \tag{181}$$

On a periodic torus choose a mean-zero combination of interval indicators. The zero mode is then removed. The datum $B = Q(-\Delta)^{-1}Q^*\lambda$ realizes this mean, because the coarse Schur operator is invertible on the transverse mean-zero sector. This is an actual constrained mean, not an unrelated forcing.

The right side of (181) is piecewise constant, so the third normal derivative has face deltas. Since $F_{\nu\mu} = \partial_\nu A_\mu$, the derivative $\partial_\nu^2 F_{\nu\mu}$ has this defect. Fine differences at mesh η approximate a delta of height $O(\eta^{-1})$ in width $O(\eta)$; its square has integrated coefficient of order η^{-1}. The Gaussian color factor is scalar, and one Cartan color supplies a nonzero invariant quadratic contribution. Thus the dimension-eight type

$$\operatorname{tr}\big[(D_\nu^2 F_{\nu\mu})^2\big] \tag{182}$$

obstructs an automatic uniform pointwise conditional insertion estimate.

This is a statement about the flat Gaussian sector and the actual sharp transverse average. It does not prove that every interacting component has this coefficient, or that every renormalized representative fails. Equations of motion and contact quotients must also be respected. The issue is not confined to that quotient: in four scalar dimensions,

$$\partial_\nu^3 \phi - \tfrac{1}{2}\partial_\nu \Delta \phi$$

is trace-free and still has a nonzero face delta in its conditional mean near a flat box face. The first term contains the full normal jump and the second removes only half. A face-dependent coarse counterterm can cancel a chosen block's coefficient, but is not automatically a translation-covariant local continuum counterterm.

38.3 Cancellation after the marked coefficients are combined

Lemma 38.2 (Exact conditional cancellation)**.** *Let X_1, X_2, Y be jointly centered Gaussian variables, and put $K = \mathbb{E}X_1X_2$, $h_i = \mathbb{E}X_iY$, $v = \mathbb{E}Y^2 > 0$, $O_i = X_i^2 - \mathbb{E}X_i^2$. Then*

$$\mathbb{E}\operatorname{Cov}(O_1, O_2 \mid Y) = 2K^2 - 2(h_1h_2/v)^2, \tag{183}$$

$$\operatorname{Cov}(\mathbb{E}[O_1 \mid Y], \mathbb{E}[O_2 \mid Y]) = 2(h_1h_2/v)^2. \tag{184}$$

Their sum is the correct full covariance $2K^2$.

Proof. The conditional cross covariance and means of the X_i are $C_{12} = K - h_1h_2/v$ and $m_i = h_iY/v$. Thus

$$\operatorname{Cov}(O_1, O_2 \mid Y) = 2C_{12}^2 + 4m_1m_2C_{12}.$$

Take its expectation using $\mathbb{E}Y^2 = v$. The conditional Wick means are $h_i^2(Y^2 - v)/v^2$; taking their covariance proves the second identity. □

Sharp averaging can make the separate terms contain divergent face factors. Taking absolute values before cancellation destroys the bound. At every finite source order the same principle is the tower identity: in a squarefree source algebra the completed coefficient equals the full Wick-pairing sum. No polynomial exponential moment is required.

38.4 A Gaussian repair for each finite operator list

Keep the Gaussian covariance and action unchanged. For a finite list of local curvature polynomials of dimensions at most D, smooth each marked field leg with a real compactly supported mollifier of physical radius σ, and Wick-order using the full covariance. Only the observables are regulated.

Free four-dimensional curvature covariances satisfy, off the diagonal,

$$|C_{r,s}(x,y)| \le C_{r,s}|x-y|^{-4-r-s}. \tag{185}$$

Differentiate the Green function proportional to $|x-y|^{-2}$ twice for the curvature legs and then $r+s$ more times. A mass or a fixed-volume infrared convention adds only smooth terms on a separated compact region.

Proposition 38.3 (Fixed-dimension separated Gaussian limit)**.** *Suppose $f_1,\dots,f_n$ have pairwise h-separated supports, and vertex i has dimension $d_i \le D$. For $\sigma < h/10$, the complete Wick correlations converge as $\sigma \downarrow 0$ and obey*

$$|S_n(f_1,\dots,f_n)| \le C_D^n (n!)^q h^{-\sum_i d_i} \prod_{i=1}^n \|f_i\|_{L^1}, \tag{186}$$

with fixed q. Connected correlations and multiple species obey the same conclusion.

Proof. Wick ordering removes all within-vertex contractions. Every cross-vertex contraction obeys (185) at distance at least a fixed fraction of h. Thus full diagrams are products of smooth cross kernels; dominated convergence removes σ after a diagram is formed.

A curvature derivative of order r has dimension $2+r$. The exponent of h^{-1} in a complete pairing is therefore $\sum_i d_i$. With at most Kn legs, K fixed by D, the pairing count satisfies

$$(Kn-1)!! \le C_D^n (n!)^{K/2}, \tag{187}$$

enlarging the fixed constant or exponent for parity and tensor choices. Integration against the tests proves the result. Connected Wick graphs are a subset of these diagrams. □

For scaled cube tests this gives $C_D^n (n!)^q h^{-bn}$ times scaled smooth norms. The order matters: regulate the legs; retain the complete conditional polynomial if a sharp disintegration is used; combine by the tower identity; estimate full Wick contractions; then remove σ. Proposition 38.1 rules out a uniform limit for every individual sharp-conditioned first insertion.

38.5 Reflection support and the limit topology

Choose the mollifier real and invariant under the desired reflections. A test at distance $t_0 > 0$ from the reflection plane has its whole determining carrier in the same half-space when $\sigma < t_0/2$. The regulated Wick polynomial is a real polynomial of positive-side Gaussian variables, so reflection-positive Gram

inequalities hold at each regulator. Off-diagonal convergence and the separated distribution bound pass them on the corresponding flat test-function space.

This constructs operator distributions on separated or flat tests. It does not assert that every high-dimensional Wick field is an ordinary random tempered distribution with finite variance on all overlapping tests. Contact extensions and reconstruction topology remain part of the statement. A Wilson version needs the same support rule: a vanishing collar is permissible; a fixed smearing across the reflection plane is not.

38.6 Smooth auxiliary Gaussian observations

For fixed D, sufficiently smooth compact averages can instead be used in an auxiliary Gaussian disintegration. If $Y_w = \phi(w)$, then $h_w = C * w$ solves an elliptic equation with smooth forcing. All derivatives needed for the finite list are locally bounded, and the face defect is absent. The observer changes, not the Gaussian covariance.

For two linear observations Q_0, Q_1 of one finite-dimensional Gaussian field, set $\Gamma_{ab} = Q_a C Q_b^*$. On the nonsingular observed subspace,

$$\mathbb{E}[Y_1 \mid Y_0] = \Gamma_{10}\Gamma_{00}^{-1}Y_0, \tag{188}$$

$$\mathrm{Cov}(Y_1 \mid Y_0) = \Gamma_{11} - \Gamma_{10}\Gamma_{00}^{-1}\Gamma_{01}. \tag{189}$$

These formulas do not identify interacting conditional laws. If the physical density contains $e^{-V(\phi)}$, that factor and all old marks must stay in the joint numerator; the Gaussian formulas alone lose the interaction. Keeping the sharp variables as additional conditions also keeps their face forcing, which must be integrated or cancelled in the completed coefficient.

38.7 The connected interacting estimate still required

For the unchanged Wilson action, the fixed-D target is concrete. Construct locally gauge-invariant regulated insertions and lower-dimensional counterterms, and retain complete joint coefficients through the frozen original maps. On separated supports prove a bound like (186) for the completed coefficients, uniformly as the observable regulator tends to zero below a fixed fraction of the separation. The cancellation in Lemma 38.2 must occur before absolute values. This would give zero-smear convergence and distribution extension without a false conditional pointwise premise.

The present absolute norm for individual conditional activities does not prove this estimate. A new norm or regrouping must retain Gaussian contractions and contact cancellations. Source-free analyticity and cold-action bounds are useful, but do not alone supply that marked estimate.

Nonlinear gauge-covariant smoothing is also not automatic. Transporting unflowed curvature along a fixed path gives an extended observable. Already in an Abelian Gaussian model the variance of a line integral grows as length divided by the microscopic cutoff, by integrating the $|x-y|^{-2}$ covariance along the path. Its exponential needs its own normalization. Free linear convolution cannot be imported unchanged.

Changing the actual averaging operation to a smooth Q instead requires an exact source-preserving joint coarea/transport identity, uniform paired-domain and locality bounds, and running scalar matching for the same bare Wilson sequence. One-loop universality is not that nonlinear theorem. Separate retuning for each operator dimension would not give one consistent full family.

38.8 The consequence for filtered normalization

Finite center/unstable/stable normalization remains useful once its analytic input is established. Automatic input fails: affine constant curvature is reproduced, but higher polynomial jets acquire cell correctors. The kernel of a naive continuum jet projection need not be invariant, and sharp high derivatives can fail the norm outright.

A general fixed-D theorem therefore needs either a corrected Gaussian operator-jet space with bounded readout and strict remainder, or the completed-distribution estimate above. A formal finite jet matrix cannot promote the clean 21-dimensional result to all dimensions. The filtered invariant curvature and covariant-derivative family is part of the Yang–Mills target in [3, Section 4]; a scalar sector alone does not supply it.

38.9 An exact nonlinear repair for smooth cylinder interactions

The cancellation extends beyond a Gaussian measure for a restricted, explicit class of interactions. Let $Z = (Z_1, \dots, Z_k)$ be finitely many smooth linear observations of the Gaussian field, with nonsingular covariance Γ. Their kernels have the derivatives required by the fixed dimension-D operator list. Suppose a nonnegative measurable function F satisfies

$$F(z) \le Be^{\theta z^T \Gamma^{-1} z/2}, \qquad 0 \le \theta < 1, \qquad Z_F := \mathbb{E}_C F(Z) \ge b > 0. \tag{190}$$

Here k, B, b, θ and the observation kernels are fixed. Define $d\mu_F = Z_F^{-1} F(Z) d\mu_C$. No polynomial or small-coupling assumption on F is imposed. Reflection positivity and Euclidean covariance are not automatic for this chosen density and are not asserted by this lemma.

Let X_i collect the regulated linear field legs at marked vertex i, and put $h_i = \mathbb{E}_C[X_i Z^T]$, $C_{ij} = \mathbb{E}_C[X_i X_j^T]$. For formal source vectors t_i, set $v(t) = \sum_i h_i^T t_i$.

Proposition 38.4 (Completed source formula for a nonlinear smooth tilt)**.** *Full source-free Wick ordering at each vertex gives*

$$\mathbb{E}_{\mu_F} \prod_i : e^{t_i^T X_i} :_C = \exp\left(\sum_{i<j} t_i^T C_{ij} t_j \right) \times \mathbb{E}_{\mu_F} \exp\Big(v(t)^T \Gamma^{-1} Z - \tfrac{1}{2} v(t)^T \Gamma^{-1} v(t) \Big). \tag{191}$$

For every fixed finite dimension-D list, the completed separated correlations have a regulator-independent bound

$$|S_{n,F}(f_1, \dots, f_n)| \le C_{D,F}^n (n!)^{q_D} h^{-b_D n} \prod_i \|f_i\|_{L^1}, \tag{192}$$

and a zero-smear limit on fixed compact separated regions. Connected correlations satisfy a bound of the same form with a fixed increase in the factorial exponent.

Proof. Condition on Z. The conditional means and covariance blocks are $h_i \Gamma^{-1} Z$ and $C_{ij} - h_i \Gamma^{-1} h_j^T$. Subtracting the unconditional self covariance in each Wick exponential removes the C_{ii} terms. The remaining observation quadratic form is $-v^T \Gamma^{-1} v/2$; the full cross blocks C_{ij} remain as displayed. Integration against F proves (191), either as a finite-order formal identity or as a finite-dimensional generating identity on its domain. An earlier sharp disintegration gives the same answer by the tower identity, so artificial face terms cancel before absolute values are taken.

For the marked curvature derivative legs, every $h_i(x)$ is uniformly smooth as the marked mollifier is removed: derivatives can be transferred to the fixed smooth observation kernels. The only singular kernels are the full cross blocks C_{ij}, controlled on separated supports by (185). The density allowance implies

$$\mathbb{E}_{\mu_F} \prod_{a=1}^{r} |l_a^T \Gamma^{-1/2} Z| \le C_F C_\theta^r r^{r/2} \prod_a |l_a|. \tag{193}$$

Indeed, its Gaussian integral is bounded by the one with precision $(1-\theta)I$, and Hölder's inequality with Gaussian moment bounds gives (193).

The finite-order expansion of (191) contains separated covariance pairings, smooth observation legs, and the moments (193). There are at most $K(D)n$ legs; their number and size have fixed factorial growth. Smooth terms are absorbed in $h^{-b_D n}$ for h bounded above. This proves (192). Dominated convergence removes the marked regulator. The cumulant partition formula adds at most a fixed factorial power. □

This is a nonlinear probability density and an exact nonperturbative fixed-D result. It does not cover the Wilson interaction, which involves arbitrarily fine local variables. A cylinder approximation would need uniform control of the observation derivatives, covariance inverses, normalization, and marked collision counterterms as its dimension grows. The source-free polymer bound does not supply these derivative-weighted estimates. Formula (191) identifies the desired completed-contraction structure without assuming a smooth-observer representation for the physical interaction.

38.10 Translated grids and protected marked points

For finite depth, globally nested changes of grid origin preserve the same bare sequence. Write $\ell_j = L^j a$ and suppose origins o_j obey $o_{j+1} - o_j \in \ell_j \mathbb{Z}^4$. For every $j \le n$,

$$o_n - o_j \in \ell_j \mathbb{Z}^4, \qquad o_j + \ell_j \mathbb{Z}^4 = o_n + \ell_j \mathbb{Z}^4. \tag{194}$$

Thus the entire hierarchy is one fine-lattice translation of the original one. Translation invariance of the Wilson measure gives equality for the same bare couplings; this does not require an interscheme tuning theorem.

At one coarsening step there are L possible next origins in each coordinate. For one anchor at most $2\delta L + 2$ choices put it within $\delta \ell_{j+1}$ of a face. All m anchors can therefore be protected if

$$m(2\delta L + 2) < L. \tag{195}$$

For example $L \ge 8m$ and $\delta \le 1/(16m)$ suffice, independently in each coordinate. A support ball is handled by adding its radius divided by ℓ_{j+1} to δ. For any fixed derivative order r, the cost δ^{-r} is polynomial in m and is compatible with a fixed factorial power.

Two limitations of this one-step rule are exact. Increasing L cannot protect a fixed margin δ independent of arbitrary m, because m arcs of fixed fractional width can cover all offsets. Also, at fixed L and $m \ge L$, after the current origin is fixed, some separated configurations defeat every next nested extension, even for an arbitrarily small positive margin. Choose one anchor in each of the L possible face phases modulo $L\ell_j$ in one coordinate. Add multiples of $L\ell_j$ to increase physical separation if necessary. Every next origin places a face through one anchor. For centered odd-L cells these are half-ℓ_j residues, realizable by fine plaquette centers. This does not rule out choosing the entire global translation in advance. The next subsection proves that this stronger choice works with fixed L and a polynomial cost in m.

At flat constant coupling the composite straight average has multiplier

$$q_\mu(p) = a_N(p_\mu)^2 \prod_{\nu \neq \mu} a_N(p_\nu). \tag{196}$$

Its transverse adjoint forcing is piecewise constant on the final blocks. The particular coarsest conditional mean has no hidden array of all finer transverse faces. Away from its piecewise-polynomial forcing interfaces, elliptic interior regularity controls every fixed derivative. Protecting points is therefore a genuine Gaussian possibility. It does not remove fine interfaces from a state still retaining fine conditional variables or running-band projectors; their removal requires completed conditional integrations.

Choosing L as a function of source order creates another obligation. The local constants and weak-coupling ceiling are given for fixed L. Their polynomial dependence on L, needed for $C^m(m!)^q$ bounds and one fixed endpoint for all m, has not been established. A large power of a fixed blocking factor preserves a subsequence of scales, but the intermediate sharp variables must actually be integrated out; retaining their old conditional representatives retains their faces. Nonlinear currents and original large-field history norms must also be compared under that regrouping. The flat multiplier identity alone is insufficient.

Separate translated grids for separated clusters likewise have no immediate joint conditional estimate. Conditioning on all global grids retains the union of their faces, so protection from only one's own grid does not suffice. A patchwork observer can place transition regions away from protected clusters, but needs uniform constraint gaps, locality, paired domains, and exact source-preserving transport through the original unmarked construction. Keeping the old sharp constraints in the patch keeps the obstruction. This is a precise possible program, not an established interacting repair.

38.11 A global grid with constant margins at every active scale

The one-step obstruction can be bypassed without changing L. Fix $L \geq 2$, microscopic spacing a, scales $\ell_j = L^j a$ for $0 \leq j \leq n$, and m arbitrary anchors in four dimensions. Let $R \geq 0$ be a fixed microscopic support or stencil radius. Allow q fixed families of coordinate face phases at each level; $q = 1$ is the ordinary cubical grid, and a fixed larger q covers additional translated knot planes. Choose the least integer $r \geq 2$ such that

$$L^r \geq 128qm(R+1)r, \qquad \delta_m = \frac{1}{128qmr}. \tag{197}$$

Then $r = O_{L,q,R}(\log(m+2))$ and $L^r \leq C_{L,q,R} m \log(m+2)$. There is a single fine-lattice translation $o \in a\{0, \dots, L^n - 1\}^4$ such that every anchor, every face family, and every level $r \leq j \leq n$ satisfy

$$\operatorname{dist}(x_i, \text{level-}j \text{ face planes}) > Ra + \delta_m \ell_j. \tag{198}$$

All grids use the same o and remain nested. The translation may depend on the tuple and the finite cutoff. No positive limiting density of good translations is required.

Finite digit proof. Choose o/a uniformly, with independent base-L digits. At level j the relevant residue is o modulo $L^j a$. Divide its fractional circle into L^r bins and enlarge every bad interval to all bins it meets. Let B_j be the event that any enlarged interval occurs for any anchor, face family, or coordinate. It depends only on digits $j-r, \dots, j-1$ in the four coordinates. One original bad interval has length $2\delta_m + 2RL^{-j}$, and enlargement adds at most $2L^{-r}$. Hence

$$\mathbb{P}(B_j) \leq 8qm\{\delta_m + (R+1)L^{-r}\} \leq \frac{1}{8r}. \tag{199}$$

The event is independent of the entire collection of events whose levels differ by at least r. Its dependency degree is at most $2r-2$. For $x = 1/(4r)$, Bernoulli's inequality gives

$$x(1-x)^{2r-2} \geq \frac{1}{8r}.$$

Here is the elementary finite local-lemma argument needed in this case. For events of dependency degree at most D with $p_i \leq x(1-x)^D$, induction over conditioning sets gives

$$\mathbb{P}\left(B_i \,\middle|\, \bigcap_{k\in S} B_k^c \right) \leq x, \qquad i \notin S. \tag{200}$$

Partition S into neighbors T of i and nonneighbors U. The numerator is bounded by $\mathbb{P}(B_i \mid \text{avoid } U) = p_i$, using independence of the complete nonneighbor collection. The chain rule and the inductive bounds make the denominator $\mathbb{P}(\text{avoid } T \mid \text{avoid } U)$ at least $(1-x)^{|T|}$. They also show that every conditioning event has positive probability. Therefore

$$\mathbb{P}\left(\bigcap_{j=r}^{n} B_j^c \right) \geq (1-x)^{n-r+1} > 0.$$

Any origin in this event satisfies (198). If $n < r$, all scales belong to the finite initial range and there are no active constraints.

Fixed derivative costs. For the flat sharp average, the adjoint constraint forcing is piecewise polynomial, with knots on finitely many final-scale face families. On a protected ball the fine harmonic lift solves a constant-coefficient lattice Poisson equation with forcing of fixed polynomial degree. At unit retained scale its curvature readout obeys

$$|\nabla_\eta^s F_j B(x)| \leq C_s \delta_m^{-s-2} \|B\|_{\mathrm{loc}}, \qquad \eta = a/\ell_j, \tag{201}$$

where the coarse norm is exponentially weighted and the constants are independent of microscopic depth. Subtract a polynomial particular solution in a ball of radius ρ comparable to δ_m. For the harmonic remainder, a cutoff Green representation and summation by parts give the four-dimensional interior bound $C_s \rho^{-s-2} \|u\|_{L^2}$. The required lattice Green derivative estimate $C_s(|x|+\eta)^{-2-s}$ follows by annular Fourier decomposition and integration by parts. The localized Schur bound controls the particular solution's coarse multiplier. When ρ is comparable to the stencil scale, elementary point evaluation and finite differences give the same bound. The clearance Ra keeps the original finite stencil on the protected side.

A derivative-of-curvature leg of order s has engineering dimension $d = 2+s$. Restoring physical units changes (201) to $C_s \delta_m^{-d} \ell_j^{-d}$ times its unit-scale coarse norm. Thus m fixed-dimension-D insertions incur a readout cost bounded by

$$(C_D \delta_m^{-D})^m \leq C_D^m [m \log(m+2)]^{Dm} \leq C_D^m (m!)^{2D}. \tag{202}$$

The exponent is deliberately generous. Fixed-degree Gaussian moments add only another fixed factorial power. The same δ_m works at every active scale; there is no product of margin losses over ultraviolet levels. For the exact orthogonal running-band Gaussian, the conditional mean is the single homogeneous harmonic lift, independently of the positive band weights. The exact-conditioning chapter proves the criterion for this statement.

Scope of the improvement. Only the first $r = O(\log(m+2))$ levels remain unprotected. Their scale ratio L^r is polynomial in $m\log(m+2)$. An independently proved direct estimate costing $(L^r)^{C_D m}$ there fits a fixed factorial order. Geometry does not prove that nonlinear estimate: an $\exp(CL^{4r})$ whole-block bound cannot be renamed a polynomial cost.

Equation (202) is a readout estimate, not a new marginal transfer theorem. It can be spent once per marked readout or against an established strict age remainder. Recharging it at every renormalization step creates a depth loss. A bounded all-depth jet transport or completed cancellation estimate is still needed for the marginal state.

The physical observable and its counterterms must be fixed independently of the chosen grid. The tuple-dependent grid is only an analysis device for the same physical correlation. Full-covariance Gaussian Wick ordering gives such a definition exactly. For the Wilson interaction its compatible definition and completed comparison remain to be proved.

Finally, (201) applies to an actual harmonic or suitably controlled critical readout. It is not an estimate on every independent (U, J) core controlling only a few low derivatives. A bounded rough current can create large higher derivatives far from every face. The grid removes the artificial face obstruction in the stated readouts; it does not supply an unrestricted high-derivative paired-domain norm.

38.12 A separate obstruction inside a low-derivative core

The last qualification has an explicit example with no block faces. On a periodic fine lattice whose number of sites in direction ν is divisible by four, choose a Cartan generator H and $\mu \neq \nu$. Set

$$A_{\mu,a}(x) = \varepsilon a^2 \cos\left(\frac{\pi x_\nu}{2a}\right) H, \qquad A_{\lambda,a} = 0 \quad (\lambda \neq \mu). \tag{203}$$

Physical differences give $\|A_a\|_\infty \le C\varepsilon a^2$, $\|\nabla A_a\|_\infty \le C\varepsilon a$, and $\|\Delta A_a\|_\infty \le C\varepsilon$. The Maxwell current is uniformly bounded by $C\varepsilon$. All commutators vanish in the chosen Cartan direction, and $\exp(iaA_a)$ lies in an arbitrarily small regular chart for small ε. Curvature Hölder norms of order at most one remain bounded.

The curvature $F_{\nu\mu,a} = D_{\nu,a}A_{\mu,a}$ has magnitude comparable to εa on a fixed positive fraction of sites, but

$$|D^2_{\nu,a}F_{\nu\mu,a}| \asymp \varepsilon a^{-1}, \qquad \int_K \operatorname{tr}\{(D^2_{\nu,a}F_{\nu\mu,a})^2\} \asymp_K \varepsilon^2 a^{-2}. \tag{204}$$

Here K is any fixed nonempty region and the invariant quadratic convention is positive. The multiplier $(e^{i\pi/2}-1)/a$ gives the powers exactly. The small-angle Wilson curvature has the same leading estimate, with higher powers of a in the correction.

Consequently a core controlling $(A, \nabla A, \Delta^* A)$, or a bounded independent current, cannot uniformly bound this insertion even away from every coarse face. These backgrounds are not constrained minimizers with smooth coarse forcing, so (201) is not contradicted. The example does not exclude an equation-of-motion or contact quotient, or a completed weak norm. It rules out extending the old independent-pair sup norm to arbitrary derivative composites merely by moving faces.

If the current core also requires a fixed C^β seminorm, replace εa^2 by $\varepsilon a^{2+\beta}$. The current has amplitude $O(\varepsilon a^\beta)$ and bounded C^β norm, whereas the third potential derivative is $O(\varepsilon a^{\beta-1})$ and diverges for $0 \le \beta < 1$. More generally, a sufficiently high derivative defeats any fixed finite amount of current regularity. The obstruction does not depend on reading the original current domain as only an L^∞ ball.

39.1 1. An exact criterion for an unchanged conditional mean

Let $C > 0$ on a finite-dimensional real Euclidean space E, and let $Q : E \to F$ be surjective. Put

$$B = QC^{1/2}, \qquad P = B^*(BB^*)^{-1}B. \tag{GC1}$$

Thus P is the orthogonal projection onto the whitened coarse subspace $\operatorname{ran} B^*$. For a further positive definite matrix A, let X be centered Gaussian with covariance $C_A = C^{1/2}AC^{1/2}$. Gaussian regression gives

$$\mathbb{E}_A[X \mid QX = v] = C^{1/2}AB^*(BAB^*)^{-1}v. \tag{GC2}$$

The conditional mean agrees with the original C-harmonic lift

$$H_C v = CQ^*(QCQ^*)^{-1}v = C^{1/2}B^*(BB^*)^{-1}v \tag{GC3}$$

for every v if and only if

$$AP = PA. \tag{GC4}$$

Here is a proof of both directions. If A preserves $\operatorname{ran} P$, the restriction of B to that subspace is an isomorphism onto F. The right side of (GC2), after removing the outer $C^{1/2}$, is the unique element of $\operatorname{ran} P$ mapped to v by B. It is therefore $B^*(BB^*)^{-1}v$. Conversely, equality of (GC2) and (GC3) implies

$$(I - P)AB^*(BAB^*)^{-1} = 0.$$

The last factor is invertible and B^* maps onto $\operatorname{ran} P$, so $(I - P)AP = 0$. Self-adjointness of A gives $PA(I - P) = 0$, proving (GC4).

The criterion is invariance of the retained Gaussian subspace. Positivity of A, comparability with the identity, or a small operator norm perturbation does not imply that invariance.

39.2 2. Positive running weights on an orthogonal filtration

Let

$$I = P_0 \geq P_1 \geq \cdots \geq P_n$$

be nested orthogonal projections arising from a compatible family of retained coarse maps. Set

$$E_j = P_j - P_{j+1} \quad (0 \leq j < n), \qquad E_n = P_n, \qquad A = \sum_{j=0}^{n} w_j E_j, \qquad w_j > 0. \tag{GC5}$$

The E_j are mutually orthogonal, sum to the identity, and commute with every P_r. Hence (GC4) holds at every retained level, without any monotonicity assumption on the w_j. In particular,

$$\mathbb{E}_A[X \mid Q_r X] = H_{C,r} Q_r X. \tag{GC6}$$

The fluctuation covariance at that level is

$$C_{A,r}^{\mathrm{fluc}} = C^{1/2}(I - P_r)A(I - P_r)C^{1/2} = \sum_{j<r} w_j C^{1/2} E_j C^{1/2}. \tag{GC7}$$

The retained conditional mean has covariance

$$C_{A,r}^{\mathrm{ret}} = C^{1/2} P_r A P_r C^{1/2} = \sum_{j\geq r} w_j C^{1/2} E_j C^{1/2}. \tag{GC8}$$

These are independent Gaussian parts and sum exactly to C_A. Thus a running Gaussian strength by itself does not force the conditional mean to retain every finer interface. Such interfaces must come from a failure of this common orthogonal-filtration description, or from the marked coefficients rather than the mean.

39.3 3. Completed Gaussian normal ordering respects this cancellation

For a formal covector t, define

$$:e^{tX}:_{C_A} = \exp\Big(tX - \tfrac{1}{2} t C_A t^*\Big).$$

Conditioning at level r and using (GC7)–(GC8) gives the exact identity

$$\mathbb{E}_A\Big[:e^{tX}:_{C_A} \mid Q_r X\Big] = \exp\Big(tH_{C,r}Q_r X - \tfrac{1}{2} t C_{A,r}^{\mathrm{ret}} t^*\Big). \tag{GC9}$$

Every fixed Wick polynomial follows by differentiating this formal identity finitely many times. In particular the normalization on the right uses the complete retained covariance, not just the newest coarse band. Successive conditioning composes because the independent band covariances add. The law of total cumulance gives the same statement in connected marked coordinates.

For spatially regularized differential insertions, one may first apply the corresponding finite linear maps to X and t. Equation (GC9) remains exact. The regulator may then be removed in completed separated Gaussian moments, provided the full off-diagonal covariance and its required derivatives have their usual limits. This last step uses the completed covariance. It does not bound each conditional coefficient separately near a sharp interface.

39.4 4. An intermediate quadratic term can destroy the invariance

Take $E = \mathbb{R}^3$, $C = I$, and nested maps

$$Q_1(x_1, x_2, x_3) = (x_1, x_2), \qquad Q_2(x_1, x_2) = x_1.$$

The original coarsest harmonic lift is $v \mapsto (v, 0, 0)$. Now multiply the original density by

$$\exp[-\tfrac{\eta}{2}(x_1 - x_2)^2], \qquad \eta > 0. \tag{GC10}$$

This is a positive quadratic interaction depending only on the intermediate coarse data. At fixed $x_1 = v$, completing the square in x_2 gives

$$\mathbb{E}_\eta[X \mid x_1 = v] = \left(v, \frac{\eta}{1+\eta} v, 0\right). \tag{GC11}$$

The conditional mean has changed even for arbitrarily small η. In whitened coordinates the new covariance is

$$A = [I + \eta(e_1 - e_2)(e_1 - e_2)^*]^{-1},$$

which does not commute with the coarsest projection $e_1 e_1^*$. The term (GC10) is not a scalar weight on one orthogonal band of the original filtration.

Consequently, using (GC6) for an actual running construction requires an identification of every retained quadratic term with the same orthogonal decomposition. A local term at a coarser stage is not automatically a multiple of the inherited Schur precision. If their difference is retained as a marked or ordinary remainder, that difference must stay in the completed comparison.

39.5 5. A sharp coarse interaction can create separated composite divergences

The following continuum Gaussian calculation makes this distinction visible without a formal perturbation series. Let ϕ be the one-dimensional massive Gaussian field with covariance

$$C = (-\partial_x^2 + \mu^2)^{-1}, \qquad \mu > 0.$$

Put $\psi = \mathbf{1}_{[0,1]}$, $Y = \phi(\psi)$, $v = \mathbb{E}Y^2 > 0$, and $h = C\psi$. The elliptic equation gives, as distributions,

$$h'' = \mu^2 h - \mathbf{1}_{[0,1]}, \qquad h''' = \mu^2 h' - \delta_0 + \delta_1. \tag{GC12}$$

The function h is continuously differentiable, and h' is bounded on compact sets.

Tilt this physical Gaussian law by the sharp coarse interaction

$$\frac{d\mathbb{P}_\eta}{d\mathbb{P}} = \frac{\exp(-\eta Y^2/2)}{\mathbb{E}\exp(-\eta Y^2/2)}, \qquad \eta > 0.$$

The resulting law is again centered Gaussian, with covariance

$$C_\eta = C - \gamma h \otimes h, \qquad \gamma = \frac{\eta}{1 + \eta v} > 0. \tag{GC13}$$

This follows either from a rank-one matrix inversion at finite regularization or from one-dimensional Gaussian conditioning and integration. In particular the tilted law is a genuine positive probability law.

Choose an even smooth compact mollifier ρ of unit integral and write $\rho_\sigma(x) = \sigma^{-1}\rho(x/\sigma)$. Define

$$X_\sigma(x) = \partial_x^3(\rho_\sigma * \phi)(x), \qquad Q_\sigma(x) = X_\sigma(x)^2 - \mathbb{E}_\eta[X_\sigma(x)^2]. \tag{GC14}$$

The subtraction in (GC14) uses the tilted covariance itself. Thus the following divergence is not an omitted one-point normal ordering.

Let $f, g \geq 0$ be smooth, with disjoint small supports near 0 and 1, respectively, and $f(0)g(1) > 0$. On these two supports the unperturbed differentiated covariance

$$K_\sigma(x, y) = \partial_x^3 \partial_y^3 (\rho_\sigma * C * \rho_\sigma)(x - y)$$

is uniformly smooth and bounded as $\sigma \downarrow 0$. Define $h_\sigma''' = \rho_\sigma * h'''$. Wick's rule gives

$$\mathrm{Cov}_\eta(Q_\sigma(f), Q_\sigma(g)) = 2 \iint f(x)g(y)\big[K_\sigma(x, y) - \gamma h_\sigma'''(x) h_\sigma'''(y)\big]^2 \, dx\, dy. \tag{GC15}$$

Equation (GC12) yields

$$\begin{aligned}\int f(x)\,[h_\sigma'''(x)]^2dx &= \sigma^{-1}f(0)\int \rho^2 + O(1),\\ \int g(y)\,[h_\sigma'''(y)]^2dy &= \sigma^{-1}g(1)\int \rho^2 + O(1).\end{aligned} \tag{GC16}$$

The mixed term in (GC15) stays bounded: the L^1 norms of h_σ''' on each of the two fixed supports are bounded, and K_σ is uniformly bounded there. The K_σ^2 term is bounded as well. Therefore

$$\boxed{\lim_{\sigma\downarrow 0}\sigma^2\,\mathrm{Cov}_\eta(Q_\sigma(f),Q_\sigma(g)) = 2\gamma^2 f(0)g(1)\left(\int\rho^2\right)^2 > 0.} \tag{GC17}$$

The divergence occurs between separated supports. Adding deterministic one-point subtractions cannot change it. We do not claim that every possible enlarged operator mixing scheme fails; (GC17) rules out the specific inference from a positive bounded coarse tilt and Wick normal ordering to uniform separated higher-derivative estimates.

The tilt in this example changes the physical measure and places a sharp defect at fixed locations. It is not a counterexample to the translation-invariant Wilson theory. It explains why an artificial coarse law cannot replace the original physical law without retaining the cancellations in their exact comparison.

39.6 6. Consequence for a grid chosen for an insertion configuration

A global translation of the analysis grid is legitimate when the physical law and renormalized inserted observables have already been fixed independently of that grid. Translation invariance then permits one to prove a bound using a favorable grid for each tuple of insertion points. The adaptive-grid lemma in the higher-jets chapter applies in this sense.

It is different to change the coarse physical interaction or the operator counterterms when choosing the favorable grid. That procedure can change the correlation being bounded, as (GC10)–(GC17) show. In particular, one cannot choose a new one-insertion normalization for each insertion count or geometry and then declare that the resulting estimates belong to one field hierarchy. A completed construction needs one grid-independent physical definition, or a proved exact comparison between the definitions used by the different analyses.

For the orthogonal Gaussian model, (GC4)–(GC9) supply such an exact cancellation. Extending it to the actual nonlinear marked Wilson map is a remaining analytic task.

40.1 An arbitrary finite-cutoff density

Let $C > 0$ on a finite-dimensional real space E, with centered Gaussian measure γ_C. Let $R \geq 0$ be measurable and $0 < Z = \int R \, d\gamma_C < \infty$, and set $d\mu = Z^{-1} R \, d\gamma_C$. Assume the moments needed for

the fixed polynomial order under consideration. An ordinary shifted-density generating function is used only when exponential moments near zero exist; compactly supported R suffices and need not be differentiable.

At vertex i choose covectors $\ell_{i,a}$ and put $J_i = \sum_a t_{i,a}\ell_{i,a}$, with formal commuting variables. Vertex normal ordering uses the same full covariance C. Then

$$\mathbb{E}_\mu \prod_i e^{J_i\phi - \frac{1}{2}J_iCJ_i^*} = e^{\sum_{i<j} J_iCJ_j^*}\mathcal{A}\left(C\sum_i J_i^*\right), \tag{205}$$

$$\mathcal{A}(v) = Z^{-1}\int R(\phi+v)\,d\gamma_C(\phi). \tag{206}$$

When ordinary shifts are admitted this follows by completing the Gaussian square:

$$\mathbb{E}_C[R(\phi)e^{J\phi - \frac{1}{2}JCJ^*}] = \mathbb{E}_C R(\phi + CJ^*). \tag{207}$$

With only finitely many moments, define the required formal jet by

$$\mathcal{A}_r(v_1,\dots,v_r) = Z^{-1}\int R(\phi) : \prod_{b=1}^{r}\langle C^{-1}v_b,\phi\rangle :_C \; d\gamma_C(\phi). \tag{208}$$

The identities then assert equality of these finite coefficients; no positive composite-source disc is assumed. Mere integrability does not justify arbitrary moments or derivatives of R. If the corresponding weak derivatives and Gaussian integrals exist, integration by parts gives

$$\mathcal{A}_r(v_1,\dots,v_r) = Z^{-1}\int D_{v_1}\cdots D_{v_r}R(\phi)\,d\gamma_C(\phi). \tag{209}$$

For nonsmooth cutoffs this includes their boundary distributions. Omitting them changes the identity. Formula (208) is the safer definition under finite moment assumptions.

The result is nonlinear in an arbitrary R. At finite cutoff it applies to a fixed chart with its Haar Jacobian, exact action weight, and chart cutoff included in R. Application to the whole gauge theory still needs its chart and coarea decomposition and estimates; no global linear gauge chart is asserted.

40.3 Finite triangular mixing and its limitations

Write Δ_C for polynomial contraction with C. For a degree-p polynomial,

$$:P:_C = e^{-\Delta_C/2}P =: e^{(\Delta_{C'}-\Delta_C)/2}P:_{C'}. \tag{211}$$

The exponential stops after $\lfloor p/2 \rfloor$ terms. It is finite triangular mixing in polynomial degree, with local lower-degree contraction counterterms. For regulated differential polynomials it gives finitely many Wick subtractions at each fixed engineering dimension. A fixed full covariance makes these independent of the analysis grid.

A change of conditioning is different. Its mean $H_Q y$ occurs in the highest retained chaos, whereas changing Wick constants affects only smaller degrees. In the sharp interval example,

$$\mathbb{E}[O_\sigma(f) \mid Y] = a_\sigma(f) v^{-2}(Y^2 - v), \qquad a_\sigma(f) \sim c_f \sigma^{-1}. \tag{212}$$

No constant or linear counterterm changes this second Hermite coefficient. In general the top conditional degree is the tensor product of the harmonic readouts; lower-chaos counterterms cannot alter it. This rules out repairing the ordinary conditioned norm by normal ordering alone. It does not rule out every enlarged same-degree derivative mixing scheme.

Multi-insertion contact counterterms also cannot repair the separated conditional cancellation. Place one test near each of the two distinct interval faces. The covariance of the two conditional means contains

$$2a_\sigma(f)a_\sigma(g)/v^2 \asymp \sigma^{-2}, \tag{213}$$

and the averaged conditional covariance has the opposite divergent term. Their sum is bounded. A distribution supported on coincident physical points pairs to zero with these disjoint tests and cannot bound either contribution separately. A Q-dependent same-chaos correction would require exact compensating transport; it cannot silently be identified with a physical local counterterm.

40.6 A covariant observer and its ribbon attachments

Choose translation-covariant paths $\gamma_{x,z}$ from x to $x+z$, with transports $P_{x,z}$, and a finite scalar kernel w. Define

$$(\mathcal{S}_w X)(x) = \sum_z w(z)\,\mathrm{Ad}\,P_{x,z} X(x+z). \tag{221}$$

This changes only the observable and is exactly gauge covariant. Comparing the path translated by a μ step with the original path followed by that step gives

$$[D_{\mu,a}, \mathcal{S}_w]X(x) = a^{-1}\sum_z w(z)\{\mathrm{Ad}[U_\mu(x)P_{x+a\hat{\mu},z}] - \mathrm{Ad}[P_{x,z}U_\mu(x+z)]\}X(x+z+a\hat{\mu}). \tag{222}$$

For coordinate paths, the strip has at most $C|z|_1/a$ plaquettes. Elementary plaquette slides give an exact telescoping sum with one factor $\mathrm{Ad}\,U_p - I$ in each term and unitary transports on both sides. Thus

$$\|[D_{\mu,a}, \mathcal{S}_w]X(x)\| \le \frac{2}{a}\sum_z |w(z)| \sum_{p\in\mathrm{strip}(x,z,\mu)} \|U_p - I\|\,\|X(x+z+a\hat{\mu})\|. \tag{223}$$

For classically smooth fields and physical radius σ, this is $C\sigma\|F\|\|X\|$. Quantum curvature has no such uniform pointwise bound. The required extra estimate is therefore a completed joint bound for these ribbon-curvature insertions, summed with their weights and retaining every physical anchor. Fixedly many derivatives create finitely many further curvature attachments and kernel derivatives. Their near-anchor local parts must use the same physical mixing scheme; far parts obey (215).

With microscopic support $R_D a$, the ribbon has only C_D terms. When σ/a grows, uniformity needs completed correlations or an independently proved smoothing theorem. The elementary bound (223) does not supply it.

40.7 The remaining analytic target

At each fixed dimension choose the physical insertions and local counterterms once, independently of source count, configuration, and grid. A uniform completed separated bound then needs:

1. a finite local expansion of near-mark attachments into the dimension-filtered types, with a strict remainder after a common physical normalization fixes their coefficients;
2. connected carrier bounds after all covariance-color and cutoff-boundary cancellations, taking absolute values only then;
3. the same bounds for covariant path and derivative commutators, allowing a common zero-smear limit and shrinking reflection collar.

The far estimate, the complete Gaussian result, and the finite smooth-cylinder nonlinear example do not establish these near-mark requirements for Wilson's ultraviolet-local density.

The adaptive-grid lemma provides one global translation with margin $c/[m\log(m+2)]$ at every active level, after $O(\log m)$ initial levels. Its harmonic readout cost fits fixed factorial growth. It does not remove rough independent currents, noncommuting retained quadratic defects, or genuine near-mark attachments. It must be used with the same physical counterterms.

41.1 1. One curvature derivative survives sharp conditioning

Work on a periodic flat lattice with physical volume pairings on forms. Let d_η, δ_η be the compatible exterior derivative and adjoint, and $\Delta_\eta = d_\eta\delta_\eta + \delta_\eta d_\eta$. For the actual flat constrained lift H_j, normalize the final coarse spacing to one and put

$$F_j = d_\eta H_j, \qquad J_j = \delta_\eta F_j, \qquad J_j B = Q_j^* \lambda_j, \qquad \lambda_j = S_j B, \quad S_j = F_j^* F_j. \tag{D6.1}$$

The last identity is the actual stationary equation after the coarse Ward identity removes the gradient gauge multiplier. Constants and coarse pure gauges give zero curvature; no inverse on a harmonic zero mode is used.

Since $d_\eta F_j = 0$, discrete integration by parts gives the exact Hodge identity

$$\|\nabla_\eta F_j B\|_2^2 = \langle F_j B, \Delta_\eta F_j B\rangle = \|J_j B\|_2^2. \tag{D6.2}$$

The physical L^2 norm of the box/line adjoint Q_j^* is uniformly bounded. The depth-independent bound on F_j bounds S_j. Therefore

$$\|F_j B\|_2 + \|\nabla_\eta F_j B\|_2 \le C\|B\|_2. \tag{D6.3}$$

This controls one curvature derivative, not two.

For a small fixed cell weight $W_v(x) = e^{\kappa\,\mathrm{dist}(x,v)}$, apply the Hodge identity to $W_v F$. Its first-order commutators cost $C\kappa W_v F$, so

$$\|W_v \nabla_\eta F_j B\|_2 \le C\|W_v J_j B\|_2 + C\kappa\|W_v F_j B\|_2. \tag{D6.4}$$

The existing final-cell exponential bounds for S_j, F_j, together with the fixed-range final-cell kernel of Q_j^*, bound this for a datum in cell v. Restoring physical final spacing ℓ gives

$$\|\chi_u^{(\ell)} \nabla_\eta F_j \chi_v\|_{2\to 2} \le C\ell^{-1} e^{-c\,\mathrm{dist}(u,v)/\ell}. \tag{D6.5}$$

Thus $(\nabla_\eta F_j)^* M_f (\nabla_\eta F_j)$ has norm at most $C\ell^{-2}\|f\|_\infty$, with exponential cell-block locality. The factor ℓ^{-2} is the engineering excess of an integrated dimension-six insertion. This is not a fine-coordinate trace estimate.

41.2 2. The actual flat force has a fractional gain

The flat composite average is normalized disjoint box averaging followed, in the bond direction, by normalized line averaging. Its adjoint is a bounded-overlap sum of translated box/tent functions. The box changes on a boundary layer of volume $C|z|$; the tent has a better Lipschitz modulus. Hence

$$\|\tau_z Q_j^*\lambda - Q_j^*\lambda\|_2^2 \le C\min\{|z|,1\}\|\lambda\|_2^2. \tag{D6.6}$$

This is uniform in the internal mesh. A fixed exponential cell weight adds only an $O(|z|^2)$ term.

The fractional difference formula gives, for $0 < s < 1/2$,

$$\|Q_j^*\lambda\|_{H^s}^2 \le C\left(1 + \frac{1}{1-2s}\right)\|\lambda\|_2^2. \tag{D6.7}$$

The small-translation radial integral is $\int_0^1 r^{-2s}\,dr$; no endpoint $s = 1/2$ assertion is made. The corresponding discrete multiplier $(1+\widehat{\Delta}_\eta)^{s/2}$ gives the same mesh-independent estimate.

The commuting flat multipliers and $dF = 0, \delta F = J$ imply

$$\|F\|_{H^{1+s}} \le C_s(\|F\|_2 + \|J\|_{H^s}), \qquad \|F_j B\|_{H^{1+s}} \le C_s\|B\|_2. \tag{D6.8}$$

Local weighted versions follow using (D6.4) and fractional commutators with the fixed cell weight. At spacing ℓ, the homogeneous part has factor ℓ^{-1-s}.

These are direct depth-j estimates for the actual flat composite average. They do not iterate a fixed-step Sobolev constant, nor establish the same estimate for a nonlinear nested group average.

41.2.1 2.1 A BV criterion for nonlinear forces

A useful weaker sufficient condition than smoothness is local $BV \cap L^\infty$. For a vector-valued J,

$$\|\tau_z J - J\|_2^2 \le 2\|J\|_\infty\|\tau_z J - J\|_1 \le 2\|J\|_\infty \mathrm{TV}(J)|z|. \tag{D6.9}$$

Thus a bounded L^2 norm and a uniform product $\|J\|_\infty \mathrm{TV}(J)$ give H^s for every $s < 1/2$, with the factor in (D6.7). Smooth spatial weights preserve this criterion with the expected lower-order term.

A covariant version follows in any controlled local gauge: $\mathrm{TV}(J) \le \mathrm{TV}_A(J) + C\|A\|_\infty\|J\|_1$. Equivalently one can prove the translation modulus using parallel transport. For kernel columns use weighted local bounds before summing; an unweighted total-volume norm is not substituted for a rooted norm.

If nonlinear interfaces at scale r have area density $O(r^{-1})$ and covariant force jumps $O(r^{1+\epsilon})$, their BV contributions are summable. The conjectured sharper jump $O(r^2 F\lambda)$ would suffice. This is ONLY a criterion: the cancellation of a possible $O(rA)$ transport term must be proved from the actual contour/mean adjoint, not assumed from gauge covariance alone.

41.3 3. A strict averaged dimension-six Taylor lemma

Let $0 < s < 1$, $F \in H^{1+s}(\mathbb{T}^4;\mathbb{R}^d)$, and $f \in C^{2+s}$. For $|v| \le 1$ define

$$B_v(f,F) = \int f(x)\langle F(x+v), F(x-v)\rangle\,dx.$$

Then

$$\begin{aligned} B_v(f,F) &= \int f|F|^2 - 2v_a v_b \int f\langle \partial_a F, \partial_b F\rangle + \tfrac{1}{2} v_a v_b \int (\partial_a\partial_b f)|F|^2 + R_v, \\ |R_v| &\le C_s |v|^{2+s} \|f\|_{C^{2+s}} \|F\|_{H^{1+s}}^2. \end{aligned} \tag{D6.10}$$

Repeated derivative indices are summed. No pointwise D^2F is needed.

Proof. Change $y = x - v$ and use polarization:

$$\begin{aligned} B_v(f,F) = \tfrac{1}{2} \int \{f(y-v) + f(y+v)\}|F(y)|^2\, dy \\ - \tfrac{1}{2} \int f(y+v)|F(y+2v) - F(y)|^2\, dy. \end{aligned} \tag{D6.11}$$

Taylor expansion of f gives the displayed second derivative and a remainder bounded by $C|v|^{2+s}\|f\|_{C^{2+s}}\|F\|_2^2$. The Fourier inequality $|e^{it} - 1 - it| \le C_s|t|^{1+s}$ gives

$$F(\cdot + 2v) - F = 2(v\cdot\nabla)F + R, \qquad \|R\|_2 \le C_s|v|^{1+s}\|F\|_{H^{1+s}}.$$

The square's remainder is $O(|v|^{2+s}\|F\|_{H^{1+s}}^2)$. Replacing $f(y+v)$ by $f(y)$ in its leading term costs $C|v|^3\|\nabla f\|_\infty\|\nabla F\|_2^2$, which is better for $s < 1$. Polarization gives the symmetric bilinear version for different curvature components. □

For lattice translations use the EXACT same polarization identity with the physical lattice sum. No equality between a lattice product sum and a product integral of trigonometric interpolants is needed. Telescoping nearest-neighbor steps along a coordinate path to $2v$, then using $\|\tau_z G - G\|_2 \le C_s|z|^s\|G\|_{H^s}$, gives

$$\tau_{2v}F - F = 2\sum_a v_a \nabla_{\eta,a} F + O_{L^2}(|v|^{1+s}\|F\|_{H^{1+s}}).$$

Here a fixed compatible forward derivative is used; reverse steps are first shifted to that convention. The shifts add only the same fractional translation remainder. The smooth physical test satisfies the ordinary central Taylor formula at its sampled sites. Thus (D6.10) holds uniformly with lattice sums and those fixed lattice derivatives. No pointwise third potential derivative is introduced.

41.3.1 3.1 A local kernel consequence

For an even symmetric matrix kernel $k(\xi)$ with $\int |k(\xi)|(1+|\xi|)^{2+s}\, d\xi < \infty$, integrate (D6.10) with $v = r\xi/2$. For large $|r\xi|$, use the elementary L^2 bound and the same weighted moment. The remainder is

$$C_s r^{2+s}\|k\|_{2+s,1}\|f\|_{C^{2+s}}\|F\|_{H^{1+s}}^2. \tag{D6.12}$$

Exponential birth kernels have the required moment; spare locality pays polynomial carrier factors. The second-order terms are the COMPLETE derivative-curvature quadratic tensor and a descendant of the curvature quadratic tensor with derivatives on the source.

A dimension-six integrated readout has an additional engineering factor r^{-2}. After its dimension-four and COMPLETE dimension-six terms are extracted, (D6.12) leaves r^s. This is a strict geometric age gain in an integrated norm, even if an artificial face crosses the support.

The midpoint/even qualification is substantive. A universal even scalar kernel has inversion symmetry after its complete covariance-promoted coefficient is formed. An asymmetric individual carrier must first be repointed, with its derivative-of-source terms retained. It is not declared even without that algebra.

41.3.2 3.2 Covariant straight translations

Let $A \in W^{1,\infty}$, and define

$$(T_v^A F)(x) = P(x, x+v)F(x+v)P(x, x+v)^{-1}$$

using straight parallel transport. It is the one-parameter translation group generated by $v \cdot D_A$, and is an L^2 isometry for a real compact connection. Integration of $(T_{tv}^A - I)(v \cdot D_A)F$ gives

$$\|T_{2v}^A F - F - 2(v \cdot D_A)F\|_2 \le C_s |v|^{1+s} (1 + \|A\|_{W^{1,\infty}})^C \|F\|_{H^{1+s}}. \tag{D6.13}$$

Here ordinary translation has an H^s-to-L^2 modulus $|v|^s$, and a $W^{1,\infty}$ multiplier preserves H^s.

Inner-product invariance and segment composition give the EXACT identity (D6.11) with $F(y+2v)$ replaced by $T_{2v}^A F(y)$. Thus (D6.10) holds with $\partial_a F$ replaced by $D_{A,a}F$. On a complex prepared core use the invariant bilinear pairing; bounded transport replaces isometry in the estimates.

There is a wholly discrete formulation. Fix a lattice path γ_v from each site to its translate by v, set T_v to its transported translation, and define $T_{-v} = T_v^{-1}$. Polarization then has $T_v^2 F$, the concatenation of γ_v and its translate, in place of $T_{2v}^A F$. This is an identity of actual lattice sums. Telescoping path bonds and the H^s translation modulus of each covariant first difference give

$$T_v^2 F - F = 2 \sum_a v_a \nabla_{\eta,a}^U F + O_{L^2}(|v|^{1+s} \|F\|_{H^{1+s}}),$$

with constants depending on the fixed $W^{1,\infty}$ connection bound. The path has length $O(|v|)$; its order/area commutators cost $O(|v|^2)$, which is better than the displayed first-difference remainder for $s < 1$. This proves the covariant averaged estimate without a continuum interpolation of the gauge field.

For a fixed short carrier path, comparison with the straight segment has a leading area-curvature term. If the actual curvature has a fixed C^β bound, that term is a local F^3 tensor and the area remainder gains r^β. The polynomial area of a fixed carrier is paid by spare locality. This does not establish a uniform statement for arbitrary nested nonlinear constraint paths.

41.5 5. Why the ordinary arbitrary-current core is insufficient

In a transverse Abelian color sector on a fixed torus take, along reciprocal integers,

$$A_\varepsilon(x) = \varepsilon^2 \sin(x_1/\varepsilon) e_2.$$

Its potential and first derivative are small, and its Laplacian is bounded, while

$$\|\nabla(dA_\varepsilon)\|_2 \asymp 1, \qquad \|dA_\varepsilon\|_{H^{1+s}} \asymp \varepsilon^{-s}. \tag{D6.20}$$

A fixed small multiplicative constant places the family in any prescribed first/Laplacian core. It represents a bounded rapidly oscillating independent current. The domain supplies no uniform fractional gain.

This also directly disproves a uniform second-order averaged remainder on that larger core. With normalized torus volume take $f = 1$, $F_\varepsilon = \varepsilon \cos(x_1/\varepsilon)$, and $v = \varepsilon e_1$. Then

$$B_v = \tfrac{1}{2}\varepsilon^2 \cos 2, \qquad \int F_\varepsilon^2 - 2\int (v \cdot \nabla F_\varepsilon)^2 = -\tfrac{1}{2}\varepsilon^2.$$

The difference is $\frac{1}{2}(1 + \cos 2)\varepsilon^2$, rather than $O(\varepsilon^{2+s})$ with a uniform constant.

Thus even this averaged D6 argument requires an ACTUAL compatible-current restriction or a completed cancellation beyond the generic independent-current norm. It requires less than a smooth germ, but more than an L^∞ current bound.

41.7 7. A nonlinear lattice Hodge lemma from exact loop identities

The fractional curvature implication can be proved directly for an ACTUAL Wilson current. It does not require identifying a projected Landau operator with a local scalar Laplacian.

Let the fine spacing be $a \le 1$. On a fixed buffered patch choose a regular gauge

$$U_\mu(x) = \exp(aA_\mu(x)), \qquad \|A\|_{W^{1,\infty}} \le M.$$

Let $P_{\mu\nu}(x)$ be its oriented plaquette product based at x, and put

$$F_{\mu\nu}(x) = a^{-2} \log P_{\mu\nu}(x), \qquad \|F\|_\infty \le K, \quad a^2 K < c. \tag{D6.22}$$

The principal logarithm is used. Opposite orientations have opposite logarithms. All norms and differences have their physical a-scaling. Real compact connections and complex connections in the same controlled chart are allowed.

41.7.1 7.1 The exact six-face word

For three distinct positive directions μ, ν, ρ, let $\gamma_{\alpha\beta\gamma}$ denote the path transport from x to $x+a(e_\mu+e_\nu+e_\rho)$ in that coordinate order. Define

$$\begin{aligned}
W_1 &= P_{\mu\nu}(x),\\
W_2 &= U_\nu(x) P_{\mu\rho}(x + ae_\nu) U_\nu(x)^{-1},\\
W_3 &= P_{\nu\rho}(x),\\
W_4 &= \{U_\rho(x) P_{\mu\nu}(x + ae_\rho) U_\rho(x)^{-1}\}^{-1},\\
W_5 &= P_{\mu\rho}(x)^{-1},\\
W_6 &= \{U_\mu(x) P_{\nu\rho}(x + ae_\mu) U_\mu(x)^{-1}\}^{-1}.
\end{aligned}$$

These are respectively the six successive path ratios in the cycle

$$\mu\nu\rho, \quad \nu\mu\rho, \quad \nu\rho\mu, \quad \rho\nu\mu, \quad \rho\mu\nu, \quad \mu\rho\nu, \quad \mu\nu\rho.$$

For example $W_1 = \gamma_{\mu\nu\rho}\gamma^{-1}_{\nu\mu\rho}$ and $W_2 = \gamma_{\nu\mu\rho}\gamma^{-1}_{\nu\rho\mu}$. Every intermediate path cancels, giving the EXACT word identity

$$W_1 W_2 W_3 W_4 W_5 W_6 = I. \tag{D6.23}$$

Write $W_l = e^{a^2 X_l}$. Its linear logarithmic sum is

$$\sum_{l=1}^{6} X_l = -a(\mathcal{D}_U F)_{\mu\nu\rho}, \quad (\mathcal{D}_U F)_{\mu\nu\rho} = \nabla^U_\mu F_{\nu\rho} - \nabla^U_\nu F_{\mu\rho} + \nabla^U_\rho F_{\mu\nu},$$

where $\nabla^U_\mu G(x) = [\mathrm{Ad}_{U_\mu(x)}\, G(x + ae_\mu) - G(x)]/a$. The convergent BCH series in (D6.23) therefore gives

$$\mathcal{D}_U F = a\, B_a(U, F). \tag{D6.24}$$

Here B_a is analytic in the six transported curvatures, starts quadratically in F, and has a uniform convergent expansion for (D6.22). This follows directly by dividing $\log \prod_l e^{a^2 X_l} - a^2 \sum_l X_l$ by a^4. Its degree-n coefficient has the additional factor a^{2n-4}, $n \geq 2$. There is no differentiated curvature hidden in this residual.

41.7.2 7.2 The exact Wilson flux equation

Define the Lie-algebra flux

$$\Phi_a(F) = a^{-2}\pi_{\mathfrak{g}} \sinh(a^2 F) = F + a^4 G_a(F). \tag{D6.25}$$

For the actual normalized Wilson current J, differentiation of a plaquette and cyclic invariance of the trace give

$$J = \mathcal{C}_U \Phi_a(F), \qquad (\mathcal{C}_U G)_\mu(x) = \frac{1}{a} \sum_{\nu \neq \mu} \left[G_{\mu\nu}(x) - \mathrm{Ad}_{U_\nu(x - ae_\nu)^{-1}}\, G_{\mu\nu}(x - ae_\nu) \right]. \tag{D6.26}$$

Fixed Lie-algebra/action conventions can be absorbed in the definition of the normalized J. To verify a term explicitly, the outgoing plaquette contributes $\pi_{\mathfrak{g}} \sinh(a^2 F_{\mu\nu}(x))$. The plaquette based at $x - ae_\nu$ contains $U_\mu(x)^{-1}$; moving its derivative around the trace gives the displayed oppositely signed transported incoming flux. Summing these incident plaquettes proves (D6.26).

The cube incidence operator $\mathcal{D}_U$ and flux operator $\mathcal{C}_U$ can be handled separately. What is needed is

$$\mathcal{D}_U = d_0 + R^d_U, \qquad \mathcal{C}_U = \delta_0 + R^\delta_U. \tag{D6.27}$$

The remainders are order zero: finite sums of shifts multiplied by $(\mathrm{Ad}_{U^{\pm 1}} - I)/a$. The representation $U = e^{aA}$ and its actual $W^{1,\infty}$ bound give

$$\|(\mathrm{Ad}_{U^{\pm 1}} - I)/a\|_{W^{1,\infty}} \leq C_M, \tag{D6.28}$$

uniformly in a. This can also be read from $(e^{a\,\mathrm{ad}A} - I)/a = \int_0^1 e^{ta\,\mathrm{ad}A} \mathrm{ad}A\, dt$. No derivative of order two of A is used.

41.7.3 7.3 The two Sobolev steps

For $0 < s < 1$, discrete fractional difference norms have the uniform product bound

$$\|UV\|_{H^s} \leq C_s \{ \|U\|_\infty \|V\|_{H^s} + \|V\|_\infty \|U\|_{H^s} \}.$$

A $W^{1,\infty}$ coefficient is a bounded H^s multiplier. The finite shifts and convergent analytic expansions above consequently give

$$\begin{aligned} \|R^d_U F\|_{H^s} + \|R^\delta_U F\|_{H^s} &\leq C_{s,M} \|F\|_{H^s}, \\ \|B_a(U, F)\|_{H^s} &\leq C_{s,M,K} \|F\|_{H^s}, \\ \|G_a(F)\|_{H^s} &\leq C_{s,K} K^2 \|F\|_{H^s}. \end{aligned} \tag{D6.29}$$

These are local estimates with a fixed spare buffer. They need only $F \in L^\infty \cap H^s$. In particular the analytic residual is not estimated using derivatives of F.

The crude difference bound is sufficient for the flux error:

$$a^4 \|\mathcal{C}_U G_a(F)\|_{H^s} \le C_{s,M,K} a^3 K^2 \|F\|_{H^s}. \tag{D6.30}$$

All powers of the fine mesh are favorable.

Equations (D6.24)–(D6.27) express the two FLAT incidences as

$$d_0 F = aB_a - R_U^d F, \qquad \delta_0 F = J - R_U^\delta F - a^4 \mathcal{C}_U G_a(F). \tag{D6.31}$$

Apply the exact flat Hodge estimate at order zero first. With nested smooth cutoffs on the buffered patch this proves

$$\|\chi_1 F\|_{H^1} \le C_{M,K,\chi_1,\chi_2} \{\|\chi_2 J\|_2 + \|F\|_{L^2(\mathrm{supp}\chi_2)}\}. \tag{D6.32}$$

Commutators with the fixed cutoffs are order zero and cost only their fixed physical derivative bounds.

Now apply the flat Hodge estimate at fractional order s to $\chi_0 F$, where $\chi_1 = 1$ on a neighborhood of its support. By (D6.29)–(D6.31) its right side contains only J in H^s and F in H^s. The latter is already bounded by (D6.32), since $s < 1$. Thus

$$\boxed{\|\chi_0 F\|_{H^{1+s}} \le C_{s,M,K,\chi_0,\chi_1,\chi_2} \{\|\chi_2 J\|_{H^s} + \|F\|_{L^2(\mathrm{supp}\chi_2)}\}.} \tag{D6.33}$$

The constant is independent of a. The same proof works for complex fields: flat Hodge is estimated with the ordinary Hermitian absolute norm, while the loop and current identities retain their holomorphic algebraic meaning.

This proves the nonlinear current-to-curvature implication required after a genuine physical BV estimate. It does NOT give an H^s estimate of an arbitrary independent current variable.

41.7.4 7.4 The old-scale normalization still matters

An inner reconstruction at old spacing r has a trivial terminal variation bound of order $r^{-1}\|\lambda\|_1$. Using its own-scale BV theorem alone can therefore lose powers when measuring the result in current-scale units. Formula (D6.33) does not repair that loss.

For an unlocalized critical minimizer, constrained-minimization consistency can identify its old inner readout with the same final critical field: $U_j(M^j U_k) = U_k$, with the precise nested constraints understood. This is consistency of MINIMIZATION, not associativity of nonlinear group averaging. In that case the final-scale current estimate can be used. An interpolated outer field need not have this identification. The full old-member argument still needs stability of its actual inner reconstruction in the stated current-scale Sobolev norm, or a completed cancellation replacing that stability. This scale-conversion issue is part of the remaining map lemma in Section 6.

41.8 Primary source check

T. Balaban, *Renormalization group approach to lattice gauge field theories. I*, Commun. Math. Phys. **109** (1987), 249–301: primary PDF. The current-domain distinction is (1.14), printed p.262. The reduced-current substitution is (3.10)–(3.17), pp.272–273, and the Ward derivative norm is (4.4)–(4.5), pp.281–282. These passages do not state the new estimate (D6.21).

42.1 1. An exact nonassociativity calculation

Let M be the local Federbush/Karcher mean on a compact Lie group, defined near the diagonal by

$$\sum_{i=1}^{n} \log(e^{-Z} e^{X_i}) = 0, \qquad M(e^{X_1}, \dots, e^{X_n}) = e^{Z}. \tag{PG1}$$

The alternate convention with the inverse mean on the right is equivalent by conjugation. The implicit derivative in Z at the origin is minus n times the identity, so the local mean is analytic. Put $\overline{X} = n^{-1} \sum_i X_i$. The BCH formula through degree three gives

$$Z = \overline{X} - \frac{1}{12n} \sum_i [X_i, [X_i, \overline{X}]] + O(X^5). \tag{PG2}$$

Indeed, the quadratic sum is proportional to $[Z, \overline{X}]$ and vanishes at $Z = \overline{X}$. At degree three the sum with two Z factors vanishes for the same reason. The remaining double commutator gives (PG2). Inversion is an isometry for the bi-invariant local metric and preserves the defining mean equation; uniqueness therefore gives $M(e^{-X_i}) = M(e^{X_i})^{-1}$. Thus Z is odd under simultaneous sign reversal and the degree-four term vanishes.

Take the four inputs $X, -X, Y, Y$. The nested pair mean is exactly

$$M\big(M(e^{X}, e^{-X}), M(e^{Y}, e^{Y})\big) = e^{Y/2},$$

whereas the direct mean satisfies

$$\log M(e^{X}, e^{-X}, e^{Y}, e^{Y}) = \frac{Y}{2} - \frac{1}{48}[X, [X, Y]] + O((X, Y)^5). \tag{PG3}$$

The difference is nonzero for suitable X,Y in any nonabelian simple factor. This is not an even-block artefact. With n entries per inner group and n outer groups, take the first inner group to contain

$X, -X$ and $n-2$ zeros, and take every other inner group to contain only Y. The nested mean is $\exp(cY)$, $c = (n-1)/n$; the direct n^2-point mean has cubic correction

$$-\frac{c}{6n^2}[X,[X,Y]]. \tag{PG4}$$

This is a calculation for the inner group mean, not a calculation of the associator of the entire outer bond map. In particular one may not flatten a composition of the inner means by invoking group-mean associativity. More generally the flat identity for the derivative of a composed averaging map does **not** imply a nonlinear associativity identity or the disappearance of all intermediate-scale forcing interfaces. Differentiating (PG3) already produces a quadratic difference in the group-mean adjoint derivative weights. This calculation does not by itself prove that the actual constrained critical field has divergent higher derivatives. Such a conclusion would require a nonzero forcing/interface contribution on that field, rather than only on arbitrary inputs, and must include any additional cancellation in the full outer map.

42.2 2. Units, current, and covariant bounded variation

Let the fine spacing be h, the next spacing be $\ell = Lh$, and let U be a real compact-group lattice connection. Write $V = M(U)$ for one actual averaged connection. A tangent field a is normalized by

$$U_b(t) = e^{tha_b}U_b, \qquad V_c(t) = e^{t\ell(\mathcal{Q}_U a)_c}V_c + O(t^2). \tag{PG5}$$

The anti-Hermitian Lie-algebra convention is used only to simplify factors of i. The adjoint $\mathcal{Q}_U^*$ uses physical-volume pairings, $h^4\sum_b$ and $\ell^4\sum_c$. Thus $\mathcal{Q}_U$ and $\mathcal{Q}_U^*$ preserve the physical amplitude units of a one-form and its dual current. For a coarse current λ, its one-step fine current is $J = \mathcal{Q}_U^*\lambda$.

Fix an Ad-invariant Euclidean norm on the Lie algebra. A one-form current $\lambda_\mu(x)$ is based at its starting site. Its forward covariant difference and covariant variation are

$$\begin{aligned}(\nabla^U_{\nu,h}\lambda_\mu)(x) &= h^{-1}\{\operatorname{Ad}U(x,x+he_\nu)\lambda_\mu(x+he_\nu) - \lambda_\mu(x)\},\\ \mathrm{TV}_{U,h}(\lambda) &= \sum_{x,\mu,\nu} h^4|\nabla^U_{\nu,h}\lambda_\mu(x)|.\end{aligned} \tag{PG6}$$

The L1 norm is $\sum_{x,\mu} h^4|\lambda_\mu(x)|$, and the sup norm is the maximum component norm. The additive variation of a transport error in the statements below is measured with this norm; there is no assertion that nonzero curvature gives exact BV contraction.

Assume all plaquettes in the fixed-size enlarged one-step patches satisfy

$$\|1 - U_p\| \le h^2\alpha, \qquad \ell^2\alpha \le c_L. \tag{PG7}$$

The patches are taken contractible, or on the universal cover before a periodic identification. This avoids confusing a large wrapping holonomy with a small contractible curvature loop. The local estimates apply on tori whose relevant patches do not wrap; wrapping terms need their usual separate filing.

The coarse current will obey the **actual coarse Ward condition**

$$d_V^*\lambda = 0. \tag{PG8}$$

Here d_V is the infinitesimal coarse gauge transformation in the same amplitude convention as (PG5), so its difference quotient is ℓ^{-1}. Condition (PG8) is essential in Section 4.

42.3 3. A positive covariant reference prolongation

There is a simple reference T_U with the following properties:

$$\begin{aligned} \|T_U\lambda\|_\infty \le \|\lambda\|_\infty, \qquad \|T_U\lambda\|_1 \le \|\lambda\|_1, \\ \mathrm{TV}_{U,h}(T_U\lambda) \le \mathrm{TV}_{V,\ell}(\lambda) + C_L\ell\alpha\|\lambda\|_1. \end{aligned} \tag{PG9}$$

The coefficient of the inherited coarse variation is **one**. No factor C_L multiplies that term.

To define T_U, first extend a coarse component λ_μ constantly to its box, after parallel transport from the coarse centre to the fine starting site along a fixed coordinate-order path. Denote this covariant box hold by I_U. Next, average $I_U\ \lambda_\mu$ along a straight segment of length ℓ in direction mu, parallel transporting every summand to the output site. This covariant line average $C_{\mu,U}$ has weights $1/L$. Set $(T_U\lambda)_\mu = C_{\mu,U}(I_U\lambda_\mu)$. At $U = 1$ this is exactly the adjoint of the normalized box average followed by the normalized length-L line average appearing in the straight flat bond average.

Both operations are positive scalar averages of isometric adjoint transports. Their L1 and sup contractions follow immediately, including the physical-volume normalization of the box hold.

For variation, consider first two adjacent fine sites in the same box. The coordinate paths to their common coarse centre, together with the fine connecting bond, enclose a ribbon of at most $C_L\ell/h$ plaquettes. Plaquette telescoping therefore bounds the transport discrepancy by $C_Lh\ell\alpha$. After division by h, its contribution is $C_L\ell\alpha$ times the carried current. At a box face, use the straight length-ℓ coarse bond between the two centres. The corresponding loop has area at most $C_L\ell^2$, so the additional transport discrepancy is $C_L\ell^2\alpha$. The face density is ℓ^{-1}, giving again $C_L\ell\alpha\|\lambda\|_1$.

The remaining face term is exactly the coarse covariant jump. Each coarse face contains L^3 fine crossings, and $h^3L^3 = \ell^3$. Hence its sum is the coarse variation with coefficient one. Initially this uses the coarse connection given by the straight fine-path product. The actual averaged V differs from that product by at most $C_L\ell^2\alpha$, by the same contractible-loop estimate and the analyticity of the local mean. Replacing the reference coarse connection by V adds only $C_L\ell\alpha\|\lambda\|_1$.

Finally, a fine covariant difference of $C_{\mu,U}$f equals the transported average of the same difference of f, plus the rectangular-ribbon commutator. Each ribbon has width h and length at most ℓ. Its contribution after division by h is at most $C_L\ell\alpha$ times the corresponding average of $|f|$. Summing proves

$$\mathrm{TV}_{U,h}(C_{\mu,U}f) \le \mathrm{TV}_{U,h}(f) + C_L\ell\alpha\|f\|_1. \tag{PG10}$$

Combining these statements proves (PG9). This argument uses no derivative of the plaquette field, and no cancellation between two signed L1 estimates.

42.4 4. The actual adjoint on a physical multiplier

For the actual symmetric contour average, there is a one-step error E_U, evaluated only on currents satisfying (PG8), such that

$$\mathcal{Q}_U^*\lambda = T_U\lambda + E_U\lambda, \qquad \|E_U\lambda\|_p \le C_L\ell^2\alpha\|\lambda\|_p, \quad p = 1,\infty. \tag{PG11}$$

This is a bound on the complete physical current, not a claim that $E_U\lambda$ itself satisfies a Ward condition.

Here is the proof, including the raw contour terms. On a fixed enlarged one-step patch, choose an ordinary rooted tree gauge. Each transformed fine bond is a contractible loop transport, and (PG7) makes every group variable needed by the one-step map lie within $C_L\ell^2\alpha$ of the identity. The same holds for the coarse V variables in this patch. A fixed L gives finitely many variables and paths. The local analytic contour/logarithm/mean formulas therefore differ from their flat first derivatives, in both row and column norms with physical-volume weights, by at most $C_L\ell^2\alpha$. The same assertion holds for T_U versus its flat value. All gauge transformations in this comparison are real compact-group transformations, hence preserve the chosen norms.

It remains to compare the **flat raw derivative**, rather than silently replacing it by the straight box-line map. Let $r(a)(y)$ be the normalized average of the line integrals of a over the coordinate-permutation paths from coarse centre y to points of its box. The flat formula for the actual raw mean is

$$\mathcal{Q}_{0,\mathrm{raw}}a = \mathcal{Q}_{0,\mathrm{box-line}}a - d_0 r(a), \qquad \|r\|_{p\to p} \le C_L\ell. \tag{PG12}$$

Indeed, the initial connector contributes $r(y)$, the final connector contributes minus $r(y+\ell e_\mu)$, and the middle straight contour gives the box-line average. The reference coarse bond in (0.4)/(0.12) cancels at first order. At the diagonal, the derivative of every admitted symmetric mean is arithmetic averaging, so the same identity holds with averaged contour variables.

Taking adjoints in (PG12) produces a connector term $-r^*d_0^*\lambda$. It vanishes for a flat co-closed current, but an actual current obeys (PG8), not necessarily $d_0^*\lambda = 0$. In the local tree gauge,

$$d_0^*\lambda = (d_0^* - d_V^*)\lambda, \qquad \|(d_0^* - d_V^*)\lambda\|_p \le C_L\ell\alpha\|\lambda\|_p. \tag{PG13}$$

The factor $\ell\alpha$ is exactly $(\ell^2\alpha)/\ell$. Multiplication by r^* costs ℓ, leaving $C_L\ell^2\alpha$. Combining this with the analytic comparison above proves (PG11). For local row estimates take a patch containing every coarse edge incident to the sites where (PG13) is used; then sum the bounded-overlap patches. No global near-identity gauge or fine-coordinate Cauchy radius is being assumed.

The group-mean part can also be seen directly. Paths with the same endpoints differ by a contractible loop of area $O(\ell^2)$, so after factoring one common transport their holonomies differ by $O(\ell^2\alpha)$. The mean differential at the diagonal is the arithmetic differential. Analyticity changes its weights by $O(\ell^2\alpha)$. This gives the curvature, rather than potential, size of the correction. The Ward calculation (PG12)–(PG13) is still needed to remove the leading raw connector contribution.

Since $h = \ell/L$ is fixed within this step, arbitrary fine differences of the error have the elementary bound

$$\mathrm{TV}_{U,h}(E_U\lambda) \le \frac{2d}{h}\|E_U\lambda\|_1 \le C_L\ell\alpha\|\lambda\|_1. \tag{PG14}$$

Thus (PG9) and (PG11) imply the actual one-step estimates

$$\boxed{\begin{aligned} \|\mathcal{Q}_U^*\lambda\|_p &\le (1+C_L\ell^2\alpha)\|\lambda\|_p, \qquad p=1,\infty,\\ \mathrm{TV}_{U,h}(\mathcal{Q}_U^*\lambda) &\le \mathrm{TV}_{V,\ell}(\lambda) + C_L\ell\alpha\|\lambda\|_1. \end{aligned}} \tag{PG15}$$

The second line has coefficient one on the inherited variation. The estimate of E is used at this step and is then discarded; only the complete physical current is passed to the next step.

42.4.1 4.1 Exact nonlinear endpoint cancellation, including the reference bond

The same conclusion has an exact endpoint formulation that makes the role of the actual outer map explicit. With normalized weights w_i summing to one, write

$$\mathscr{A}(C_i;R) = \exp\!\left(\sum_i w_i \log(C_iR^{-1})\right) R, \qquad R=U(c), \qquad V=\mathscr{A}(C_i;R). \tag{PG15a}$$

The C_i are centre-to-centre contours. In version (0.4) the index i includes the two endpoint-path permutations independently. In version (0.12), each endpoint transport is the group mean of its endpoint paths. Formula (PG15a), rather than a Karcher mean of the C_i, is the actual outer block map.

For $C_i = C$ for every i, one has $\mathscr{A}(C,\dots,C;R) = C$ identically in the reference R. Consequently the left-trivialized differential at this diagonal is $\sum_i w_i\,\delta C_iC_i^{-1}$, and its reference-bond differential is exactly zero. If all relative C_iR^{-1} lie in the $O(\ell^2\alpha)$ chart, analyticity gives

$$\delta V V^{-1} = \sum_i w_i \delta C_i C_i^{-1} + \mathscr{R}_U(\delta C_iC_i^{-1}, \delta R R^{-1}), \quad \|\mathscr{R}_U\| \le C_L\ell^2\alpha. \tag{PG15b}$$

This includes the reference variation. Its physical path variation costs $O(\ell\|a\|)$; division by ℓ in (PG5) therefore puts it in the same $O(\ell^2\alpha)$ error as the other terms.

Write a typical contour as $C_x = P_xW_x(P'_{x'})^{-1}$, with the two P's the appropriate endpoint transports. Its exact differential is

$$\delta C_xC_x^{-1} = r_x + \mathrm{Ad}\,P_x(\delta W_xW_x^{-1}) - \mathrm{Ad}\,C_x r'_{x'}, \qquad r_x = \delta P_xP_x^{-1}. \tag{PG15c}$$

Replace $\mathrm{Ad}\,C_x$ by $\mathrm{Ad}\,V$ only in the last term. The error is $O(\ell^2\alpha)$, by the contractible-loop estimate. Averaging the endpoint variations defines the same site function $r_U(a)(y)$ at the two endpoints, with norm $C_L\ell\|a\|$. Its difference is exactly $-\ell d_V r_U(a)$, before the physical division by ℓ. Thus

$$\mathcal{Q}_U = \mathcal{B}_U - d_V r_U + \mathcal{R}_U, \qquad \|\mathcal{R}_U\|_{p\to p} \le C_L\ell^2\alpha, \quad p=1,\infty. \tag{PG15d}$$

Here B_U is the normalized transported middle-line term. Its adjoint differs from the fixed-coordinate-path T_U in Section 3 by $O(\ell^2\alpha)$, since both endpoint transports have the same endpoints and differ by a contractible loop. All statements include the finite physical-volume conversion for adjoints. Pairing (PG15d) with a multiplier satisfying $d_V^*\lambda = 0$ cancels the endpoint term **exactly** before taking absolute values. This proves (PG11) directly and is consistent with the flat-Ward proof (PG12)–(PG13).

In particular the error can be chosen as the full-space linear operator

$$\widetilde{E}_U = \mathcal{B}_U^* - T_U + \mathcal{R}_U^*. \tag{PG15e}$$

Its row and column bounds hold on all coarse currents; only the identity $\mathcal{Q}_U^*\lambda = T_U\lambda + \widetilde{E}_U\lambda$ requires the actual Ward condition. Schur's inequality therefore gives $\|\widetilde{E}_U\|_{2\to 2} \le C_L\ell^2\alpha$, and the positive transported T_U is an L^2 contraction for real links. This proves the L^2 counterpart of (PG15). It avoids an unjustified interpolation argument on a Ward-constrained subspace. The same full-space extension follows from the flat proof by replacing $-r^*d_0^*$ with $-r^*(d_0^* - d_V^*)$ before taking operator norms.

42.5 5. Iteration through the genuine hierarchy

Let $U_{s+1} = M(U_s)$, with spacings h_s and $h_{s+1} = Lh_s$, for $s = 0, \dots, k-1$. Suppose the scale-normalized plaquette curvatures of U_s are bounded by α. Let λ_k be a coarse multiplier satisfying $d_{U_k}^*\lambda_k = 0$ and set

$$\lambda_s = \mathcal{Q}_{U_s}^*\lambda_{s+1}. \tag{PG16}$$

Gauge covariance of the original map gives the exact infinitesimal intertwining

$$\mathcal{Q}_{U_s}d_{U_s} = d_{U_{s+1}}S_s, \qquad d_{U_s}^*\lambda_s = S_s^*d_{U_{s+1}}^*\lambda_{s+1} = 0, \tag{PG17}$$

where S_s samples gauge generators at the retained coarse centres. Thus the Ward hypothesis is preserved by the **actual chain**. There is no claim that a Duhamel error column is separately co-closed.

Write $\ell_s = h_{s+1}$ and $\ell_* = h_k$. Applying (PG15) gives

$$\begin{aligned} \|\lambda_s\|_p &\le \exp\left(C_L\alpha\sum_{v=s}^{k-1}\ell_v^2\right)\|\lambda_k\|_p, \\ \mathrm{TV}_{U_0,h_0}(\lambda_0) &\le \mathrm{TV}_{U_k,h_k}(\lambda_k) + C_L\alpha\sum_{s=0}^{k-1}\ell_s\|\lambda_{s+1}\|_1 \\ &\le \mathrm{TV}_{U_k,h_k}(\lambda_k) + C_L\alpha\ell_* e^{C_L\alpha\ell_*^2}\|\lambda_k\|_1. \end{aligned} \tag{PG18}$$

The constants are independent of k. Geometric summation uses $\sum_s \ell_s \le \ell_*/(1-L^{-1})$ and $\sum_s \ell_s^2 \le \ell_*^2/(1-L^{-2})$. In particular the intermediate forcing interfaces are retained and paid, rather than assumed to disappear.

For a genuine constrained critical orbit, gauge invariance of the action and the Lagrange multiplier equation $DS(U_0) = (DM_{U_0}^k)^*\lambda_k$ imply the terminal Ward condition: apply both sides to a fine gauge variation and use the surjectivity of centre sampling. The complete current λ_0 in (PG18) is therefore the actual physical Euler current. A terminal bound on λ_k is still required when this estimate is used in an old-member norm; it cannot be inferred merely by writing down a multiplier equation. At a fixed terminal lattice its variation is bounded by $C\ell_*^{-1}$ times its L1 norm, so a bounded terminal multiplier gives a uniform variation bound per fixed physical volume.

42.5.1 5.1 Intermediate curvature control

The uniform intermediate-curvature premise has its own fixed-L local estimate. Let F_s denote the physical logarithmic plaquette curvature. The flat identity (PG12) implies

$$d_c \mathcal{Q}_{0,\mathrm{raw}} = \mathcal{Q}_0^{(2)} d, \qquad \|\mathcal{Q}_0^{(2)}\|_{\infty\to\infty} \le 1, \tag{PG19}$$

because the connector is exact and disappears under curl. The two-form average $Q_0^{(2)}$ is the normalized box average followed by normalized length-L line averages in the two plaquette directions. The identity follows by telescoping the coarse differences over their L fine bonds.

In a local tree gauge, all dimensionless arguments of the finite one-step map have size $O_L(\ell_s^2 \|F_s\|_\infty)$. Taylor's formula after (PG19) therefore gives

$$\|F_{s+1}\|_\infty \le \|F_s\|_\infty + C_L \ell_s^2 \|F_s\|_\infty^2. \tag{PG20}$$

The difference between logarithmic plaquette curvature and $h_s^{-2}(1 - U_p)$ is absorbed in the same quadratic remainder. If $\|F_0\|_\infty = \alpha_0$ and $C_L \alpha_0 \ell_*^2$ is sufficiently small, a bootstrap in (PG20) gives $\|F_s\|_\infty \le 2\alpha_0$ for every s. The leading coefficient is one; replacing it by C_L would invalidate this all-depth conclusion.

Equations (PG18)–(PG20) concern genuine averages of the same real field on regular, nonwrapping patches. They do not assert a corresponding bound for independently modified random-walk operators or for all large-field history reconstructions.

42.5.2 5.2 Local weights without a depth-dependent loss

Let a positive spatial weight satisfy $w(x)/w(y) \le \exp(\kappa|x-y|/H)$, where the final retained spacing $\ell_* \le H$. The one-step reference, error, and connector carriers have diameter at most $C_L \ell_s$. Conjugating each L^p estimate by w therefore costs at most $\exp(C_L \kappa \ell_s / H)$. Define weighted variation by inserting $w(x)$ in the sum (PG6), without differentiating the weight. The inherited coarse jump has the same factor, and the additive curvature term uses $\|w\lambda\|_1$.

Multiplying these factors over the genuine hierarchy costs

$$\exp\left(C_L \kappa H^{-1} \sum_s \ell_s \right) \le e^{C_L \kappa}. \tag{PG20a}$$

Thus the real or complex BV estimates have weighted versions with no microscopic-depth loss. They can be applied to exponentially localized terminal multiplier columns when such column bounds have actually been proved. This observation does not derive localized terminal kernels or arbitrary parameter-tree estimates from an unweighted global L1 bound.

42.6 6. What BV supplies, and what it does not

In a bounded local gauge, covariant BV controls ordinary BV with the usual additional term $C\|A\|_\infty \|J\|_1$. For a real bounded current J, translation and interpolation give

$$\|\tau_z J - J\|_2^2 \le 2\|J\|_\infty \,\mathrm{TV}(J)|z|. \tag{PG21}$$

Consequently J has uniform local H^s norm for every fixed $0 < s < 1/2$, with the factor proportional to $(1-2s)^{-1}$ in the standard difference-quotient integral. This is sufficient input for the integrated quadratic D6 remainder developed in the separate dimension-six note, provided the relevant curvature

Hodge/elliptic estimate and marked-readout identification are made on the same actual map. It is much weaker than arbitrary pointwise higher regularity.

For comparison, an all-jet interior implication has the following precise form. On $B(x,2\rho)$, suppose a fixed local gauge gives a uniformly elliptic small-field Wilson equation with smooth forcing J and coefficients in the same local analytic chart. If, after scaling the ball to unit size, ρA is uniformly small in $C^{1,\beta}$ and $\rho^3 J$ is bounded in $C^{r-1,\beta}$, then the usual local discrete elliptic bootstrap gives

$$\rho^{r+2}\|\nabla_A^r F\|_{L^\infty(B(x,\rho))} \le C_{r,\beta}\big(\|\rho A\|_{C^{1,\beta}(B_2)} + \|\rho^3 J\|_{C^{r-1,\beta}(B_2)}\big), \quad r \ge 1. \tag{PG22}$$

Here the norms on the right are in rescaled coordinates. The premise includes the chosen local elliptic gauge and smoothness of the **actual forcing**. It is not an assertion that the original projected Landau operator is already a local scalar Laplacian. One proves (PG22) by the uniform discrete Schauder estimate, starting with the small principal perturbation of the lattice Laplacian, and then differentiating and bootstrapping. The nonlinear terms use only derivatives already controlled in the preceding step. A common complex parameter ball with these bounds would also give the corresponding parameter Cauchy estimates.

Neither (PG18) nor the original low-derivative paired core supplies the $C^{r-1,\beta}$ forcing premise for all r. Face clearance only provides an interior region on which an already controlled forcing may be differentiated. It does not increase its regularity by itself.

42.7 7. Why fixed-order face defects do not imply all-jet bounds

Suppose, only as a general diagnostic, that forcing errors at scale ℓ_s have amplitude at most $C\ell_s^p$. If the elliptic inverse improves by two derivatives, the expected r-th potential derivative cost is $C_r\ell_s^{p+2-r}$. Even a fixed positive face margin changes only C_r by a power of the margin; it does not alter this scale exponent. Hence a finite p is enough only below a corresponding finite derivative threshold.

This limitation is sharp already for smooth periodic scalar forcings. With $\ell_s = L^{-s}$, set

$$f_s(x) = \ell_s^p \cos(2\pi x_1/\ell_s), \qquad u_s = (-\Delta)^{-1} f_s = (2\pi)^{-2}\ell_s^{p+2}\cos(2\pi x_1/\ell_s). \tag{PG23}$$

The r-th derivative has size $C_r\ell_s^{p+2-r}$. There are no block faces in this example. Thus a summable BV correction is not an all-derivative estimate. Formula (PG23) is not being presented as an actual Wilson critical field; it only identifies the extra scale gain that an all-jet argument would need.

For the nonlinear contour map, a derivative of an adjoint transport can create a curvature insertion along a ribbon through the entire path. Local smoothness near its endpoint does not alone bound that derivative if the path crosses a rough remote region. The exact lattice path/ribbon identities are in the completed-insertions note. An all-jet norm must therefore control the relevant path dependence or prove an actual cancellation/factorization of its remote part. Nonassociativity (PG3) rules out assuming such a factorization merely from the flat composition law.

43.2 2. The exact endpoint terms of the contour derivative

The outer one-step operation in Bałaban I (0.4), and in its symmetrized-contour version (0.12), is the **reference-log mean**

$$\mathcal{A}(P_1,\dots,P_n;R)=\exp\left(\frac{1}{n}\sum_i \log(P_iR^{-1})\right)R. \tag{LR6a}$$

Here R is the reference coarse-bond path transport. The separate symmetric, bi-equivariant group mean of I (0.10) is used inside the endpoint contour transports of (0.11). These two operations must not be identified. The complete center-to-center contours have the form

$$C_{y,x,\mu}=P_{y,x}\,W_{x,x+\ell e_\mu}\,P^{-1}_{y+\ell e_\mu,x+\ell e_\mu}. \tag{LR6}$$

Here W is the straight line of L fine bonds. In (0.4), P is the specified endpoint path transport; in (0.12), P is its internally averaged endpoint transport from (0.11). The latter is within $O_L(F\ell^2)$ of any fixed shortest coordinate path on the same regular patch.

Assume the real links are regular on the bounded neighborhood of these contours and

$$|U(\partial p)-1|\le F\eta^2, \qquad F\ell^2\le c. \tag{LR7}$$

Every two contours in (LR6) with the same endpoints differ by a contractible loop of area at most $C_d\ell^2$. Elementary plaquette filling and unitary conjugation imply

$$|C_{y,x,\mu}V_\mu(y)^{-1}-1|\le C_LF\ell^2, \qquad V=MU. \tag{LR8}$$

The estimate for the outer mean follows directly from (LR6a). Its right-trivialized differential at $P_i = R$ is $n^{-1}\sum_i \dot{P}_i P_i^{-1}$: the two occurrences of $\dot{R}R^{-1}$ cancel. More generally, if $|P_i R^{-1} - 1| \le \epsilon$, analytic differentiation of (LR6a) gives

$$\dot{\mathcal{A}}\mathcal{A}^{-1} = \frac{1}{n}\sum_i \dot{P}_i P_i^{-1} + \mathcal{R}, \qquad |\mathcal{R}| \le C\epsilon\left(\frac{1}{n}\sum_i |\dot{P}_i P_i^{-1}| + |\dot{R}R^{-1}|\right). \tag{LR8a}$$

Thus the reference variation has not been suppressed. Its coefficient vanishes on the diagonal and is $O_L(F\ell^2)$ on the regular patch. Since $|\dot{R}R^{-1}| \le C\ell\|A\|_\infty$, division by ℓ puts it in the same error class as the contour-weight corrections. For the *inner* symmetric endpoint mean, permutation symmetry and $M(P,\dots,P) = P$ give the arithmetic differential on the diagonal; its off-diagonal differential is controlled by the same finite-step analytic estimate.

Define the site-valued linear map

$$(r_U A)(y) = \mathrm{av}_x \frac{1}{i}\frac{d}{dt} P_{y,x}(e^{i\eta t A}U)P_{y,x}(U)^{-1}\Big|_{t=0}. \tag{LR9}$$

For an averaged endpoint P, the derivative in this formula is the derivative of that entire endpoint mean. It remains a finite-step sum of transported fine field legs with uniformly bounded analytic coefficients, so its operator norms satisfy $\|r_U\| \le C_L\ell$.

Differentiation of one whole contour gives the exact identity

$$\frac{1}{i}\dot{C}C^{-1} = \frac{1}{i}\dot{P}P^{-1} + \mathrm{Ad}(P)\frac{1}{i}\dot{W}W^{-1} - \mathrm{Ad}(C)\frac{1}{i}\dot{P}'P'^{-1}. \tag{LR10}$$

Replace only the last $\mathrm{Ad}(C)$ by $\mathrm{Ad}(V_\mu(y))$. Its error is $O_L(F\ell^2)$ times an endpoint path variation. After dividing by ℓ, averaging, and including both the outer contour-weight and reference-variation errors in (LR8a), replace the endpoint transport in the middle term by a fixed coordinate-order endpoint path; its $O_L(F\ell^2)$ difference is also put in the error. Denote this chosen positive long-line map by $Q_{\mathrm{long},U}$. This proves

$$Q_U = Q_{\mathrm{long},U} - D_V r_U + \mathcal{E}_U, \qquad \|\mathcal{E}_U\|_{\infty\to\infty} + \|\mathcal{E}_U\|_{1\to 1} \le C_L F\ell^2. \tag{LR11}$$

The placement of the adjoint norms is fixed by the volume measures in (LR1); both adjoint bounds have the same estimate after changing C_L. The finite ratio L, finite path lengths, and bounded local overlap are used here. No constant depends on the total number of earlier steps.

For a physical coarse multiplier, the exact Ward equation is

$$D_V^*\lambda = 0. \tag{LR12}$$

Consequently the endpoint term cancels before taking absolute values:

$$Q_U^*\lambda = T_U\lambda + E_U\lambda, \qquad T_U = Q_{\mathrm{long},U}^*, \quad E_U = \mathcal{E}_U^*,$$

$$\|E_U\lambda\|_\infty \le C_L F\ell^2\|\lambda\|_\infty, \qquad \|E_U\lambda\|_{1,\eta} \le C_L F\ell^2\|\lambda\|_{1,\ell}. \tag{LR13}$$

In this decomposition T_U may be chosen to use a fixed coordinate-order endpoint path. Replacing an internally averaged endpoint transport by that path changes the middle-segment adjoint by $O_L(F\ell^2)$, which is included in E_U. This is a statement about the complete derivative on a constrained input. It is not a Duhamel expansion in which $E_U\lambda$ is subsequently propagated as though it were separately constrained.

The Ward condition is legitimate for an actual constrained critical background. Infinitesimal gauge covariance of the complete average gives

$$Q_U D_U = D_V R, \qquad D_U^* Q_U^* = R^* D_V^*, \tag{LR14}$$

where R restricts the fine gauge parameter to coarse sites. Gauge invariance of the Wilson action and its Euler equation $J = Q_U^* \lambda$ imply (LR12), since R^* is injective. The complete output in (LR13), after recombination, again obeys its fine Ward equation. This is the property used at the next step.

43.3 3. Coefficient-one covariant variation and its iteration

Put

$$\mathrm{TV}_{\ell,V}(\lambda) = \ell^{d-1} \sum_{y,\mu,\nu} |\mathrm{Ad}(V_\nu(y))\lambda_\mu(y + \ell e_\nu) - \lambda_\mu(y)|. \tag{LR15}$$

The transported long-segment adjoint T_U is the same positive average of shifted block holds as (LR3), with each value conjugated along the corresponding prefix of the path (LR6). For a fixed shift and a fixed path permutation, compare adjacent fine points.

- If their held coarse cell is unchanged, the two prefixes, joined by the fine edge, bound a ribbon of area at most $C_d \eta \ell$. This produces an error at most $C_d F \eta \ell \, |\lambda(y)|$.
- If their held cells are neighboring coarse cells, insert the coarse transport $V_\nu(y)$. The principal difference is exactly the term occurring in (LR15). The discrepancy of the two open paths from that coarse transport is at most $C_L F \ell^2$, by (LR8) and a plaquette filling in their bounded union of cells.

For the principal differences, the interface count in (LR5) applies without change. The same-cell ribbon errors, after the fine variation factor η^{d-1} is summed, give $C_L F \ell \|\lambda\|_1$. The coarse-interface errors give the same bound, since $\ell^{d-1} F \ell^2 = F \ell \, \ell^d$. Averaging over shifts and permutations is positive and preserves the principal coefficient one. Thus

$$\mathrm{TV}_{\eta,U}(T_U \lambda) \le \mathrm{TV}_{\ell,V}(\lambda) + C_L F \ell \|\lambda\|_{1,\ell}. \tag{LR16}$$

The shortest coordinate paths must be chosen consistently under translation; the symmetrized fixed-permutation paths in I (0.4) have this property. Sign changes of an individual coordinate segment have zero length at the transition and do not introduce an extra discontinuity. The estimate is local on a contractible chart containing the contours. A noncontractible wrapping carrier must be treated separately, as in the original expansion.

For the error in (LR13), no regularity assumption is needed:

$$\mathrm{TV}_{\eta,U}(E_U \lambda) \le C_d \eta^{-1} \|E_U \lambda\|_{1,\eta} \le C_L F \ell \|\lambda\|_{1,\ell}. \tag{LR17}$$

Unitarity is used in the first inequality. Combining the complete output gives

$$\begin{aligned} \|Q_U^* \lambda\|_{1,\eta} &\le (1 + C_L F \ell^2) \|\lambda\|_{1,\ell}, \\ \|Q_U^* \lambda\|_{\infty} &\le (1 + C_L F \ell^2) \|\lambda\|_{\infty}, \\ \mathrm{TV}_{\eta,U}(Q_U^* \lambda) &\le \mathrm{TV}_{\ell,V}(\lambda) + C_L F \ell \|\lambda\|_{1,\ell}. \end{aligned} \tag{LR18}$$

Now iterate the actual adjoint steps, always recombining (LR13) before the next step and using (LR14). If $\ell_j = L^{-j} \ell_0$ and the fine and averaged plaquettes satisfy one uniform physical curvature bound F, then

$$\|\lambda_j\|_{1,\ell_j} + \|\lambda_j\|_\infty \le \exp\left(\frac{C_L F \ell_0^2}{1 - L^{-2}} \right) (\|\lambda_0\|_{1,\ell_0} + \|\lambda_0\|_\infty),$$

$$\mathrm{TV}_{\ell_j,U_j}(\lambda_j) \le \mathrm{TV}_{\ell_0,U_0}(\lambda_0) + C_L F \frac{\ell_0}{1-L^{-1}} \exp\left(\frac{C_L F \ell_0^2}{1-L^{-2}}\right) \|\lambda_0\|_{1,\ell_0}. \tag{LR19}$$

In particular there is no factor C^j. One can use the trivial terminal estimate $\mathrm{TV}_{\ell_0,U_0}(\lambda_0) \le C_d \ell_0^{-1} \|\lambda_0\|_{1,\ell_0}$. This also makes clear the scale and volume normalization.

43.3.1 3.1 The actual complex hierarchy: a nonunitarity budget

The path estimate needed to complexify this argument can be proved directly from the actual averaging formula. It need not be posited as a new common-gauge hypothesis.

In a fixed faithful unitary matrix representation put

$$N(W) = \max\{\|W\|, \|W^{-1}\|\}, \qquad K_s = h_s^{-1} \sup_b \log N(U_s(b)), \qquad F_s = h_s^{-2} \sup_p \|\log U_s(\partial p)\|. \tag{LR19a}$$

Here $h_{s+1} = Lh_s$, $U_{s+1} = M(U_s)$, and the principal plaquette logarithms lie in their fixed small chart. The quantity N is submultiplicative. Thus a path of physical length at most $C\ell$, made out of h_s-bonds, has

$$N(P) \le e^{CK_s\ell}, \qquad \|\mathrm{Ad}(P)\| \le e^{CK_s\ell}. \tag{LR19b}$$

The constants in the two exponents can differ. Original complex links $U_s(b) = e^{ih_s A'(b)} \bar{U}_s(b)$ with $\bar{U}_s$ unitary obey $K_s \le C\|A'\|_\infty$ at the initial scale. Only this initial assertion is required below.

Set $\ell = h_{s+1}$. Plaquette filling on a one-step patch now uses (LR19b), and bounds every contour relative to its straight reference by

$$\|P_i R^{-1} - 1\| \le C_L \ell^2 F_s e^{CK_s\ell}. \tag{LR19c}$$

The same estimate holds with internally averaged endpoints: factor one reference endpoint path, use the small relative loops, and apply the analytic mean in that neighborhood. Its reference path has the same bound (LR19b).

For the outer formula (LR6a), write $M(U_s) = e^Z R$. The straight reference contains L fine bonds, so $N(R) \le e^{K_s\ell}$, while $\|Z\| \le C_L \ell^2 F_s e^{CK_s\ell}$. Since $N(e^Z R) \le e^{\|Z\|} N(R)$, one obtains

$$K_{s+1} \le K_s + C_L \ell F_s e^{CK_s\ell}. \tag{LR19d}$$

There is no fixed coefficient multiplying K_s.

The curvature estimate has the analogous normalization:

$$F_{s+1} \le e^{CK_s\ell} F_s + C_L \ell^2 F_s^2 e^{CK_s\ell}. \tag{LR19e}$$

For completeness, take a tree gauge on a fixed one-step patch. Every tree gauge transport and its inverse has norm bounded by (LR19b). Every transformed non-tree bond is a contractible curvature loop and is $O_L(\ell^2 F_s e^{CK_s\ell})$ from the identity. At the identity the coarse curl of the raw contour average is the positive box-and-two-line average of the fine curl; the endpoint gradient has zero curl. Its physical sup norm is at most one. Taylor's quadratic remainder is $O_L(\ell^4 F_s^2)$ before dividing by the coarse plaquette area ℓ^2, with the bounded transport factors included. Replacing linear curl by the fine logarithmic plaquette curvature contributes the same remainder. Undoing the local tree gauge only inserts another factor $e^{CK_s\ell}$. This proves (LR19e) with a changed exponent constant. The leading coefficient is exponential in $K_s\ell$, and equals one at $K_s = 0$; it is not an arbitrary C_L.

Let $H = h_k$ be the largest physical scale under consideration. There is a fixed $\epsilon_L > 0$ such that

$$K_0 H + F_0 H^2 \le \epsilon_L \tag{LR19f}$$

implies, for every finite hierarchy ending at or before H,

$$F_s \le 2F_0, \qquad K_s \le K_0 + C_L F_0 H. \tag{LR19g}$$

Indeed, under these two bootstrap bounds, sum (LR19d) and take logarithms of the multiplicative version of (LR19e). Use $\sum \ell \le H/(1 - L^{-1})$ and $\sum \ell^2 \le H^2/(1 - L^{-2})$. Choose the constant in the bound for $K_s - K_0$ first, then choose ϵ_L so that the resulting exponent for F_s/F_0 is less than $\log(3/2)$. This improves both bootstrap bounds and keeps every relative logarithm/mean argument inside its analytic chart. The constants do not depend on the number of levels. The same argument applies uniformly to a complex parameter family whose initial K_0, F_0 bounds are uniform.

The one-step current inequalities in the complex case become

$$\begin{aligned} \|Q_U^* \lambda\|_p &\le e^{CK_s\ell}(1 + C_L F_s \ell^2)\|\lambda\|_p, \\ \mathrm{TV}_{\eta,U}(Q_U^* \lambda) &\le e^{CK_s\ell}\mathrm{TV}_{\ell,V}(\lambda) + C_L F_s \ell e^{CK_s\ell}\|\lambda\|_1. \end{aligned} \tag{LR19h}$$

All adjoints here are the bilinear analytic adjoints, not complex-conjugate operations on the parameter dependence. The endpoint and Ward identities remain algebraic. Equations (LR19g)–(LR19h) give a common depth-independent complex BV bound, since $\sum K_s \ell$ and $\sum F_s \ell^2$ are bounded. An arbitrary independent complex current J is still not covered: the current must be the complete Ward-compatible adjoint transport. This result also does not assert that independently wall-localized propagators coincide with such an actual averaged hierarchy.

43.3.2 3.2 Fractional consequence

On a local gauge chart with bounded connection, covariant variation and ordinary variation differ by at most $C\|A\|_\infty \|\lambda\|_1$. The piecewise-constant embedding of a uniformly bounded field with bounded variation obeys

$$\|\tau_h J - J\|_2^2 \le 2\|J\|_\infty \,\mathrm{TV}(J)|h|. \tag{LR20}$$

Integrating the fractional difference quotient gives a uniform H^s bound for every $0 < s < 1/2$. This is the useful force regularity for the proposed dimension-six integrated readout. It supplies neither C^∞ force nor all-order spatial jets.

The assertion proved here is for actual, unlocalized critical forces on a uniform hierarchy (and a local regular chart). Mixed determining-set interfaces and the independently interpolated background maps require the further verification in the next section. They are not silently included in (LR19). In particular the terminal variation in (LR19) is measured at the actual terminal spacing. If an inner reconstruction is stopped at an old spacing r, the elementary terminal bound is $Cr^{-1}\|\lambda_{\mathrm{terminal}}\|_1$; it is not a current-scale bound independent of r.

There is an exact favorable special case. If $U_k(V)$ is the genuine full k-step constrained minimum, nested constraints and uniqueness imply

$$U_j(M^j U_k(V)) = U_k(V) \tag{LR20a}$$

on their common domain, modulo the allowed gauge transformation. Indeed every configuration with the same j-step average as $U_k(V)$ also has its k-step average equal to V. A smaller action on the

former constraint set would contradict the defining k-step minimum; uniqueness identifies the critical orbits. Thus the full k-scale force estimate can be used for this exact nested readout. Arbitrary scalar interpolations or independently wall-localized outer data need not satisfy this identity. For those data an additional Sobolev stability theorem for the inner reconstruction is needed; replacing its own terminal spacing by the current one would be an unjustified gain.

43.3.3 3.3 An interior version of the physical-current estimate

The endpoint cancellation can be used locally. This is useful for an actual mixed determining set and is different from localizing a propagator by wall parameters.

Consider a uniform chain of actual fixed-L averaging maps between the fine spacing a and terminal spacing r. Each one-step derivative and endpoint map has a kernel supported within physical distance $c_L\ell$ of its input, where ℓ is that step's coarse spacing. This follows directly from the finite paths and block means in the averaging definition. Put

$$c_* = \frac{4c_L L}{L-1}.$$

Let a terminal multiplier λ be defined on a coarse region D, and obey the coarse Ward equation on D. Suppose the fine region B satisfies

$$\operatorname{dist}(B, D^c) > c_* r. \tag{LR20b}$$

Then the force on B obtained by the actual adjoint chain is independent of an extension of λ outside D, and obeys the interior versions of (LR19):

$$\begin{aligned} \|J\|_{p,B} &\le C\|\lambda\|_{p,D}, \qquad p = 1, \infty, \\ \mathrm{TV}_B(J) &\le C\big(\mathrm{TV}_D(\lambda) + Fr\|\lambda\|_{1,D}\big). \end{aligned} \tag{LR20c}$$

In the complex case the constants contain the bounded path factor in (LR19g), with the same fine-cutoff independence.

Here is a local proof that does not apply the Ward estimate to an arbitrary cutoff error. Extend λ by zero outside D, with a collar large enough to include the coarse differences used in the displayed norms. Its Ward defect is confined to the exterior collar. The exact chain identity $D^*_{\mathrm{fine}} Q^*_U = R^* D^*_{\mathrm{coarse}}$ transports that defect only to the associated sampled sites. The endpoint terms attached to those defects are supported within distance $c_L\ell$ of them. Subsequent averaging factors can enlarge this support by at most $\sum_{\ell \le r} c_L \ell$, which is less than $c_* r/2$. They therefore do not meet B. On the nested interior regions containing the dependency cone of B, all endpoint terms cancel exactly. Apply (LR17)–(LR18) there and sum the geometric scale gains. This proves (LR20c). No global Ward-compatible extension has been assumed, and no estimate of its exterior defect is needed.

For a true mixed critical background, this lemma applies only after identifying the actual local multiplier and its bounds. The nested collars of the variational paper's (144), (148)–(150) are a useful interface: shell constraints whose complete averaging carriers miss B cannot contribute to the physical current there. In a region meeting two adjacent terminal layers, a single additional fixed-L comparison is allowed; it does not give a factor once per old depth. This observation reduces the mixed-domain question to the local terminal multiplier and the finitely many interface stencils. It does not yet assert their required quantitative bounds or apply to the wall-modified nonlinear maps of §4.

43.6 6. The nonlinear chart has a separate compatibility condition

The protected inverse identity (LR25) is linear. The actual formula II (1.17) also uses a nonlocal nonlinear change of variables. A local equation for each linear block does not, by itself, establish the physical Euler equation after that change of variables.

Here is the exact finite-dimensional algebra. Write a nonlinear constraint as

$$\mathcal{F}(A) = QA + C(A), \qquad QH = I, \qquad A = \Psi_H(A') = A' - HC(A). \tag{LR28}$$

Assume the chart is invertible on the stated small domain. For the physical action gradient $J = S'(A)$, the gradient in the new coordinates is

$$\widetilde{J} = (D\Psi_H)^* J, \qquad J = (I + C'(A)^* H^*)\widetilde{J}. \tag{LR29}$$

These stars are bilinear transposes on a complex analytic domain. If the transformed fixed-point equation gives $\widetilde{J} = Q^*\lambda + R$, then

$$J = \mathcal{F}'(A)^*(\lambda + H^*R) + (I - Q^*H^*)R. \tag{LR30}$$

Even when R vanishes on a protected set, the last term in (LR30) need not vanish there: H^*R can carry an exterior residual back into an interior coarse multiplier. An assertion that each localized inverse has an interior equation must therefore be accompanied either by the compatible full nonlinear equation or by a bound for this transported residual.

An elementary example makes the logical point concrete. In $\mathbb{R}^3$, with coordinates (x, y, z), take

$$\mathcal{F}(x, y, z) = y + \gamma x^2, \qquad S(x, y, z) = \tfrac{1}{2}(x^2 + y^2 + z^2) - tx, \qquad Hc = (0, c, ac). \tag{LR31}$$

Thus $QH = I$. On the constraint chart with $QA' = b$,

$$A' = (x, b, z'), \qquad A = (x, b - \gamma x^2, z' - a\gamma x^2).$$

Use a bounded auxiliary covariance $G_s = \mathrm{diag}(1, 0, s)$, and choose $a = a_0(1 - s)$. At $s = 1$ this is the ordinary constrained stationary construction. At $s = 0$ its fixed-point equation imposes $\widetilde{J}_x = 0$ and $z' = 0$, while the exterior transformed gradient $\widetilde{J}_z = -a\gamma x^2$ is retained. For small nonzero t, the implicit-function theorem gives a small nonzero x. The physical gradients then satisfy

$$J_y = b - \gamma x^2, \qquad J_x - 2\gamma x J_y = -2a^2\gamma^2 x^3 \neq 0. \tag{LR32}$$

There is no multiplier that makes the physical Euler equations hold on both protected coordinates x, y, although the transformed x-equation holds and all auxiliary maps are analytic with uniform low bounds. This example does not claim that the actual Wilson propagators have this defect; it shows that the arbitrary-operator low-norm replacement argument in II (1.5) alone cannot rule it out.

A sufficient constructive repair is to localize one augmented inverse and solve the nonlinear KKT system directly in the field and multiplier together. If

$$\Xi_s = \Xi_{\mathrm{ref}} - \mathbb{G}_s(\mathbb{L}\Xi_{\mathrm{ref}} + \mathcal{N}(\Xi_s)) \tag{LR33}$$

and the protected rows obey $\mathbb{L}\mathbb{G}_s = I$, then those rows of the *full* nonlinear equation $\mathbb{L}\Xi_s + \mathcal{N}(\Xi_s) = 0$ hold exactly. This follows by applying $\mathbb{L}$ to (LR33); it does not multiply independently localized block identities. A fixed-point theorem on a fractional current hull would then give a precise invariant-domain statement. Identifying the original II (1.2)–(1.17) construction with this jointly localized system, or proving an adequate H^s estimate for the residual (LR30), remains necessary. No such identification is being inferred from the paper's low-norm analyticity theorem.

43.7 7. The retrieved variational paper identifies the compatibility

The full primary variational paper, CMP **102** (1985), 277–309, has now been retrieved. Its equations make the distinction in §6 concrete.

The nonlinear constraint in its (20), (44) is the adapted $Q_j(U_0, \eta A)$ after the gauge-coordinate transformation. One must keep it distinct from the raw group average and from the flat box-and-line operator. The operator H in (45) is the quadratic constrained minimizer for the original Hessian; it is a right inverse to the adapted linear Q and obeys the specified Landau condition. The linearizing transformation is precisely (LR28), with the layer normalization inserted into Q and C.

For the gauge-restricted quadratic form, equations (99)–(111) first construct the constrained covariance and solve the nonlinear stationary equation. Physical gauge invariance is then used in (123)–(128) to remove the restriction on variations in the Landau direction. Only after that step are the improved regularity operators introduced. In normalized notation their identities are

$$G = \Delta_a^{-1}, \qquad H_0 = GQ^*(QGQ^*)^{-1}, \qquad P_0 = I - H_0 Q, \tag{LR34}$$

$$\widetilde{G} = GP_0^* = G - GQ^*(QGQ^*)^{-1}QG. \tag{LR35}$$

These are the identities in (129)–(143), with the explicit own-layer factor on the paper's B absorbed in our coarse variable. The line preceding (143) visibly omits the final QG in its displayed subtraction; (LR35) is the exact algebraic consequence of its (131).

The resulting alternative equation uses

$$A_0 = -\widetilde{G}J + \widetilde{G}\Delta^{(2)}(A_0 + H_0 B) - \widetilde{G}V'(A_0 + H_0 B). \tag{LR36}$$

Immediately after (143), the paper explains that this equation alone need not impose the Landau condition and therefore does not replace the first existence equation. Section G again gives the two representations separately: (174)–(175) and (179)–(180). They agree for the physical minimizer.

Consequently the wall-localized version requires more than one unqualified assertion that all substituted operators have the same bounds. A successful identification must preserve the gauge-coordinate identities used to pass between the two representations, or must estimate the defect of that passage. The source gives improved covariant-Laplacian bounds for the physical maps; it does not identify an arbitrary independently localized replacement with a physical constrained solution. This is a precise operator-level issue, rather than a general absence of a differentiability theorem.